



CLC

Educational Institute

DSAT FOUNDATION COURSE

Workbook Practice — Student Edition

435 Questions — 15 per Skill, Every Skill on the PSAT Math Test

All 29 skills across Algebra · Problem-Solving & Data Analysis · Advanced Math · Geometry & Trigonometry — 5 Easy + 5 Medium + 5 Hard per skill

Every skill opens with its formula and fastest-solving approach. No worked solutions or answers are shown — every question is solved by the student, with a Desmos hint where one genuinely helps.

Prepared and Compiled by:
The CLC Digital SAT Team

CLC Educational Institute



How to Use This Workbook

What's In Every Skill Section

- A Formula & Fastest Approach header — the core formula(s) for this skill and the one habit that solves most of its questions quickest.
- 15 questions: 5 Easy to build confidence, 5 Medium with a genuine twist each, and 5 Hard combining two ideas at once.
- A Desmos hint under any question where the calculator genuinely helps — telling you what to type in, not the answer.
- Ruled work space under every question for you to solve it by hand.

How to Practice

- Read the Formula & Fastest Approach header before starting a skill's questions.
- This edition has no worked solutions or answer key — solve every question fully on your own before moving on.
- Work Easy, then Medium, then Hard, in order — each tier is calibrated to build on the one before it.
- For full worked solutions, Tips, Traps, and Process of Elimination on every question, use the companion full-solution Workbook Practice with your tutor.
- This workbook assumes you already know how to use the Digital SAT's built-in Desmos calculator — see the companion's Desmos primer if you need a refresher.



PART 1 — ALGEBRA

1. Lines

Formula & Rules for This Skill

Slope: $m = (y_2 - y_1) / (x_2 - x_1)$ · Slope-intercept form: $y = mx + b$ ·
Point-slope form: $y - y_1 = m(x - x_1)$ · Parallel lines share slope;
perpendicular slopes multiply to -1.

Fastest Way to Solve: Before doing anything else, decide what the question actually needs — slope, an intercept, or a full equation — then pick the form of the line that gets you there in one step.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Lines — Question 1 (Easy)

What is the slope of the line that passes through the points (2, -3) and (6, 5)?

Desmos: Type $(5 - (-3)) / (6 - 2)$ directly into Desmos as an expression and read the computed value as the slope.

Lines — Question 2 (Easy)

The equation of a line is $y = -3x + 7$. What is the y-intercept of this line?

Desmos: Type $y = -3x + 7$ into Desmos and look at the point where the graph crosses the y-axis.



Lines — Question 3 (Easy)

A taxi ride in Athens costs a flat fee of 3 euros plus 1.5 euros per kilometer traveled. Write an equation for the total cost C , in euros, of a ride of d kilometers.

Desmos: Type $y = 1.5x + 3$ into Desmos and trace to whole-number x -values to confirm the cost at those distances.

Lines — Question 4 (Easy)

What is the x -intercept of the line $2x + 5y = 20$?

Desmos: Type $2x + 5y = 20$ into Desmos and trace along the graph to where it crosses the x -axis.

Lines — Question 5 (Easy)



A line has slope $\frac{4}{3}$. What is the slope of a line that is perpendicular to it?

Desmos: Type $y = \frac{4}{3}x$ and $y = -\frac{3}{4}x$ together in Desmos and confirm visually that the two lines meet at a right angle.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Lines — Question 6 (Medium)

A line is given by the equation $6x - 2y = 10$. What is the slope of this line?

Desmos: Type $6x - 2y = 10$ directly into Desmos; trace two points on the resulting line to confirm the slope is 3.

Lines — Question 7 (Medium)

A line has slope 5 and y-intercept 3. If the point $(2, k)$ lies on this line, what is the value of k ?

Desmos: Type $y = 5x + 3$ into Desmos and trace to $x = 2$ to read the y-value directly.



Lines — Question 8 (Medium)

Line p has equation $y = (2/3)x - 1$. Line q is perpendicular to line p and passes through the point $(4, 5)$. What is the equation of line q in slope-intercept form?

Desmos: Type $y = (2/3)x - 1$ and $y = -3/2x + 11$ together in Desmos and confirm the lines cross at a visible right angle near $(4, 5)$.

Lines — Question 9 (Medium)

A ferry departs Piraeus heading toward Aegina. Five minutes after departure, the ferry is 2 kilometers from Piraeus; twenty minutes after departure, it is 8 kilometers from Piraeus. The ferry can carry up to 600 passengers. Assuming the ferry travels at a constant speed, how many kilometers per minute is it traveling?

Desmos: Type $(8-2)/(20-5)$ directly into Desmos as an expression to get the speed as a decimal instantly.



Lines — Question 10 (Medium)

For what value of k are the lines $4x + ky = 12$ and $8x + 6y = 5$ parallel?

Desmos: Type $4x + ky = 12$ in Desmos with k as a slider, and adjust it until the line looks parallel to $8x + 6y = 5$.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Lines — Question 11 (Hard)

Line m is given by $ax - 4y = 9$, where a is a constant. Line n is given by $3x + 6y = -12$. If line m is perpendicular to line n , what is the value of a ?

Desmos: Type $3x + 6y = -12$ to see its slope, then type $ax - 4y = 9$ with a as a slider and adjust until the two lines look perpendicular.

Lines — Question 12 (Hard)

A line passes through the points $(2, p)$ and $(p, 8)$, and has a slope of 2. Given that p is positive, what is the value of p ?

Desmos: Type $(8-x)/(x-2)$ and $y=2$ as separate expressions in Desmos and find where they intersect — that x -value is p .

Lines — Question 13 (Hard)

A tech startup in Tallinn tracks its total costs as a linear function of units produced. When it has produced 50 units, its total cost is 4500 euros. When it has produced 130 units, its total cost is 8900 euros. What is the startup's fixed cost — the cost when zero units are produced — in euros?

Desmos: Type $y = 55x + 1750$ into Desmos and trace to $x = 0$ to confirm the fixed cost reads 1750 on the y -axis.

Lines — Question 14 (Hard)

Line r passes through the point $(1, 2)$ and is parallel to the line $9x - 3y = 15$. For what value of x does line r cross the line $y = 26$?

Desmos: Type $y = 3x - 1$ and $y = 26$ into Desmos and trace to their intersection to read $x = 9$ directly.



Lines — Question 15 (Hard)

Points A(-2, 5), B(4, k), and C(10, 17) all lie on the same line. What is the value of k?

Desmos: Type the line through A and C as $y = 1(x - (-2)) + 5$ in Desmos, then trace to $x = 4$ to confirm $y = 11$ for point B.



2. Systems of Linear Equations

Formula & Rules for This Skill

Substitution: solve one equation for a variable and plug it into the other. **Elimination:** add/subtract equations (after scaling) to cancel a variable. **Same slope + same intercept = infinite solutions; same slope + different intercept = no solution.**

Fastest Way to Solve: Glance at both equations first — if one variable already matches or cancels easily, use elimination; if one equation is already solved for a variable, use substitution.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Systems of Linear Equations — Question 1 (Easy)

Solve the system of equations: $y = 3x - 4$ and $2x + y = 11$. What is the solution (x, y) ?

Desmos: Type $y = 3x - 4$ and $2x + y = 11$ as two separate equations in Desmos and read the coordinates of their intersection point.

Systems of Linear Equations — Question 2 (Easy)

Solve the system of equations: $4x + 3y = 18$ and $4x - y = 2$. What is the solution (x, y) ?

Desmos: Type $4x + 3y = 18$ and $4x - y = 2$ into Desmos as two lines and read the coordinates of their intersection point.

Systems of Linear Equations — Question 3 *(Easy)*

How many solutions does the system of equations $y = 2x + 5$ and $y = 2x - 3$ have?

Desmos: Type $y = 2x + 5$ and $y = 2x - 3$ into Desmos and observe that the two lines are parallel and never touch.

Systems of Linear Equations — Question 4 *(Easy)*

A cinema in Lisbon sells adult tickets for 8 euros and child tickets for 5 euros. One evening, the cinema sold a total of 50 tickets and collected 340 euros. How many adult tickets were sold?

Desmos: Type $y = 50 - x$ and $y = (340 - 8x)/5$ into Desmos and trace to their intersection — the x-value is the number of adult tickets.

Systems of Linear Equations — Question 5 *(Easy)*



The sum of two numbers is 19. Three times the first number minus the second number is 21. What is the first number?

Desmos: Type $y = 19 - x$ and $y = 21 - 3x$ into Desmos and trace to their intersection point to read off x .

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Systems of Linear Equations — Question 6 (Medium)

The system $5x - ay = 14$ and $2x + y = 10$ has solution $(x, y) = (4, 2)$. What is the value of a ?

Desmos: Type $y = 20 - 2x$ into Desmos and trace to where $y = 14$; the x -value there is the value of a .

Systems of Linear Equations — Question 7 (Medium)

Solve the system: $3(x + 2) = y + 11$ and $2y - 4x = 6$. What is the value of $x + y$?

Desmos: Type $3(x+2) = y + 11$ and $2y - 4x = 6$ into Desmos exactly as written and trace to their intersection to confirm x and y .

Systems of Linear Equations — Question 8 *(Medium)*

For what value of k does the system $6x + 9y = 12$ and $4x + ky = 8$ have infinitely many solutions?

Desmos: Type $6x + 9y = 12$ into Desmos, then type $4x + 6y = 8$ as a second line and confirm they draw as the exact same line.

Systems of Linear Equations — Question 9 *(Medium)*

A bookstore in Rotterdam sells paperback novels for 9 euros and hardcover novels for 15 euros. The store, which first opened in 2015, sold a combined total of 40 books in one day and brought in 456 euros. How many hardcover novels did the store sell that day?

Desmos: Type $y = 40 - x$ and $y = (456 - 15x)/9$ into Desmos and trace to their intersection — the x -value there is the number of hardcovers.

**Systems of Linear Equations — Question 10 (Medium)**

The solution (x, y) to the system $x - y = 1$ and $2x + y = 11$ also satisfies the equation $kx + y = 15$. What is the value of k ?

Desmos: Type $y = x - 1$ and $y = 11 - 2x$ into Desmos, trace to confirm the intersection $(4, 3)$, then compute $k = (15 - 3)/4$ by hand.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Systems of Linear Equations — Question 11 (Hard)

Ioanna invested a total of 5,000 euros in two accounts at a bank in Patras, one paying 4% annual interest and the other paying 7% annual interest. Both accounts were opened on the same day. After one year, the two accounts together earned 290 euros in interest. How much more interest did the 7% account earn than the 4% account?

Desmos: Type $y = 5000 - x$ and $y = (290 - 0.04x)/0.07$ into Desmos and trace to their intersection to confirm the two principal amounts before computing the interest difference by hand.

Systems of Linear Equations — Question 12 (Hard)



A ferry leaves the dock in Piraeus heading toward Aegina at a constant speed of 18 kilometers per hour. Twenty minutes later, a speedboat leaves the same dock along the same route at a constant speed of 30 kilometers per hour. Let t be the number of hours after the speedboat departs, and let d be the distance in kilometers from Piraeus. How many kilometers from Piraeus does the speedboat catch up to the ferry?

Desmos: Type $y = 30x$ and $y = 18(x + 1/3)$ into Desmos as two lines in terms of $x = t$ and $y = d$, and trace to their intersection point to read off both t and d .

Systems of Linear Equations — Question 13 (Hard)

For what value of k does the system $kx + 4y = 9$ and $3x + 2y = 5$ have no solution?

Desmos: Type $6x + 4y = 9$ and $3x + 2y = 5$ into Desmos and confirm the two lines appear parallel and never cross.

Systems of Linear Equations — Question 14 (Hard)

Solve the system: $(1/2)x + (1/3)y = 7$ and $x - y = 9$. What is the value of xy ?

Desmos: Type $(1/2)x + (1/3)y = 7$ and $x - y = 9$ into Desmos exactly as fractions and trace to the intersection point to read x and y , then multiply them by hand.



Systems of Linear Equations — Question 15 (Hard)

A vineyard in the Douro Valley blends Wine A and Wine B to make 60 liters of a blend that is 18% alcohol by volume, using twice as many liters of Wine A as Wine B. Wine B is 9% alcohol. What is the alcohol percentage of Wine A?

Desmos: Type $y = 60 - x$ and $y = 0.5x$ into Desmos (representing b in terms of a) and trace to their intersection to confirm $a = 40$ and $b = 20$, then solve the concentration equation by hand.



3. Linear Inequalities

Formula & Rules for This Skill

Solve like an equation, but flip the inequality sign whenever you multiply or divide by a negative number. Compound inequalities: isolate the variable in the middle of all three parts.

Fastest Way to Solve: Every time you multiply or divide both sides by a negative, immediately flip the inequality sign before doing anything else.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Linear Inequalities — Question 1 (Easy)

Solve the inequality for x : $-7x + 4 > -24$.

Desmos: Type $y1 = -7x + 4$ and $y2 = -24$, then trace to see the first graph stays above the second exactly for $x < 4$.

Linear Inequalities — Question 2 (Easy)

A parking garage in Bergen charges a flat entry fee of 2 euros plus 3 euros per hour. Sofia does not want to spend more than 20 euros in total. What is the greatest number of full hours she can park?

Desmos: Type $y1 = 2 + 3x$ and $y2 = 20$, then find the greatest whole x where the first graph is still at or below the second.



Linear Inequalities — Question 3 (Easy)

Solve the compound inequality for x : $-8 \leq 4x + 4 < 16$.

Desmos: Type $y1 = 4x + 4$ in Desmos and trace to see it stays between -8 (inclusive) and 16 (exclusive) exactly when x is between -3 and 3.

Linear Inequalities — Question 4 (Easy)

Three times a number, increased by 8, is at least 29. What is the smallest value the number could be?

Desmos: Type $y1 = 3x + 8$ and $y2 = 29$ in Desmos and check where the first graph meets or rises above the second — it happens at $x = 7$.

Linear Inequalities — Question 5 (Easy)



Solve the inequality for x : $5(x - 3) < 2x + 9$.

Desmos: Type $y1 = 5(x-3)$ and $y2 = 2x+9$, then trace to see the first graph stays below the second exactly for $x < 8$.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Linear Inequalities — Question 6 (Medium)

The inequality $-ax + 6 \leq 21$ has solution set $x \geq -5$. What is the value of a ?

Desmos: Type $y1 = -3x + 6$ and $y2 = 21$ (using the solved value $a = 3$), then confirm the first graph stays at or below the second exactly for $x \geq -5$.

Linear Inequalities — Question 7 (Medium)



A catering company in Bratislava charges a flat setup fee of 45 euros plus 8 euros per guest for a buffet dinner. The event hall also requires a separate, refundable 30 euro cleaning deposit that does not count toward the food budget. If Nina's food budget for the buffet is 325 euros, what is the greatest number of guests she can invite?

Desmos: Type $y1 = 45 + 8x$ and $y2 = 325$ in Desmos and find the greatest whole x where the first graph is still at or below the second.

Linear Inequalities — Question 8 (Medium)

A tank at a bottling plant in Porto must contain no less than 60% of its 800-liter capacity for the pump to run safely. The tank currently holds 350 liters and is being filled at a rate of 25 liters per minute. After how many full minutes of filling will the tank first satisfy the safe-running condition?

Desmos: Type $y1 = 350 + 25x$ and $y2 = 480$, then find the smallest whole x where the first graph reaches or passes the second — it lands at $x = 6$.

Linear Inequalities — Question 9 (Medium)



A hiking club in Innsbruck requires a day-trip group to have between 12 and 20 people, inclusive, to qualify for the group cable-car discount. Right now 5 people are signed up, and each flyer the club posts brings in exactly 3 more hikers. What is the range of the number of flyers the club can post so the total number of hikers falls within the discount range?

Desmos: Type $y1 = 5 + 3x$, $y2 = 12$, and $y3 = 20$ in Desmos and find the whole x -values where $y1$ stays between $y2$ and $y3$.

Linear Inequalities — Question 10 (Medium)

The greatest integer that satisfies $4x - 9 < 23$ is called m . What is the value of $2m - 5$?

Desmos: Type $y1 = 4x - 9$ and $y2 = 23$, trace to confirm the graphs meet at $x = 8$, then use $x = 7$ (the last integer before 8) as m .

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Linear Inequalities — Question 11 (Hard)



The organizing committee for a conference in Riga has booked a hall that can seat between 150 and 210 people, inclusive, for the closing dinner. So far, 42 people have confirmed, and each batch of invitations the committee sends brings in exactly 9 more confirmed guests. What is the range of the number of batches the committee can send while keeping the confirmed total within the hall's capacity?

Desmos: Type $y1 = 42 + 9x$, $y2 = 150$, and $y3 = 210$ in Desmos and find the whole x -values where $y1$ stays between $y2$ and $y3$.

Linear Inequalities — Question 12 (Hard)

Two internet providers serve Krakow. NetOne charges a 60 zloty setup fee plus 25 zloty per month. FiberLux charges no setup fee but 40 zloty per month, and also throws in a free router worth 80 zloty. For how many months must Marek subscribe for NetOne to be the cheaper total cost?

Desmos: Type $y1 = 60 + 25x$ and $y2 = 40x$ in Desmos and find where the first graph drops below the second — it crosses at $x = 4$, so NetOne is cheaper for whole months after that.

Linear Inequalities — Question 13 (Hard)

The compound inequality $-4 < 5 - kx \leq 11$ has solution set $-2 \leq x < 3$. What is the value of k ?

Desmos: Type $y1 = 5 - 3x$, $y2 = -4$, and $y3 = 11$ in Desmos and confirm the graph of $y1$ stays strictly above $y2$ and at or below $y3$ exactly for $-2 \leq x < 3$.

Linear Inequalities — Question 14 (Hard)

A print shop in Vilnius charges 0.12 euros per flyer plus a one-time 8 euro design fee. Customers who order more than 200 flyers also get a free coffee voucher, which does not affect the price. Elza needs to spend strictly under 50 euros total. What is the greatest number of flyers she can order?

Desmos: Type $y1 = 8 + 0.12x$ and $y2 = 50$ in Desmos and find the greatest whole x where the first graph stays strictly below the second.

Linear Inequalities — Question 15 (Hard)

Suppose $4 \leq x \leq 10$ and $1 \leq y \leq 3$. What is the full range of possible values of $x - 2y$?

Desmos: There isn't a single graph for a two-variable range like this — instead, type the two extreme cases $4 - 2 \cdot 3$ and $10 - 2 \cdot 1$ directly into Desmos as expressions to confirm the endpoints -2 and 8 .





4. Absolute Value Equations & Inequalities

Formula & Rules for This Skill

$|x|$ = distance from 0. If $|\text{expression}| = k$ ($k > 0$), split into two cases: $\text{expression} = k$ AND $\text{expression} = -k$, then solve both.

Fastest Way to Solve: Isolate the absolute value expression completely on one side first, then split into the positive and negative cases — never split before isolating.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Absolute Value Equations & Inequalities — Question 1 (Easy)

Solve for x: $|x - 5| = 12$

Desmos: Type $y = \text{abs}(x - 5)$ and $y = 12$ as two separate graphs, then find the two x-values where they cross.

Absolute Value Equations & Inequalities — Question 2 (Easy)

Solve for x: $|3x + 6| = 21$

Desmos: Type $y = \text{abs}(3x + 6)$ and $y = 21$, then find the two x-values where the graphs intersect.



Absolute Value Equations & Inequalities — Question 3 (Easy)

Solve for x: $|x - 4| < 9$

Desmos: Type $y = \text{abs}(x - 4)$ and $y = 9$, then look at the x-values where the V-graph dips below the horizontal line.

Absolute Value Equations & Inequalities — Question 4 (Easy)

Solve for x: $|x| - 7 = 15$

Desmos: Type $y = \text{abs}(x) - 7$ and $y = 15$, then find where the graphs cross.

Absolute Value Equations & Inequalities — Question 5 (Easy)

Solve for x: $|2x + 5| \geq 13$

Desmos: Type $y = \text{abs}(2x + 5)$ and $y = 13$, then see where the V-graph is at or above the line.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Absolute Value Equations & Inequalities — Question 6 (Medium)

The equation $|x - k| = 8$ has solution set $\{3, 19\}$. What is the value of k ?

Desmos: Type $y = \text{abs}(x - 11)$ and $y = 8$, and confirm the graphs cross at $x = 3$ and $x = 19$.

Absolute Value Equations & Inequalities — Question 7 (Medium)

Solve for x : $\sqrt{(x + 6)^2} = 15$

Desmos: Type $y = \sqrt{(x+6)^2}$ and $y = 15$, then find both x -values where the two graphs meet.

**Absolute Value Equations & Inequalities — Question 8** (Medium)

Solve for x: $4|2x - 3| - 5 = 27$

Desmos: Type $y = 4 \cdot \text{abs}(2x - 3) - 5$ and $y = 27$, then find the two x-values where the graphs intersect.

Absolute Value Equations & Inequalities — Question 9 (Medium)

A bakery in Vienna produces 800 loaves per day. Each loaf of their signature rye bread is supposed to weigh 900 grams, but the packaging line allows a tolerance so that the actual weight w , in grams, satisfies $|w - 900| \leq 15$. What is the range of acceptable weights, in grams, for a loaf?

Desmos: Type $y = \text{abs}(x - 900)$ and $y = 15$, then read the x-interval where the V-graph is at or below the line.

Absolute Value Equations & Inequalities — Question 10 (Medium)

Solve for x: $|5x + 1| + 8 = 3$

Desmos: Type $y = \text{abs}(5x + 1) + 8$ and $y = 3$, and note the graphs never meet since the first graph never dips below 8.



Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Absolute Value Equations & Inequalities — Question 11 (Hard)

Solve for x : $|x^2 - 9| = 16$

Desmos: Type $y = \text{abs}(x^2 - 9)$ and $y = 16$, then read off the two x -intercepts where the graphs meet.

Absolute Value Equations & Inequalities — Question 12 (Hard)

A rail company in Zurich rejects a section of track whenever its corrected gauge g , in millimeters, differs from the standard 1435 mm by more than 3 mm: $|g - 1435| > 3$. The corrected gauge is the raw field measurement m minus a fixed 2 mm calibration offset, so $g = m - 2$. Find the two raw measurement values of m , in millimeters, at which the track sits exactly on the boundary between acceptable and rejected.

Desmos: Type $y = \text{abs}(x - 2) - 1435$ and $y = 3$, then read the two x -values (raw measurements m) where the graphs cross.



Absolute Value Equations & Inequalities — Question 13 (Hard)

Solve for x: $|x - 3| = |2x + 1|$

Desmos: Type $y = \text{abs}(x - 3)$ and $y = \text{abs}(2x + 1)$ as two separate graphs, then find the two x-values where they intersect.

Absolute Value Equations & Inequalities — Question 14 (Hard)

How many integer values of x satisfy $-3|x + 2| + 10 \geq 1$?

Desmos: Type $y = -3 \cdot \text{abs}(x + 2) + 10$ and $y = 1$, then trace where the curve is at or above the line and count the integer x-values in that closed interval.

Absolute Value Equations & Inequalities — Question 15 (Hard)



The solution set of $|2x - a| \leq b$ is the interval $[-3, 9]$, where a and b are constants. What is the value of $a - b$?

Desmos: Type $y = \text{abs}(2x - 6)$ and $y = 12$ (the solved values $a=6$, $b=12$) and confirm the shaded solution region runs from $x = -3$ to $x = 9$.



5. Exponent Rules

Formula & Rules for This Skill

$$a^m \cdot a^n = a^{m+n}$$

Fastest Way to Solve: Rewrite every term with the same base first — exponent rules only combine terms that already share a base.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Exponent Rules — Question 1 (Easy)

Simplify the expression , writing your answer as a single power of x.

Desmos: Type and as two separate graphs and confirm the curves overlap exactly.

Exponent Rules — Question 2 (Easy)

Simplify the expression .

Desmos: Type and as two graphs and confirm they trace the same curve.



Exponent Rules — Question 3 (Easy)

Simplify the expression .

Desmos: Type 2^{-1} and 2^{-1} and confirm the two graphs overlap.

Exponent Rules — Question 4 (Easy)

Evaluate .

Desmos: Type 2^{-1} directly into the Desmos calculator to confirm the value is 2.

Exponent Rules — Question 5 (Easy)

Rewrite 0.015625 as a fraction in simplest form.

Desmos: Type 0.015625 into the Desmos calculator and confirm it returns the decimal 0.015625, which equals $\frac{1}{64}$.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Exponent Rules — Question 6 (Medium)

If for all values of x , what is the value of k ?

Desmos: Type $11+4$ into Desmos to double-check it equals 15, confirming $k = 11$ satisfies the exponent equation.

Exponent Rules — Question 7 (Medium)

Simplify the expression , writing your answer as a single term.

Desmos: Type and as two graphs and confirm they overlap for positive x .



Exponent Rules — Question 8 (Medium)

Simplify by rewriting both numbers as powers of 3, and give the value as a single number.

Desmos: Type directly into Desmos to confirm the value is exactly 1.

Exponent Rules — Question 9 (Medium)

A bank in Zurich uses the expression in one of its internal compounding formulas, but no banking details are needed to answer this: if , what is the value of ?

Desmos: Type into Desmos to instantly confirm the value 15625.

Exponent Rules — Question 10 (Medium)

Simplify , writing your answer with only positive exponents.

Desmos: Plug in a test value like $x = 2$, $y = 3$ into both the original and simplified expressions in the calculator to confirm the results match.



Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Exponent Rules — Question 11 (Hard)

If $x = 2$, what is the value of y ?

Desmos: Type 2^6 directly into Desmos to confirm the value 64 numerically.

Exponent Rules — Question 12 (Hard)

If $x = 2$ and $y = 3$, what is the value of z ?

Desmos: Type 2^3 to confirm it equals 8, and 3^2 to confirm it equals 9, then solve $x+y=4$ with $x=2$ by hand.



Exponent Rules — Question 13 (Hard)

If $2^x = 14$ and $x > 0$, what is the value of 2^{x+7} ?

Desmos: Graph $y = 2^x$ and $y = 14$ in Desmos, check the positive intersection is at $x = 7$, then compute 2^{x+7} by hand.

Exponent Rules — Question 14 (Hard)

Simplify $2^{x+7} - 2^{x+2}$ completely, then evaluate the result when $x = 2$.

Desmos: Type $2^{x+7} - 2^{x+2}$ directly into Desmos to confirm the value 0.125, which equals $\frac{1}{8}$.

Exponent Rules — Question 15 (Hard)

A lab in Reykjavik models a bacterial culture's mass in grams after t hours as $M(t) = 45 \cdot 2^{bt}$, where b is a constant. If $M(2) = 45$, what is $M(4)$?

Desmos: Type $M(t) = 45 \cdot 2^{bt}$ and trace to $x = 4$ to confirm the output is 405.





6. Radicals, Rational Exponents & Scientific Notation

Formula & Rules for This Skill

- $\sqrt[n]{a}$ -th root of a
- $a^{\frac{m}{n}}$ -th root of a , raised to the m power.

Fastest Way to Solve: Convert between radical and rational-exponent form whenever one form makes the arithmetic or the exponent rules easier to apply.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Radicals, Rational Exponents & Scientific Notation — Question 1 (Easy)

Simplify to simplest radical form.

Desmos: Type $\sqrt[5]{12.247}$ and separately $12.247^{\frac{1}{5}}$ in the calculator — both give about 12.247, confirming the simplification.

Radicals, Rational Exponents & Scientific Notation — Question 2 (Easy)

Evaluate .

Desmos: Type $16^{\frac{1}{4}}$ directly into the calculator to confirm it evaluates to 2.



Radicals, Rational Exponents & Scientific Notation — Question 3 (Easy)

Simplify .

Desmos: Type $\sqrt[3]{85}$ and $2\sqrt[3]{10.625}$ — both return about 9.487, confirming the match.

Radicals, Rational Exponents & Scientific Notation — Question 4 (Easy)

Multiply and write the result in proper scientific notation.

Desmos: Type 6×10^7 into the calculator and confirm it shows 60000000, which is 6×10^7 .

Radicals, Rational Exponents & Scientific Notation — Question 5 (Easy)

Solve for x: .

Desmos: Type $y = \sqrt{x}$ and $y = 11$, then find where the two graphs intersect — the x-coordinate is 36.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Radicals, Rational Exponents & Scientific Notation — Question 6 (Medium)

If the fifth root of x cubed equals 8 (that is, $\sqrt[5]{x^3} = 8$), what is the value of x ?

Desmos: Type $\sqrt[5]{x^3} = 8$ and trace where the graph crosses zero near $x = 32$ to confirm.

Radicals, Rational Exponents & Scientific Notation — Question 7 (Medium)

A bacterium has a mass of 4×10^{-12} kg. A sample cultured in a lab in Uppsala over 3 days contains 3000 such bacteria. What is the total mass of the sample, in kg, written in scientific notation?

Desmos: Type $4 \times 10^{-12} \times 3000$ into the calculator to confirm it evaluates to 0.000012, which is 1.2×10^{-5} .



Radicals, Rational Exponents & Scientific Notation — Question 8 (Medium)

If $\sqrt{a}$ for some positive integer a , find a . Then use that value to compute $\sqrt{a}$.

Desmos: Type $\sqrt{4}$ into the calculator to confirm it equals 4, then compute $\sqrt{4}$ by hand.

Radicals, Rational Exponents & Scientific Notation — Question 9 (Medium)

Solve for x : $\sqrt{x-5} = x-5$. Be sure to check for extraneous solutions.

Desmos: Type $\sqrt{x-5}$ and $y = x - 5$, then trace to see they intersect only at $x = 10$, confirming $x = 2$ is extraneous.

Radicals, Rational Exponents & Scientific Notation — Question 10 (Medium)

Simplify $\sqrt{x^2}$ to a single term (assume $x > 0$).

Desmos: Type $\sqrt{x^2}$ and $y = x$ on the same graph — they should overlap exactly for $x > 0$.



Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Radicals, Rational Exponents & Scientific Notation — Question 11 (Hard)

A cube-shaped storage tank in Rotterdam has a volume of 0.000064 cubic meters. Since it's a cube, its edge length equals the cube root of its volume. What is the edge length of the tank, in meters, written in scientific notation?

Desmos: Type $\sqrt[3]{0.000064}$ into the calculator and confirm it equals 0.000064 , which matches 0.000064 .

Radicals, Rational Exponents & Scientific Notation — Question 12 (Hard)

Solve for x : .

Desmos: Type $y = \sqrt{x}$ and $y = 1$, then trace to find they meet at $x = 16$.

**Radicals, Rational Exponents & Scientific Notation — Question 13** (Hard)

A satellite orbits at a distance of _____ meters from a ground station in Reykjavik. A signal travels at _____ meters per second. What is the total round-trip time for a signal to travel from the station to the satellite and back, in seconds, written in scientific notation?

Desmos: Type _____ into the calculator to confirm it evaluates to 0.15.

Radicals, Rational Exponents & Scientific Notation — Question 14 (Hard)

Solve for x : _____, where x is a real number and $x < 0$.

Desmos: Type _____ and trace near $x = -125$ and $x = 125$ — both are zeros, confirming two real solutions exist before the restriction is applied.

Radicals, Rational Exponents & Scientific Notation — Question 15 (Hard)

Simplify _____ to the form _____, where a , b , and c are integers.

Desmos: Type _____ and _____ into the calculator — both give about 3.646, confirming the simplification.





PART 2 — PROBLEM-SOLVING AND DATA ANALYSIS

7. Ratios & Proportions

Formula & Rules for This Skill

$a/b = c/d \rightarrow$ cross-multiply: $ad = bc$. Scale factor: new value = original value $\times$ (scale factor).

Fastest Way to Solve: Set up the ratio with matching units in matching positions (both numerators the same type, both denominators the same type) before cross-multiplying.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Ratios & Proportions — Question 1 (Easy)

A bookstore in Lisbon stocks fiction and non-fiction books in the ratio 7:4. If the store has 176 books in total, how many are non-fiction?

Desmos: Type $y = 11x - 176$ in Desmos and find where the graph crosses zero to confirm $x = 16$, then multiply by 4.

Ratios & Proportions — Question 2 (Easy)

A bakery in Vienna uses 3 eggs for every 8 pastries it bakes. If the bakery needs to bake 96 pastries for a wedding order, how many eggs are needed?

Desmos: Type $y1 = (3/8)x$ in Desmos and trace the graph to $x = 96$ to confirm the output is 36.



Ratios & Proportions — Question 3 (Easy)

At a technology firm in Helsinki, the ratio of engineers to designers is 5:2. If there are 28 designers, how many engineers are there?

Ratios & Proportions — Question 4 (Easy)

A train travels from Munich to a nearby town at a constant speed, covering 84 kilometers in 4 hours. At this same speed, how many kilometers will it cover in 9 hours?

Desmos: Type $y1 = 21x$ in Desmos and read the y-value at $x = 9$ to confirm the distance is 189 kilometers.

Ratios & Proportions — Question 5 (Easy)



A recipe for lemonade calls for sugar and lemon juice in a ratio of 3:5. If a chef uses 18 cups of sugar, how many cups of lemon juice are needed?

Desmos: Type $y1 = (5/3)x$ in Desmos and read the y-value at $x = 18$ to confirm 30 cups of lemon juice.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Ratios & Proportions — Question 6 (Medium)

A moving company in Warsaw needs 6 movers to load a truck in 5 hours. If the company wants the same truck loaded in only 2 hours, working at the same rate, how many movers are needed?

Desmos: Type $y1 = 30/x$ in Desmos and read the y-value at $x = 2$ to confirm 15 movers are needed.

Ratios & Proportions — Question 7 (Medium)



A paint supplier in Krakow mixes blue and white paint in a ratio of 5:12, using a formula the company has kept unchanged since 1998. If a customer buys 30 liters of blue paint, how many total liters of the blue-white mixture will the supplier prepare?

Desmos: Type $y1 = (12/5)x$ in Desmos, evaluate at $x = 30$ to get 72 liters of white, then add 30 for the total mixture.

Ratios & Proportions — Question 8 (Medium)

The variable m varies directly with n . When $n = 8$, $m = 30$. What is the value of m when $n = 20$?

Desmos: Type $y1 = (15/4)x$ in Desmos and read the y -value at $x = 20$ to confirm $m = 75$.

Ratios & Proportions — Question 9 (Medium)

A hostel in Prague reports that the ratio of guests to staff on duty is 9:2. After 15 more guests check in, with the number of staff unchanged, the new ratio of guests to staff becomes 12:2. How many staff are on duty?

Desmos: Type $y = 3x - 15$ in Desmos and trace where it crosses zero to confirm $k = 5$, then double it for the staff count.



Ratios & Proportions — Question 10 (Medium)

At a music festival in Bruges, the ratio of volunteers to security staff is 3:4, and the ratio of security staff to medics is 2:5. If there are 150 medics on site, how many volunteers are there?

Desmos: Type $y_1 = 3x$, $y_2 = 4x$, and $y_3 = 10x$ in Desmos, then find x where $y_3 = 150$ to instantly read off $y_1 = 45$.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Ratios & Proportions — Question 11 (Hard)

A radiator holds a mixture of water and coolant in the ratio 5:1, for a total of 72 liters. A mechanic tops off the radiator by adding pure coolant only, until the ratio of water to coolant becomes 3:1. How many liters of pure coolant did the mechanic add?

Desmos: Type $y = 60/(12 + x) - 3$ in Desmos and trace to find where it crosses zero — that x -value is the coolant added.



Ratios & Proportions — Question 12 (Hard)

At a logistics company in Zurich, the ratio of full-time to part-time employees is 7:3, and there are 45 part-time employees. The company then hires only full-time employees, which raises the full-time count by 40%, while the number of part-time employees does not change. How many full-time employees does the company have after the hiring?

Desmos: Type $y = 1.4(7x)$ in Desmos and evaluate at $x = 15$ (from $3x = 45$) to confirm the new full-time count is 147.

Ratios & Proportions — Question 13 (Hard)

At a school in Zagreb, the ratio of boys to girls in a particular grade is 3:4. Six boys transfer to a different school and six girls transfer in from another school, with no other change in enrollment. After the transfers, the ratio of boys to girls becomes 1:2. How many students are in the grade after the transfers?

Desmos: Type $y = 2(3x - 6) - (4x + 6)$ in Desmos and trace to where it crosses zero to confirm $k = 9$, then compute $7x$ at that point for the total.

Ratios & Proportions — Question 14 (Hard)

A caterer in Bruges is preparing trays of spanakopita, where each full batch of filling requires exactly 5 kg of spinach and 2 kg of feta cheese (a ratio of 5:2). The caterer has 40 kg of spinach and 15 kg of feta cheese on hand, and can only make whole batches while keeping this exact ratio. What is the greatest number of kilograms of spinach that will be left over after making as many whole batches as possible?

Desmos: Type $y = 40 - 5 \cdot \text{floor}(15/2)$ in Desmos (evaluating $\text{floor}(15/2) = 7$) to confirm the leftover spinach is 5 kg.

Ratios & Proportions — Question 15 (Hard)

The variable z varies directly with x and inversely with y . When $x = 6$ and $y = 4$, $z = 15$. What is the value of z when $x = 10$ and $y = 5$?

Desmos: Type $y1 = 10(x/5)$ in Desmos and read the value at $x = 10$ to confirm $z = 20$.



8. Probability

Formula & Rules for This Skill

$P(\text{event}) = \text{favorable outcomes} / \text{total outcomes}$. Independent events: $P(A \text{ and } B) = P(A) \times P(B)$. Mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$.

Fastest Way to Solve: Decide first whether the events are independent, dependent, or mutually exclusive — that choice determines whether you multiply or add.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Probability — Question 1 (Easy)

A standard six-sided die is rolled once. What is the probability that the result is a number greater than 4?

Desmos: Type 2/6 into Desmos to confirm it simplifies to 1/3.

Probability — Question 2 (Easy)

A bag contains only red, blue, and green marbles. The probability of drawing a red marble is 0.35, and the probability of drawing a blue marble is 0.28. What is the probability of drawing a green marble?

Desmos: Type 1-0.35-0.28 directly into Desmos to get 0.37 in one step.



Probability — Question 3 *(Easy)*

A fair coin is flipped, and a fair six-sided die is rolled. What is the probability that the coin shows heads and the die shows a 6?

Desmos: Type $(1/2) * (1/6)$ into Desmos to confirm the product is $1/12$.

Probability — Question 4 *(Easy)*

A spinner is divided into 8 equal sections numbered 1 through 8. What is the probability that a single spin lands on a number that is either 3 or 7?

Desmos: Type $1/8 + 1/8$ into Desmos to confirm the sum simplifies to $1/4$.

Probability — Question 5 *(Easy)*



A Digital SAT prep class has 60 students total. In Grade 10, 18 of the 30 students own a graphing calculator. In Grade 11, 24 of the 30 students own a graphing calculator. If one student from the entire class is selected at random, what is the probability that the student owns a graphing calculator?

Desmos: Type $42/60$ into Desmos to confirm it reduces to $7/10$.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Probability — Question 6 (Medium)

A drawer contains 5 blue socks and 3 black socks, with no other colors. Two socks are pulled from the drawer at random, one after another, without looking and without putting the first sock back. What is the probability that both socks pulled are blue?

Desmos: Type $(5/8) * (4/7)$ into Desmos to confirm the product simplifies to $5/14$.

Probability — Question 7 (Medium)



At a school in Ljubljana, 90 students were surveyed about music and sports participation. Of the 90 students, 45 play a sport and 45 do not. Among the 45 students who play a sport, 21 also play a musical instrument. Among the 45 students who do not play a sport, 15 also play a musical instrument. If a randomly selected student is known to play a sport, what is the probability that this student also plays a musical instrument?

Desmos: Type $21/45$ into Desmos to confirm it simplifies to $7/15$.

Probability — Question 8 (Medium)

A quality control station tests batteries, and each battery has a 0.1 probability of being defective, independent of the others. If 3 batteries are tested, what is the probability that at least one of them is defective?

Desmos: Type $1-0.9^3$ into Desmos to get 0.271 directly.

Probability — Question 9 (Medium)



A box used for a Warsaw charity raffle contains 40 tickets, numbered 1 through 40. The raffle was organized by a group of 12 volunteers. One ticket is drawn at random. What is the probability that the number on the ticket is a multiple of 5?

Desmos: Type $8/40$ into Desmos to confirm it reduces to $1/5$.

Probability — Question 10 (Medium)

A jar contains only red and yellow candies. The probability of randomly selecting a yellow candy is $3/10$. If there are 21 yellow candies in the jar, how many red candies are in the jar?

Desmos: Type $21/(3/10)$ into Desmos to find the total of 70, then subtract 21 to get 49.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Probability — Question 11 (Hard)



A committee jar in Reykjavik holds 10 name cards: 6 for men and 4 for women. Two cards are drawn at random, one after another, without replacement, to select two committee co-chairs. What is the probability that at least one of the two co-chairs selected is a woman?

Desmos: Type $1 - (6/10) * (5/9)$ into Desmos to confirm the result is $2/3$.

Probability — Question 12 (Hard)

At a firm in Zurich, 200 employees were surveyed about remote work and years of experience. Overall, 120 employees work remotely and 80 do not. Of the 90 employees with 5 or more years of experience, 50 do not work remotely. If one employee with 5 or more years of experience is selected at random, what is the probability that this employee works remotely?

Desmos: Type $40/90$ into Desmos to confirm it reduces to $4/9$.

Probability — Question 13 (Hard)



Aegean Freight ships each package through three independent regional hubs before final delivery, and the company operates a fleet of 45 delivery trucks. Each hub has a 0.05 probability of delaying a package, independent of the other hubs. What is the probability that a randomly chosen package is delayed by at least one of the three hubs?

Desmos: Type $1 - 0.95^3$ into Desmos to get 0.142625 directly.

Probability — Question 14 (Hard)

A basket at a Seoul bakery contains 12 pastries: 7 filled with custard and 5 filled with red bean. A customer reaches in and randomly selects pastries one at a time, without replacement, eating each one before choosing the next. What is the probability that the first two pastries chosen are both filled with red bean, and the third is filled with custard?

Desmos: Type $(5/12) * (4/11) * (7/10)$ into Desmos to confirm the product simplifies to $7/66$.

Probability — Question 15 (Hard)



At a language institute in Montevideo, 60% of students are enrolled in the morning session and the rest are enrolled in the evening session. Among morning-session students, 30% are studying for the Digital SAT. Among evening-session students, 45% are studying for the Digital SAT. If one student is selected at random from the entire institute, what is the probability that the student is studying for the Digital SAT?

Desmos: Type $0.6 * 0.3 + 0.4 * 0.45$ into Desmos to confirm the total is 0.36.



9. Unit Conversions & Rates

Formula & Rules for This Skill

Multiply by a conversion factor written as a fraction equal to 1, arranged so the unit you don't want cancels out.

Fastest Way to Solve: Write every conversion factor as a fraction and check that the unit you're removing appears once on top and once on bottom before multiplying.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Unit Conversions & Rates — Question 1 (Easy)

A delivery van in Rotterdam travels at a constant 60 kilometers per hour. How many kilometers does it travel in 24 minutes?

Desmos: Type $y = 60x$ and evaluate at $x = 0.4$ (24 minutes as a fraction of an hour) to confirm the distance.

Unit Conversions & Rates — Question 2 (Easy)

A cheetah sprints at 25 meters per second. What is its speed in kilometers per hour?

Desmos: Type $25 * 3.6$ in the calculator to instantly confirm the converted speed.



Unit Conversions & Rates — Question 3 (Easy)

A water tank in Lisbon is filled by a pump at a constant rate of 45 liters every 3 minutes. At this rate, how many liters does the pump deliver in 10 minutes?

Desmos: Type $y = 15x$ and evaluate at $x = 10$ to check the total liters delivered.

Unit Conversions & Rates — Question 4 (Easy)

A recipe calls for 2.5 kilograms of flour. A bag of flour weighs 500 grams. How many bags are needed?

Desmos: Type $2500 / 500$ in the calculator to confirm the number of bags.

Unit Conversions & Rates — Question 5 (Easy)

1 gallon is equivalent to about 3.8 liters. A fuel tank holds 20 gallons. About how many liters is that?

Desmos: Type $20 * 3.8$ in the calculator to get the exact number of liters.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Unit Conversions & Rates — Question 6 (Medium)

A train travels 180 kilometers in 2 hours and 30 minutes at a constant speed. What is the train's speed in meters per second?

Desmos: Type $72 * (5/18)$ in the calculator to confirm the speed in meters per second.

Unit Conversions & Rates — Question 7 (Medium)

A shop in Zurich sells cheese for 8.40 francs per 600 grams. If Elena buys a wheel of cheese weighing 1.05 kilograms, how much does she pay, in francs?

Desmos: Type $y = 0.014x$ and evaluate at $x = 1050$ to confirm the total price.



Unit Conversions & Rates — Question 8 (Medium)

Pipe A fills a pool at a rate of 300 liters every 5 minutes. Pipe B fills the same pool at a rate of 240 liters every 4 minutes. If both pipes run together at these constant rates, how many liters do they add to the pool in 12 minutes?

Desmos: Type $y = 120x$ and evaluate at $x = 12$ to confirm the combined total.

Unit Conversions & Rates — Question 9 (Medium)

A metal alloy used in a bridge support in Rotterdam has a density of 7.8 grams per cubic centimeter. A support beam made of this alloy has a volume of 0.5 cubic meters. Given that the beam must also pass a corrosion-resistance rating of at least 200 hours in salt-spray testing, what is the mass of the beam, in kilograms? (1 cubic meter = 1,000,000 cubic centimeters)

Desmos: Type $0.5 * 1000000 * 7.8 / 1000$ in the calculator to get the mass in kilograms directly.

Unit Conversions & Rates — Question 10 (Medium)



On a map of Athens, a distance of 4 centimeters represents an actual distance of 3 kilometers. Two landmarks are 9.5 centimeters apart on the map. What is the actual distance between them, in kilometers?

Desmos: Type $9.5 \times \frac{3}{4}$ in the calculator to get the actual distance directly.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Unit Conversions & Rates — Question 11 (Hard)

A courier drives from Bruges to Ghent, a distance of 45 kilometers, at a constant speed of 90 kilometers per hour. She then drives back the same 45-kilometer route at a constant speed of 20 meters per second due to traffic. What is her average speed for the entire round trip, in kilometers per hour?

Desmos: Type $(2 \times 90 \times 72) / (90 + 72)$ in the calculator to confirm the average speed using the harmonic-mean shortcut.

Unit Conversions & Rates — Question 12 (Hard)

A chemical plant releases wastewater at a constant rate of 8 milliliters per second into a treatment pond. The pond has a safety limit of 250 liters per day. If the plant runs continuously at this rate, by how many liters does it exceed the daily safety limit?

Desmos: Type $8 * 86400 / 1000 - 250$ in the calculator to get the excess in liters directly.

Unit Conversions & Rates — Question 13 (Hard)

A car in Munich consumes diesel at a rate of 6.5 liters per 100 kilometers. Diesel costs 1.60 euros per liter, but the price rises by 5% right before Petra fills up. If Petra drives 350 kilometers, how much does the diesel for this trip cost, in euros, rounded to the nearest cent?

Desmos: Type $350 * 6.5/100 * 1.60 * 1.05$ in the calculator to get the total cost directly.

Unit Conversions & Rates — Question 14 (Hard)

Machine A stamps parts at a constant rate of 1 part every 4 seconds. Machine B stamps parts at a constant rate of 540 parts per hour. If both machines run together at these rates, how many seconds does it take them to stamp a batch of 100 parts?

Desmos: Type $100 / (0.25 + 540/3600)$ in the calculator to get the combined time in seconds directly.



Unit Conversions & Rates — Question 15 (Hard)

A blueprint for a park in Rotterdam uses a scale where 1 centimeter on the blueprint represents 25 meters of actual distance. On the blueprint, a rectangular garden measures 6 centimeters by 4 centimeters. What is the actual area of the garden, in square meters?

Desmos: Type $24 * 25^2$ in the calculator to confirm the actual area using the squared-scale-factor shortcut.



10. Statistics — Mean, Median, Mode, Range, Standard Deviation

Formula & Rules for This Skill

Mean = sum of values / number of values. **Median** = middle value (average the two middle values if the count is even). **Range** = max - min.

Fastest Way to Solve: Re-sort the data in order before finding the median, and check whether the question wants mean, median, or mode before calculating anything.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 1 (Easy)

A rowing team in Patras records the number of kilometers each of its 6 members ran in training this week: 8, 5, 10, 7, 9, 9. What is the mean number of kilometers run by a team member?

Desmos: Type `mean(8,5,10,7,9,9)` directly into the calculator to get 8 instantly.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 2 (Easy)

The prices (in euros) of 7 lamps at a home store in Vienna are: 42, 18, 55, 61, 18, 30, 27. What is the median price?

Desmos: Type `median(42,18,55,61,18,30,27)` into the calculator to get 30 directly.



Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 3 (Easy)

In a survey, 9 students in Krakow were asked how many hours they slept last night: 6, 7, 7, 8, 9, 7, 6, 8, 7. What is the mode of the data set?

Desmos: Type $L1=\{6,7,7,8,9,7,6,8,7\}$ then $\text{sort}(L1)$ to group equal values together, making the most frequent value easy to count by eye.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 4 (Easy)

A weather station in Reykjavik recorded the daily high temperatures (in degrees Celsius) over 6 days: 4, -2, 7, 1, -5, 3. What is the range of the temperatures?

Desmos: Type $\text{max}(4,-2,7,1,-5,3) - \text{min}(4,-2,7,1,-5,3)$ to get the range, 12, directly.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 5 (Easy)



Two vending machines, A and B, each dispensed an average of 5 bags of chips per hour over a 5-hour workday. Machine A's hourly counts were 5, 5, 5, 5, 5. Machine B's hourly counts were 2, 4, 5, 6, 8. Both machines have the same mean of 5. Which machine has the greater standard deviation?

Desmos: Type $\text{stdev}(5,5,5,5,5)$ and $\text{stdev}(2,4,5,6,8)$ in two separate cells to see that Machine A returns 0 while Machine B returns a clearly positive value.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 6 (Medium)

The mean score on a 5-question quiz for a class of 12 students in Lisbon was 7.5 out of 10. After the teacher discovers a grading error, one student's score is corrected from 6 to 9. What is the new mean score for the class?

Desmos: Type $(7.5 * 12 - 6 + 9) / 12$ directly into the calculator to get 7.75 in one line.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 7 (Medium)



A data set of 6 values, listed in increasing order, is: 4, 9, x , 15, 18, 22. If the median of the data set is 13, what is the value of x ?

Desmos: After solving, type `median(4,9,11,15,18,22)` to double check it returns 13.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 8 (Medium)

A small bakery in Ghent tracked how many croissants each of its 20 morning customers bought. The results are: 5 customers bought 1 croissant, 8 customers bought 2 croissants, 4 customers bought 3 croissants, and 3 customers bought 4 croissants. What is the mode of the number of croissants bought?

Desmos: Type `L1={1,1,1,1,1,2,2,2,2,2,2,2,2,2,3,3,3,3,4,4,4}` then `sort(L1)` to see the block of 2's is the longest run, confirming the mode is 2.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 9 (Medium)

In a shot put competition in Vilnius, one athlete's six throws had a range of 3.4 meters. The shortest throw was 9.8 meters, and the athlete's average throw length was 11.1 meters. What was the length of the athlete's longest throw?

Desmos: Type `y = 3.4 + 9.8` to confirm the sum, 13.2, as a quick arithmetic check.



Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 10 (Medium)

A data set of exam scores has a standard deviation of 6 points. The teacher first adds 4 bonus points to every student's score, and then, as a separate curve, multiplies every one of those already-boosted scores by 1.5. What is the standard deviation of the final, twice-adjusted scores?

Desmos: Test the rule with a simple list, e.g. $L1=\{2,4,6\}$: compare $\text{stdev}(L1)$ to $\text{stdev}(L1+4)$ (equal) and to $\text{stdev}(1.5*L1)$ (scaled by 1.5), to confirm both transformation rules.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 11 (Hard)

A DSAT prep class in Rotterdam has two sections. Section A has 8 students with a mean quiz score of 82. Section B has 12 students with a mean quiz score of 91. What is the mean quiz score of all 20 students combined?

Desmos: Type $(8*82+12*91)/20$ directly into the calculator to get 87.4 in one line.



Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 12 (Hard)

A data set of 7 values, in increasing order, is: 3, 6, 8, x , 15, 19, 24. The mean of the data set equals the median of the data set. What is the value of x ?

Desmos: Type $y = (75+x)/7 - x$ and trace to find where the graph crosses zero — that x -value, 12.5, is the answer.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 13 (Hard)

At an amusement park in Porto, the number of rides visitors went on is recorded for a group of 24 people: 2 people went on 1 ride, 5 people went on 2 rides, 7 people went on 3 rides, 7 people went on 4 rides, and 3 people went on 5 rides. The park manager claims the data set has a single mode. Is the manager correct, and if so, what is it?

Desmos: Type $L1=\{1,1,2,2,2,2,2,3,3,3,3,3,3,3,4,4,4,4,4,4,4,5,5,5\}$ then $\text{sort}(L1)$ to see the two equally long blocks of 3's and 4's, confirming both are modes.



Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 14 (Hard)

A cyclist in Bergen logs her daily distances (in km) for 7 days: 20, 22, 21, 19, 23, 20, 65. The 65 km entry was an all-day charity ride, far outside her normal routine. If she removes that single outlier when analyzing her typical week, by how many km does the range decrease?

Desmos: Type $(65-19) - (23-19)$ into the calculator to get the decrease, 42, in a single line.

Statistics — Mean, Median, Mode, Range, Standard Deviation — Question 15 (Hard)

Two students, Elena and Marco, each recorded 4 quiz scores. Elena's scores were 70, 70, 90, 90. Marco's scores were 78, 82, 80, 80. Both students have the same mean score. Without doing a full standard deviation calculation, which student has the larger standard deviation, and what is the sum of the squared deviations from the mean for that student's data set?

Desmos: Type $\text{stdev}(70,70,90,90)$ and $\text{stdev}(78,82,80,80)$ in separate cells to confirm Elena's result is larger than Marco's.



11. Data Displays — Histograms, Dot Plots, Box Plots

Formula & Rules for This Skill

Frequency = how many data points fall in a category or interval. **Box plot five-number summary:** minimum, Q1, median, Q3, maximum.

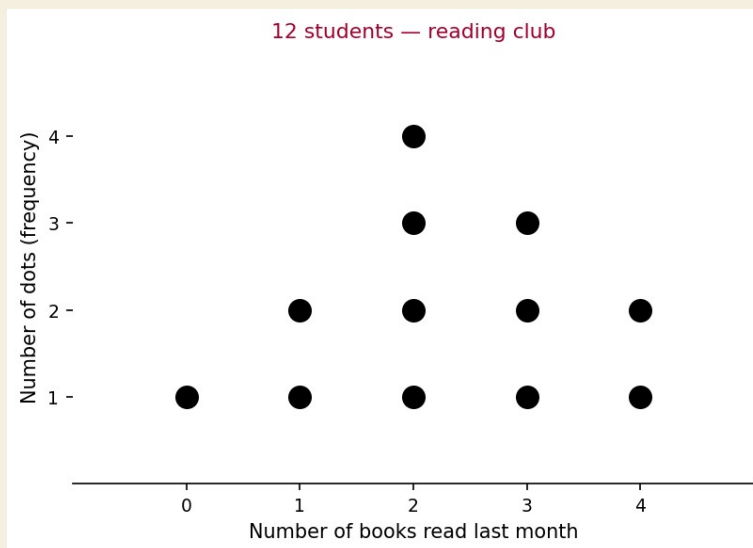
Fastest Way to Solve: Read the axis labels and scale of the display carefully before reading off any value — most mistakes here come from misreading the scale, not the data.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Data Displays — Histograms, Dot Plots, Box Plots — Question 1 (Easy)

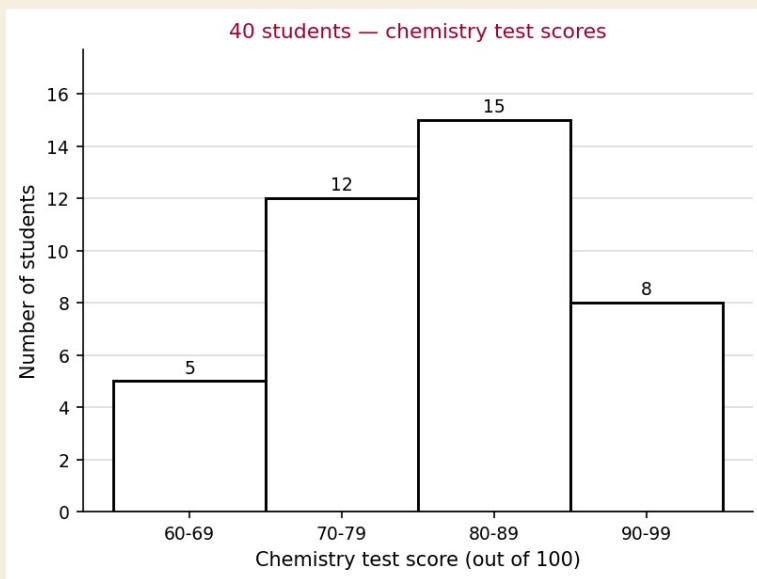
A dot plot shows the number of books read last month by 12 students in a reading club: 0 books - 1 student, 1 book - 2 students, 2 books - 4 students, 3 books - 3 students, 4 books - 2 students. What is the mean number of books read per student?



Desmos: Type $L1=\{0,1,1,2,2,2,2,3,3,3,4,4\}$ listing each value once per dot, then type $\text{mean}(L1)$ to get 2.25 directly.

Data Displays — Histograms, Dot Plots, Box Plots — Question 2 (Easy)

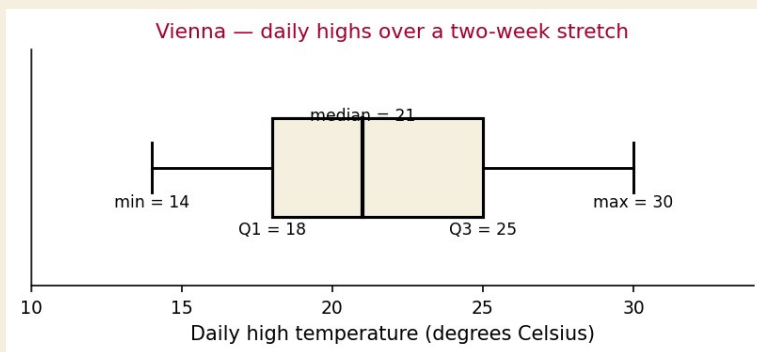
A histogram shows the scores (out of 100) of 40 students on a chemistry test, grouped into ten-point bins: 60-69 - 5 students, 70-79 - 12 students, 80-89 - 15 students, 90-99 - 8 students. How many students scored at least 80 points?



Desmos: Type $L1=\{5,12,15,8\}$ and $\text{sum}(L1)$ to confirm the bins total 40, then add the last two entries for 23.

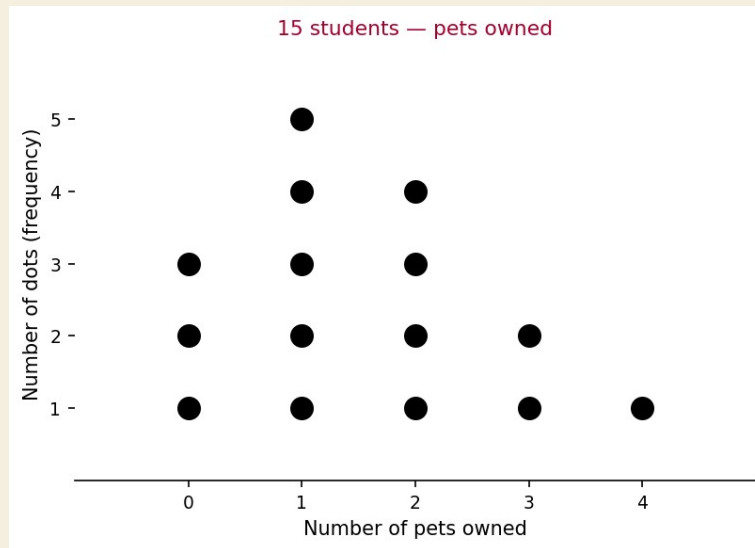
Data Displays — Histograms, Dot Plots, Box Plots — Question 3 (Easy)

A box plot summarizes the daily high temperatures (in degrees Celsius) recorded in Vienna over a two-week stretch: minimum = 14, $Q1 = 18$, median = 21, $Q3 = 25$, maximum = 30. What is the interquartile range (IQR) of these temperatures?



Data Displays — Histograms, Dot Plots, Box Plots — Question 4 (Easy)

A dot plot shows the number of pets owned by 15 students: 0 pets - 3 students, 1 pet - 5 students, 2 pets - 4 students, 3 pets - 2 students, 4 pets - 1 student. What is the sum of the mode and the range of this data set?

**Data Displays — Histograms, Dot Plots, Box Plots — Question 5** (Easy)

A histogram shows the weights (in kilograms) of 50 packages at a shipping warehouse, grouped into bins: 0-4 kg - 10 packages, 5-9 kg - 20 packages, 10-14 kg - 15 packages, 15-19 kg - 5 packages. What percentage of the packages weigh at least 10 kg?



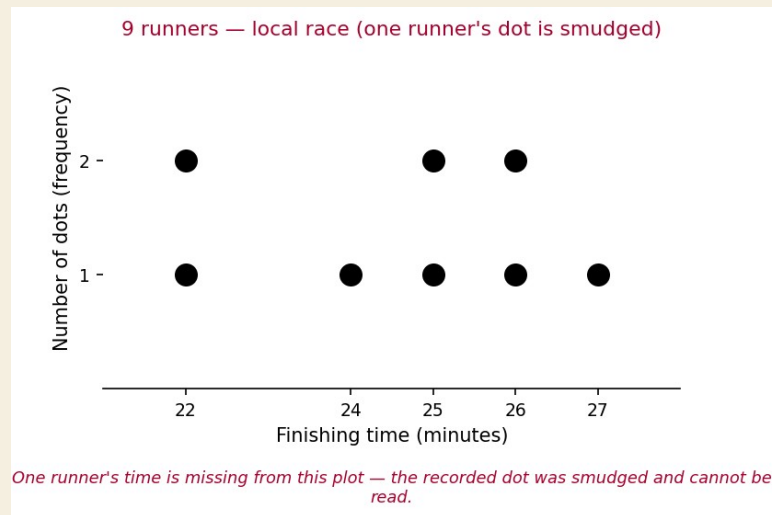
Desmos: Type 20/50 directly into the calculator to instantly confirm the decimal 0.4, then convert to 40%.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

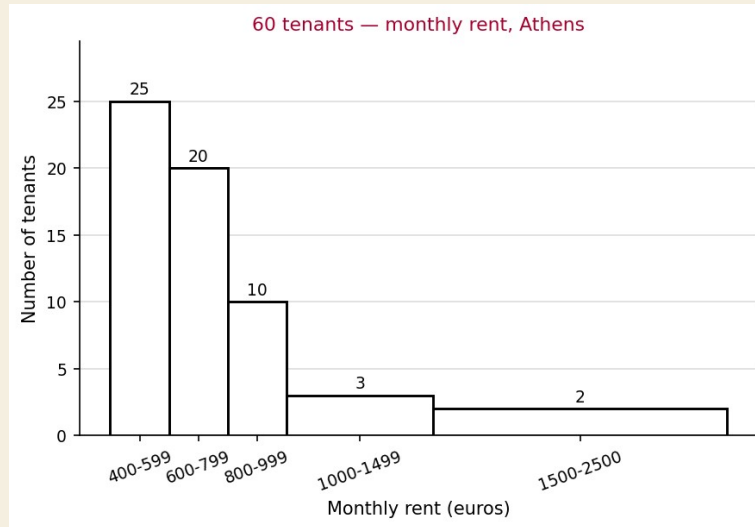
Data Displays — Histograms, Dot Plots, Box Plots — Question 6 (Medium)

A dot plot shows the finishing times, in minutes, of 9 runners in a local race: 22, 22, 24, 25, 25, 26, 26, 27, and one runner's time is missing because the recorded dot is smudged. If the mean finishing time for all 9 runners is 25 minutes, what was the smudged runner's time?

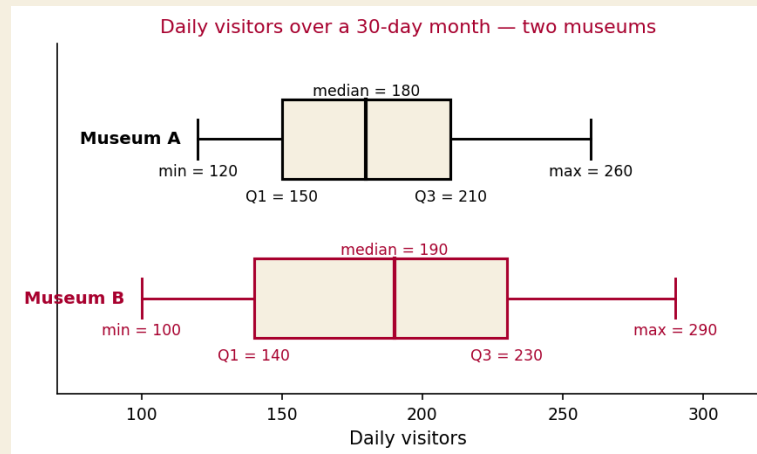


Desmos: Type $L1=\{22,22,24,25,25,26,26,27,x\}$ with $\text{mean}(L1)=25$, then use the calculator's solver to find $x = 28$.

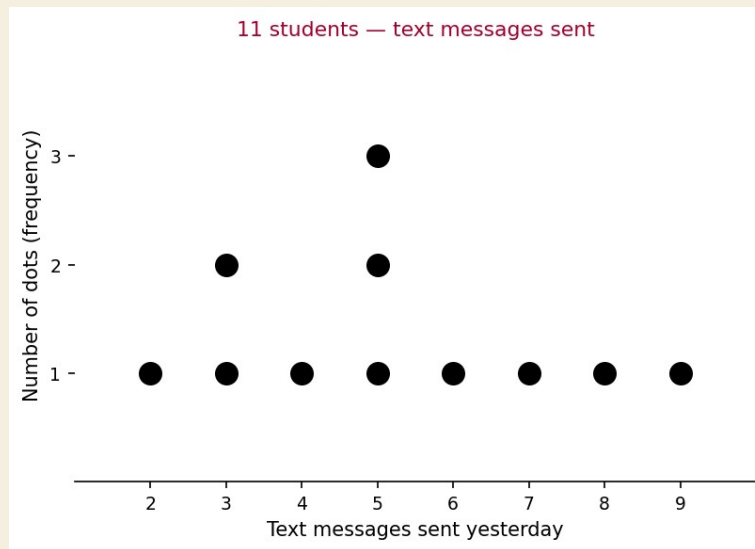
A histogram shows the monthly rent, in euros, paid by 60 tenants in Athens, grouped into bins: 400-599 - 25 tenants, 600-799 - 20 tenants, 800-999 - 10 tenants, 1000-1499 - 3 tenants, 1500-2500 - 2 tenants. Based on the shape of this distribution, which is greater: the mean monthly rent or the median monthly rent?



Box plots compare the number of daily visitors to two museums over a 30-day month. Museum A: minimum = 120, Q1 = 150, median = 180, Q3 = 210, maximum = 260. Museum B: minimum = 100, Q1 = 140, median = 190, Q3 = 230, maximum = 290. Both museums charge 8 euros for admission. Which museum has the larger interquartile range of daily visitors, and what is that IQR?

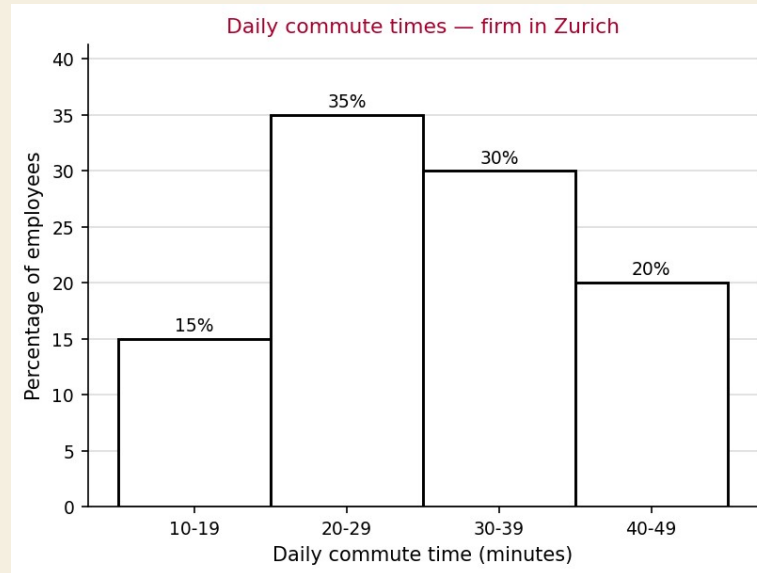


A dot plot shows the number of text messages sent yesterday by 11 students in a class of 28 total students: 2, 3, 3, 4, 5, 5, 5, 6, 7, 8, 9. What is the interquartile range of the number of text messages sent by these 11 students?



Desmos: Type $L1=\{2,3,3,4,5,5,5,6,7,8,9\}$, then type $\text{quartile1}(L1)$ and $\text{quartile3}(L1)$ to confirm $Q1 = 3$ and $Q3 = 7$.

A histogram shows the distribution of daily commute times, in minutes, for employees at a firm in Zurich, given in percentages: 10-19 min - 15%, 20-29 min - 35%, 30-39 min - 30%, 40-49 min - 20%. Exactly 21 employees have a commute between 20 and 29 minutes. How many employees have a commute of 40 minutes or more?



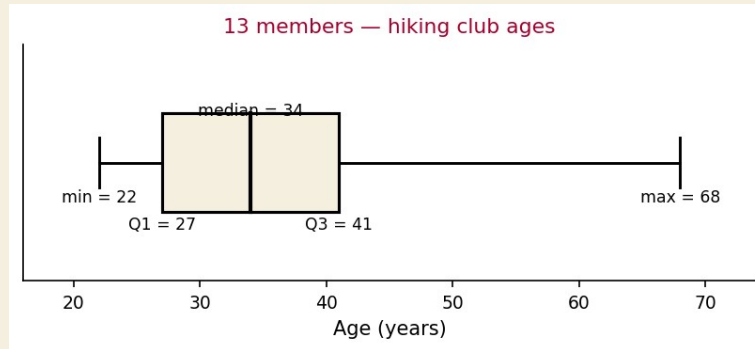
Desmos: Type $21/0.35$ to get the total of 60, then type $0.20 * 60$ to get 12 directly.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

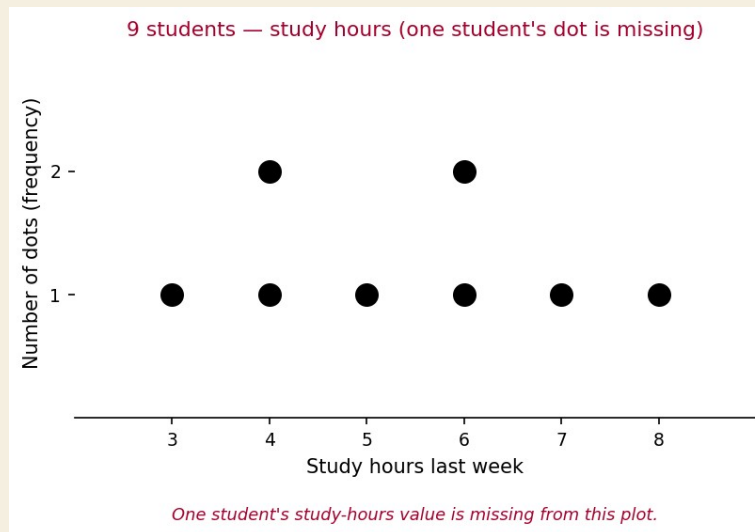
Data Displays — Histograms, Dot Plots, Box Plots — Question 11 (Hard)

A box plot summarizes the ages, in years, of 13 members of a hiking club: minimum = 22, Q1 = 27, median = 34, Q3 = 41, maximum = 68. A value is considered an outlier if it lies more than 1.5 times the interquartile range above Q3 (or below Q1). After confirming that the maximum age of 68 is an outlier and that member leaves the club, the next-oldest remaining member is 45 years old. What is the range of the ages of the remaining 12 members?



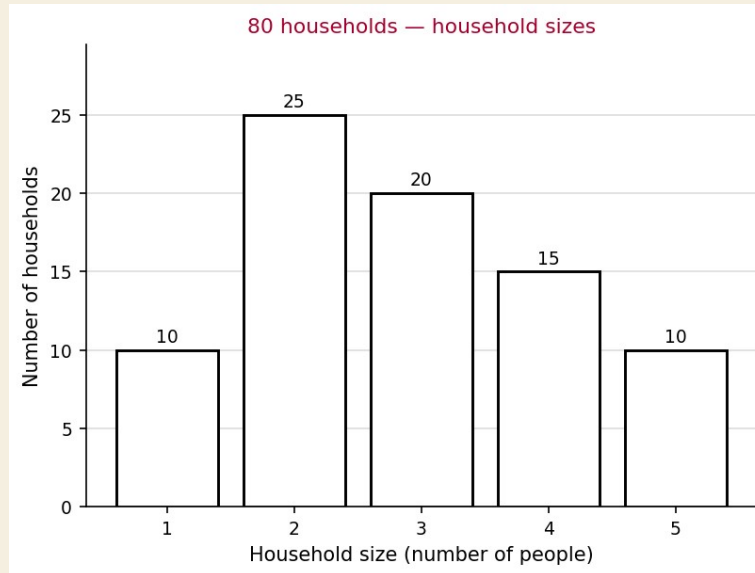
Desmos: Type $41 + 1.5 * 14$ to instantly confirm the upper outlier fence is 62 before comparing it to 68.

A dot plot shows the number of hours 9 students spent studying last week: 3, 4, 4, 5, 6, 6, 7, 8, and one student's hours are missing from the plot. The mean study time for all 9 students is 6 hours. What is the median study time for all 9 students, including the missing value?



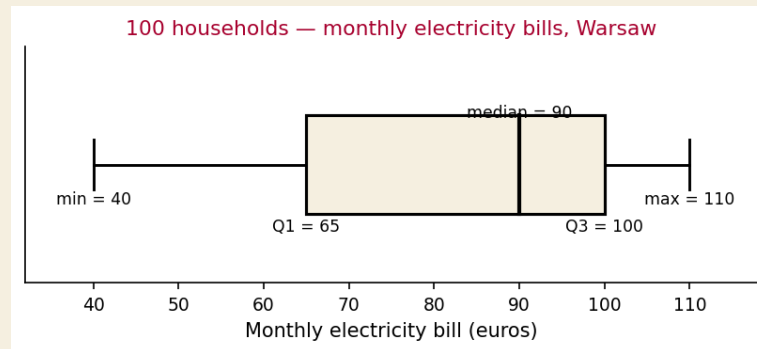
Desmos: Type $L1=\{3,4,4,5,6,6,7,8,x\}$, set $\text{mean}(L1)=6$ and solve for x , then compute $\text{median}(L1)$ using $x = 11$ to confirm the median is 6.

A histogram shows the household sizes recorded in a survey of 80 households: 1 person - 10 households, 2 people - 25 households, 3 people - 20 households, 4 people - 15 households, 5 people - 10 households. If a household from this survey is selected at random, given that the household has at least 3 people, what is the probability that it has exactly 4 people?



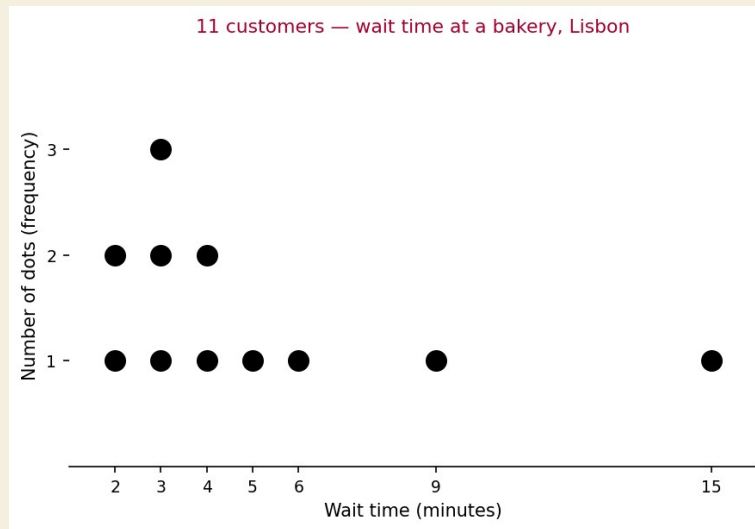
Desmos: Type 15/45 into the calculator and confirm the decimal is about 0.333, matching $\frac{1}{3}$.

A box plot for the monthly electricity bills, in euros, of 100 households in Warsaw has minimum = 40, $Q1 = 65$, median = 90, $Q3 = 100$, maximum = 110. A homeowner claims the distribution is roughly symmetric because the median is close to halfway between the minimum and maximum. Using the spacing of the quartiles instead, find $(\text{median} - Q1) - (Q3 - \text{median})$ to show whether the lower half or the upper half of the middle 50% of the data is more spread out.



Desmos: Type $(90-65)-(100-90)$ directly into the calculator to get 15 in one step.

A dot plot shows the number of minutes 11 customers waited in line at a bakery in Lisbon: 2, 2, 3, 3, 3, 4, 4, 5, 6, 9, 15. A market researcher claims that because the median wait time is under 5 minutes, the mean wait time must also be under 5 minutes. Find the actual mean wait time, in minutes, and determine whether the researcher's claim is correct.



Desmos: Type $L1=\{2,2,3,3,3,4,4,5,6,9,15\}$, then type $\text{mean}(L1)$ to get $56/11 \approx 5.09$ directly.

12. Scatterplots & Lines of Best Fit

Formula & Rules for This Skill

Line of best fit: $y = mx + b$, fit to minimize distance from all points.
Residual = actual y - predicted y . **Positive association:** slope > 0 ;
negative: slope < 0 .

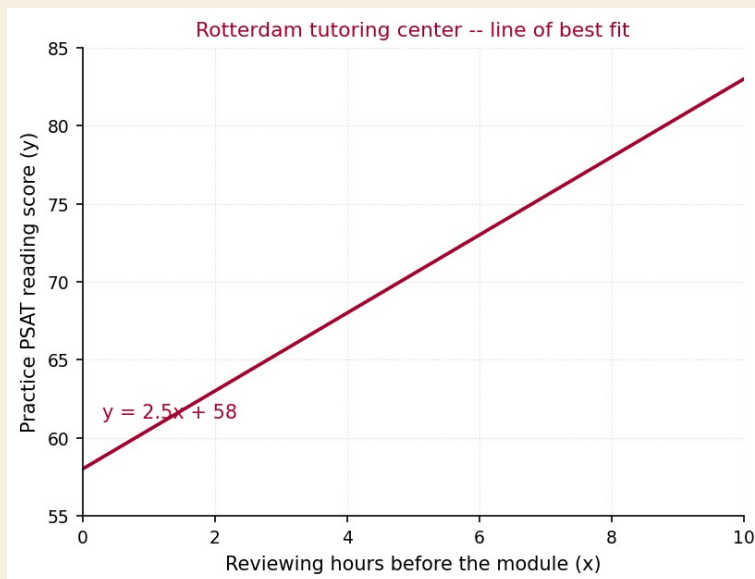
Fastest Way to Solve: Use the line of best fit's equation to predict values, and check the sign of the slope to describe the association before answering.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Scatterplots & Lines of Best Fit — Question 1 (Easy)

A tutoring center in Rotterdam recorded, for a group of students, the number of hours each student spent reviewing before a practice PSAT reading module (x) and the student's score out of 100 on that module (y). The line of best fit for the scatterplot of this data is $y = 2.5x + 58$. Using this line, what score does the model predict for a student who reviews for 6 hours?



Desmos: Type $y1 = 2.5x1 + 58$ into the calculator, then trace the graph to $x1 = 6$ to confirm the predicted score.

Scatterplots & Lines of Best Fit — Question 2 (Easy)

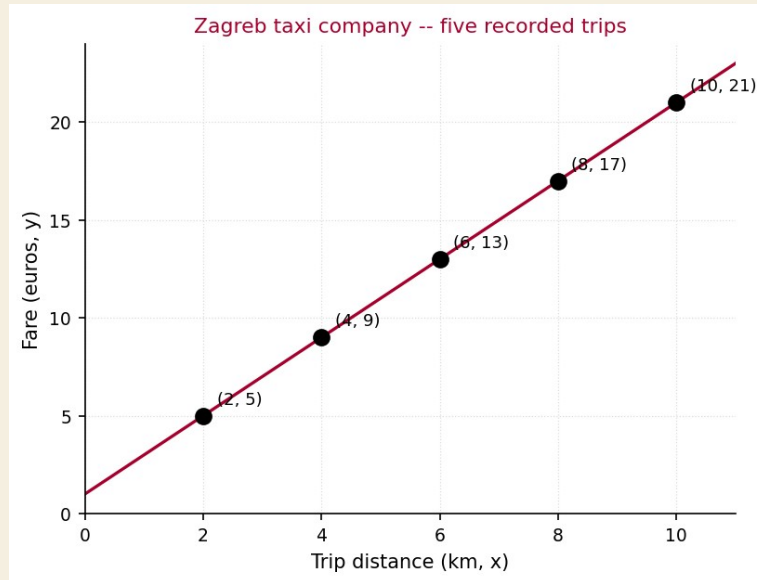
A used-car dealership in Lisbon models the relationship between a car's age in years (x) and its predicted resale value in euros (y) with the line of best fit $y = -1200x + 18000$. Based on this line, by how many euros does the model predict the resale value decreases for each additional year of the car's age?



Desmos: Type $y1 = -1200x1 + 18000$ and trace between two consecutive whole-number x -values to see the 1200-unit drop in y directly.

Scatterplots & Lines of Best Fit — Question 3 (Easy)

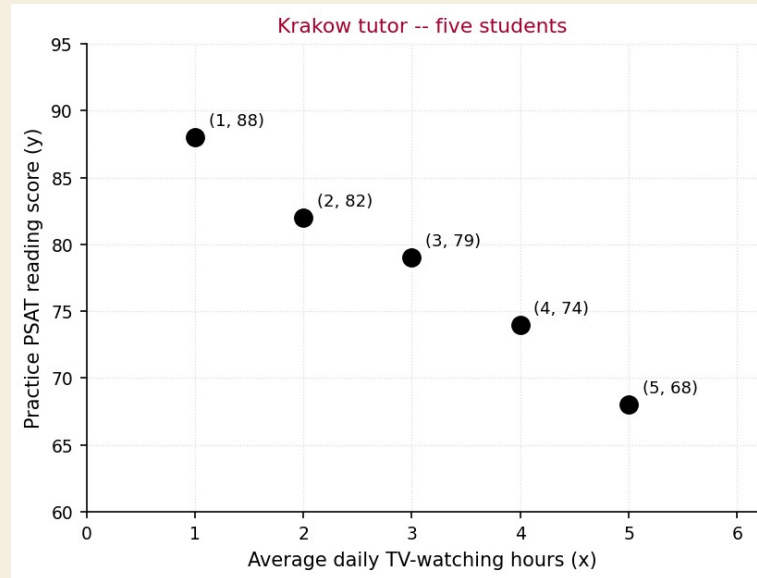
A taxi company in Zagreb recorded fares for several trips of different distances. The scatterplot shows these five data points, all falling exactly on the line of best fit: (2 km, 5 euros), (4 km, 9 euros), (6 km, 13 euros), (8 km, 17 euros), and (10 km, 21 euros). What is the slope of the line of best fit for this data?



Desmos: Enter the five points as a table (x1, y1 columns), then type $y1 \sim mx1 + b$ beneath it — Desmos will regress the line and report $m = 2$ directly.

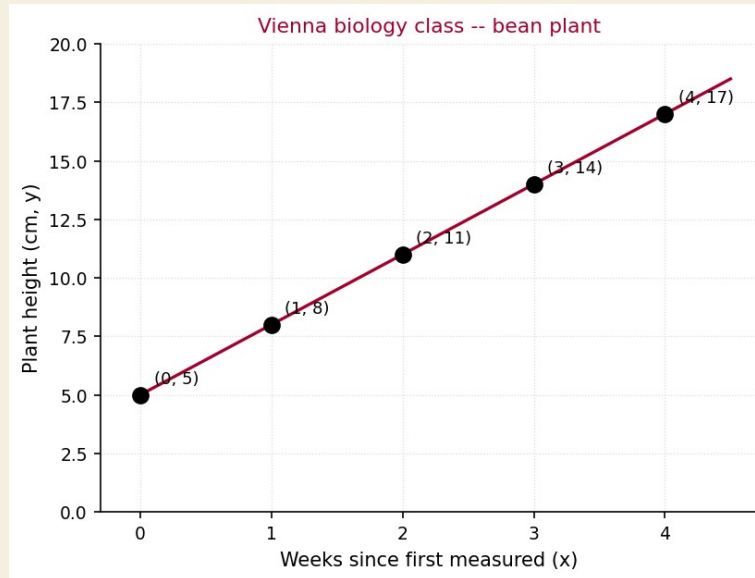
Scatterplots & Lines of Best Fit — Question 4 (Easy)

A tutor in Krakow recorded five students' average daily TV-watching hours (x) and their scores (y) on a practice PSAT reading section: (1 hour, 88), (2 hours, 82), (3 hours, 79), (4 hours, 74), and (5 hours, 68). Based on this scatterplot, what kind of correlation does the data show between TV hours and test score?



Desmos: Enter the five points as a table and look at the plotted points directly — a clear downward drift from left to right confirms the negative correlation without needing to type a regression command.

A biology class in Vienna is tracking the height of a bean plant over time. They recorded these five measurements, which fall exactly on the line of best fit: (0 weeks, 5 cm), (1 week, 8 cm), (2 weeks, 11 cm), (3 weeks, 14 cm), and (4 weeks, 17 cm). What is the y-intercept of the line of best fit for this data?



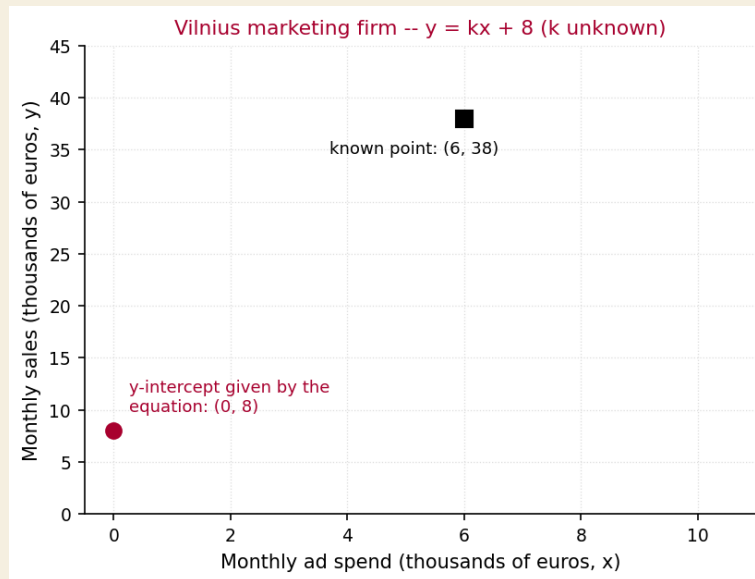
Desmos: Enter the five points as a table, then type $y_1 \sim mx_1 + b$ — the constant term Desmos reports is the y-intercept, 5.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Scatterplots & Lines of Best Fit — Question 6 (Medium)

A marketing firm in Vilnius models the relationship between monthly advertising spend in thousands of euros (x) and monthly sales in thousands of euros (y) with a line of best fit of the form $y = kx + 8$. The model predicts sales of 38 thousand euros when spend is 6 thousand euros. Using this information, what does the model predict for monthly sales when advertising spend is 10 thousand euros?



Desmos: Type $y1 = 5x1 + 8$ (using the k you solved for) and trace to $x1 = 10$ to confirm the predicted 58.

Scatterplots & Lines of Best Fit — Question 7 (Medium)

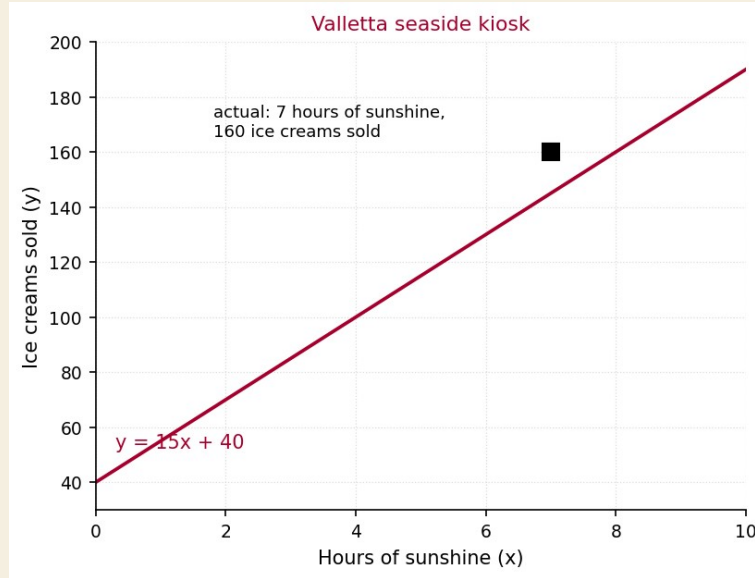
A secondhand bicycle shop in Ghent models the relationship between a bicycle's age in years (x) and its resale price in euros (y) with the line of best fit $y = -8x + 150$. According to this model, after how many years is a bicycle's predicted resale price 54 euros?



Desmos: Type $y1 = -8x1 + 150$, then trace the graph to where $y1 = 54$ to read off $x1 = 12$.

Scatterplots & Lines of Best Fit — Question 8 (Medium)

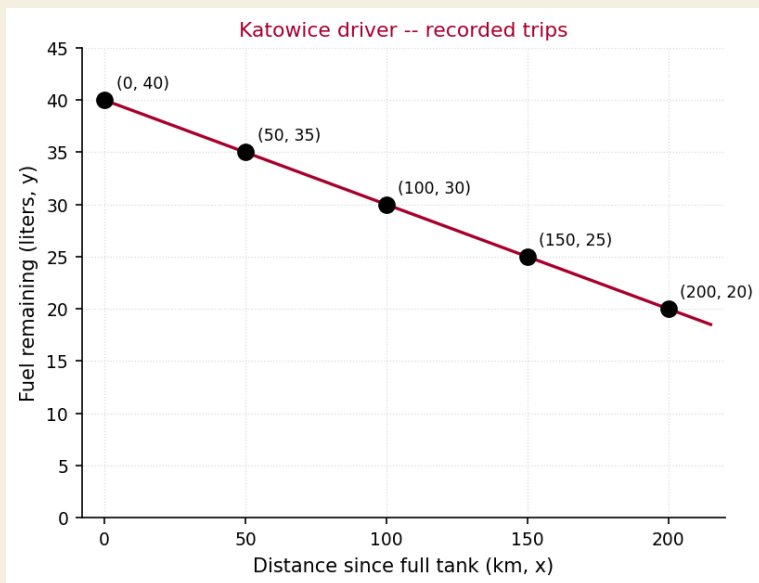
A researcher in Valletta models the relationship between hours of sunshine in a day (x) and the number of ice creams sold at a seaside kiosk (y) with the line of best fit $y = 15x + 40$. On a day with 7 hours of sunshine, the kiosk actually sold 160 ice creams. What is the residual (actual minus predicted) for that day?



Desmos: Type $y1 = 15x1 + 40$, trace to $x1 = 7$ to see the predicted value, then subtract it from 160 by hand to get the residual.

Scatterplots & Lines of Best Fit — Question 9 (Medium)

A driver in Katowice recorded, over several trips, how many kilometers the car had driven since a full tank (x) and how many liters of fuel remained (y). The scatterplot's five recorded points fall exactly on the line of best fit: (0 km, 40 L), (50 km, 35 L), (100 km, 30 L), (150 km, 25 L), and (200 km, 20 L). The car's fuel tank has a total capacity of 45 liters. Based on the line of best fit, how many liters of fuel does the model predict remain after the car has driven 320 km?



Desmos: Enter the five points as a table, then type $y1 \sim mx1 + b$ to get the regression line automatically, and trace it to $x1 = 320$ to read off 8.

Scatterplots & Lines of Best Fit — Question 10 (Medium)

A staffing agency in Bratislava models the relationship between an employee's years of experience (x) and their monthly salary in euros (y) with a line of best fit given by $300x - y = -2200$. According to this model, what monthly salary, in euros, does the line predict for an employee with 9 years of experience?



Desmos: Type $y1 = 300x1 + 2200$ (the rearranged form) and trace to $x1 = 9$ to confirm the predicted 4900.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Scatterplots & Lines of Best Fit — Question 11 (Hard)

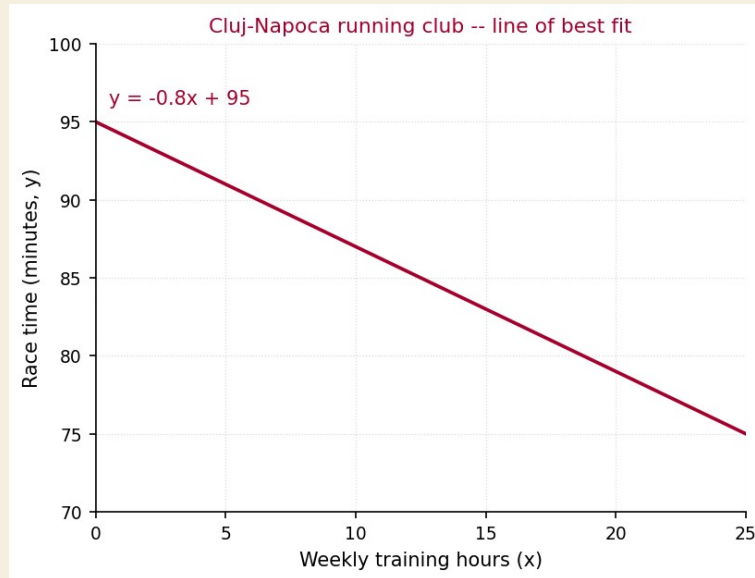
An online retailer based in Espoo, Finland tracks the number of customer service calls it receives per day (x) against that day's average customer satisfaction score out of 100 (y). The scatterplot's five recorded points fall exactly on the line of best fit: (10 calls, 80), (20 calls, 70), (30 calls, 60), (40 calls, 50), and (50 calls, 40). According to the line of best fit for this data, by what percent does the predicted satisfaction score decrease from a day with 10 calls to a day with 30 calls?



Desmos: Enter the five points as a table, type $y1 \sim mx1 + b$ to get the regression line, then trace to $x1 = 10$ and $x1 = 30$ to read off both predicted scores before finding the percent decrease by hand.

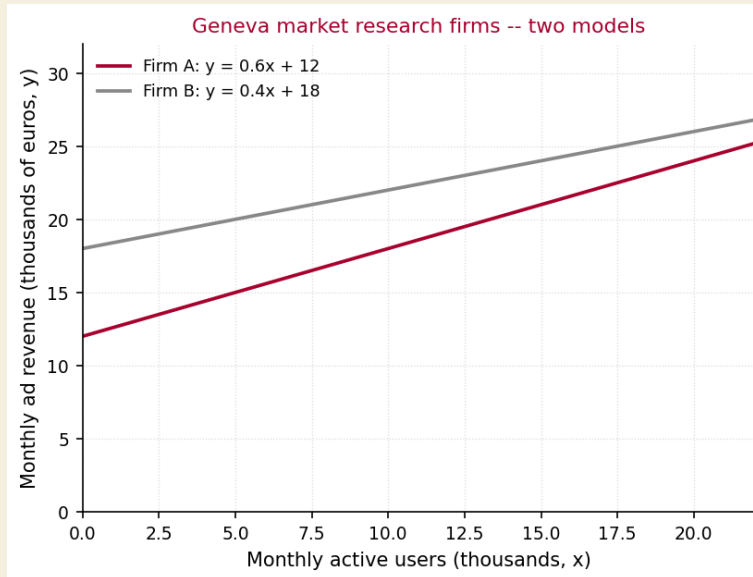
Scatterplots & Lines of Best Fit — Question 12 (Hard)

A running club in Cluj-Napoca models the relationship between a member's weekly training hours (x) and their race time in minutes (y) with the line of best fit $y = -0.8x + 95$. For one runner who trains 15 hours per week, the residual (actual race time minus predicted race time) was -3.5 minutes. What was that runner's actual race time, in minutes?



Desmos: Type $y1 = -0.8x1 + 95$, trace to $x1 = 15$ to confirm the predicted 83, then add the residual by hand to find the actual time.

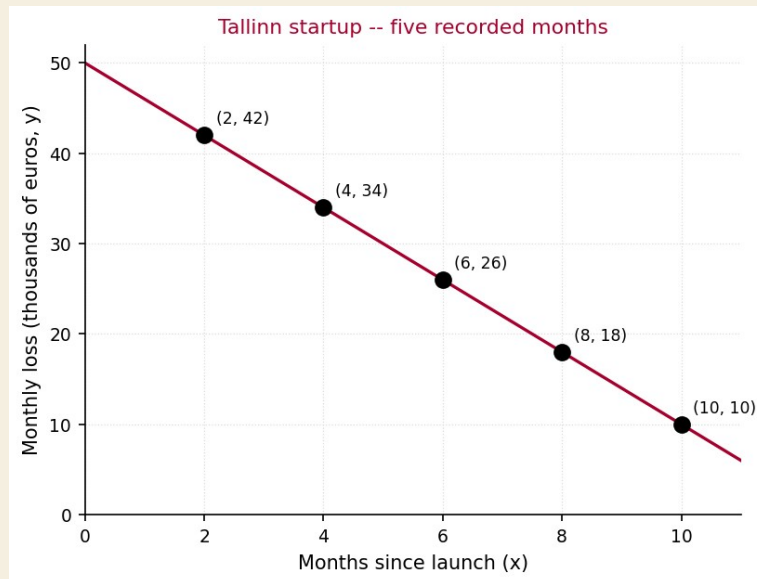
Two market research firms in Geneva separately modeled the relationship between an app's number of monthly active users, in thousands (x), and its monthly ad revenue, in thousands of euros (y), using different survey samples. Firm A's line of best fit is $y = 0.6x + 12$. Firm B's line of best fit is $y = 0.4x + 18$. At what number of monthly active users, in thousands, do the two firms' models predict the same ad revenue?



Desmos: Type $y1 = 0.6x1 + 12$ and $y2 = 0.4x1 + 18$ as two separate lines, then use the intersection point Desmos displays to read $x1 = 30$ directly.

Scatterplots & Lines of Best Fit — Question 14 (Hard)

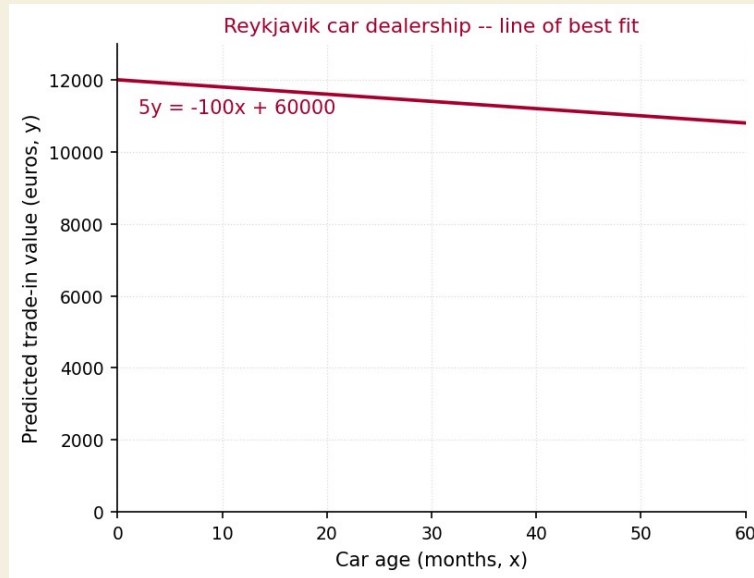
A startup based in Tallinn is tracking its monthly loss as it grows. The scatterplot's five recorded points fall exactly on the line of best fit: (2 months since launch, 42 thousand euros lost), (4 months, 34 thousand euros lost), (6 months, 26 thousand euros lost), (8 months, 18 thousand euros lost), and (10 months, 10 thousand euros lost). If this linear trend continues, in which whole month after launch does the company first have a monthly loss of 0 or less (that is, first reach breakeven)?



Desmos: Enter the five points as a table, type $y_1 \sim mx_1 + b$ to get the regression line, then trace near $y_1 = 0$ to see the crossing point falls between $x_1 = 12$ and $x_1 = 13$.

Scatterplots & Lines of Best Fit — Question 15 (Hard)

A car dealership in Reykjavik models the relationship between a car's age in months (x) and its predicted trade-in value in euros (y) with the line of best fit $5y = -100x + 60000$. According to this model, what is the predicted trade-in value, in euros, of a car that is 4 years old?



Desmos: Type $y1 = -20x1 + 12000$ (the simplified form), then trace to $x1 = 48$ to confirm the predicted 11040.



13. Sampling, Bias & Margin of Error

Formula & Rules for This Skill

A random sample fairly represents its population; margin of error shrinks as sample size grows. Bias comes from how the sample was chosen, not from its size alone.

Fastest Way to Solve: Ask whether every member of the population had an equal, random chance of being chosen — if not, look for the specific source of bias.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Sampling, Bias & Margin of Error — Question 1 (Easy)

The city council of Rotterdam wants to estimate what percent of ALL residents support building a new bicycle bridge over the river. Instead of contacting a random sample of residents, they post a survey link only on the council's own social media page, which is followed mostly by residents who already follow local politics closely. Of the residents who click the link and respond, 82% say they support the bridge. Explain why this 82% is probably not an accurate estimate of support among all residents.

Sampling, Bias & Margin of Error — Question 2 (Easy)

A pollster in Reykjavik surveys a random sample of adults about a proposed carbon tax and finds that 44% support it, with a reported margin of error of 3 percentage points at the 95% confidence level. Find the interval of plausible values for the true percentage of all adults who support the tax.



Sampling, Bias & Margin of Error — Question 3 (Easy)

A research firm wants to estimate average daily screen time for ALL adults in Lisbon. They conduct the survey by randomly dialing residential landline telephone numbers. Explain why this method is likely to produce a biased estimate, even though the landline numbers themselves are chosen randomly.

Sampling, Bias & Margin of Error — Question 4 (Easy)

A survey using simple random sampling of 500 residents produces a margin of error of 4 percentage points. If the same organization repeats the survey using proper simple random sampling of 1,000 residents instead, will the new margin of error most likely be greater than, less than, or equal to 4 percentage points?

Desmos: Graph $y = 1/\sqrt{x}$ and note that it decreases as x increases — this mirrors why a larger random sample produces a smaller margin of error.



Sampling, Bias & Margin of Error — Question 5 (Easy)

A teacher wants to select a simple random sample of 5 students from her class of 30 to interview about the course. Method A: she assigns each student a number from 1 to 30 and uses a random number generator to select 5 distinct numbers. Method B: she selects the first 5 students who arrive at class the next day. Identify which method produces a valid simple random sample, and explain the flaw in the other method.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Sampling, Bias & Margin of Error — Question 6 (Medium)

A single poll asks 800 registered voters in Valencia about a two-candidate mayoral race. Candidate Marin gets 51%, and Candidate Ibanez gets 46%, each with a margin of error of 3 percentage points at the 95% confidence level. A local newspaper reports 'Marin has a clear lead.' Using the margins of error, determine whether the poll actually supports that claim.

Desmos: Plot the four boundary values 48, 54, 43, 49 on a Desmos number line to see at a glance that the intervals overlap only between 48 and 49.

Sampling, Bias & Margin of Error — Question 7 (Medium)

A market research company in Bratislava wants a very small margin of error for a survey estimating the percent of ALL smartphone users in the city who prefer a particular messaging app. Instead of contacting a random sample of city residents, they survey every customer who visits their own retail phone shop over a month — a large group of 4,000 people — reasoning that such a big sample will make the margin of error very small and the result very accurate. Explain the flaw in this reasoning.

Sampling, Bias & Margin of Error — Question 8 (Medium)

A city survey in Ljubljana randomly selects residents by phone to ask about a proposed new tram line. Half the residents are asked: 'Do you support the new tram line, which will reduce car traffic and pollution in the city center?' The other half are asked simply: 'Do you support the new tram line?' The first group supports the tram line at a notably higher rate than the second group. Explain what is causing the difference in results, given that both groups were selected the same random way.

Sampling, Bias & Margin of Error — Question 9 (Medium)



A statistics agency randomly selects 3,000 households across Norway to mail a detailed survey about household income and spending habits. Only 400 households (about 13%) return the completed survey, and the agency uses just those 400 responses to estimate national averages. Even though the original 3,000 households were selected using a proper random sampling method, explain why the final results might still be biased.

Sampling, Bias & Margin of Error — Question 10 (Medium)

Estonia's national education ministry wants to estimate the percent of ALL secondary school students in the country who have reliable internet access at home. To save money, they distribute the survey exclusively through an app that students already need internet access to download and use, and they collect a very large sample of 50,000 student responses. A ministry official argues that because the sample size is so large, the margin of error will be tiny and the estimate will be trustworthy. Explain the flaw in the official's argument.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Sampling, Bias & Margin of Error — Question 11 (Hard)



In March, a national poll of 1,000 adults in Hungary found that 52% approved of a new transportation law, with a margin of error of 3 percentage points. In June, a second independent poll of a different random sample of 1,000 adults found 56% approval, with the same 3-point margin of error. A columnist claims 'support for the law clearly increased between March and June.' Using the margins of error, evaluate whether the polls actually establish that support increased.

Desmos: Plot the four boundary values 49, 55, 53, 59 on a Desmos number line to see at a glance that the two intervals overlap only between 53 and 55.

Sampling, Bias & Margin of Error — Question 12 (Hard)

A university in Milan draws a proper simple random sample of 300 students from its own list of currently enrolled students and surveys them about study habits, finding that 68% study primarily in the evening. The sampling method itself was entirely valid. A newspaper then reports: '68% of Milan's young adults study primarily in the evening.' Explain what is wrong with the newspaper's claim, given that the sampling method itself was valid.

Sampling, Bias & Margin of Error — Question 13 (Hard)



A research firm in Zurich reports that a survey of 100 randomly selected commuters produced a margin of error of 6 percentage points. The firm then runs a new, larger survey — still using simple random sampling of the same population — this time with 900 commuters. Assuming margin of error scales as $1/\sqrt{\text{sample size}}$, what is the new margin of error, in percentage points?

Desmos: Graph $y = 6/\sqrt{x/100}$ and trace to $x = 900$ to read off the new margin of error directly.

Sampling, Bias & Margin of Error — Question 14 (Hard)

A logistics company has three offices: Athens with 200 employees, Vienna with 500 employees, and Warsaw with 300 employees, for a total of 1,000 employees. The HR director wants to survey a sample of 100 employees company-wide to estimate average commute time, and wants the sample to reflect the same proportions across the three cities as the overall workforce. Using proportional stratified sampling, how many employees should be sampled from each city?

Sampling, Bias & Margin of Error — Question 15 (Hard)



A news article about Valletta states: 'Our online poll of 5,000 self-selected readers found that 61% support the new stadium proposal, with a margin of error of just 1.4 percentage points.' A reader concludes that because the margin of error is so small, the 61% figure must be very close to the true opinion of the entire city's population. Explain what's wrong with the reader's reasoning.

**PART 3 — ADVANCED MATH****14. Functions — Notation, Domain, Range & Graphs**

Formula & Rules for This Skill

$f(x)$ means "the output of function f when the input is x ." Domain = allowed inputs; Range = possible outputs. To evaluate $f(a)$, substitute a for every x .

Fastest Way to Solve: Treat $f(x)$ as an instruction, not a multiplication — substitute the given input for every x in the function's rule, one at a time.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Functions — Notation, Domain, Range & Graphs — Question 1 (Easy)

If $f(x) = 2x^2 - 5x + 1$, what is the value of $f(3)$?

Desmos: Type $f(x) = 2x^2 - 5x + 1$ in Desmos, then trace to $x = 3$ to read the y -value directly.

Functions — Notation, Domain, Range & Graphs — Question 2 (Easy)

What is the domain of the function $g(x) = 7/(x + 4)$?

Desmos: Type $y = 7/(x+4)$ in Desmos and note the vertical break in the graph (asymptote) at $x = -4$.



Functions — Notation, Domain, Range & Graphs — Question 3 (Easy)

What is the domain of the function $h(x) = \sqrt{x - 6}$?

Desmos: Type $y = \sqrt{x-6}$ in Desmos and observe that the graph only appears starting at $x = 6$ and extends rightward.

Functions — Notation, Domain, Range & Graphs — Question 4 (Easy)

What is the range of the function $f(x) = (x - 3)^2 + 8$?

Desmos: Type $y = (x-3)^2 + 8$ in Desmos and confirm the lowest point (vertex) on the graph has a y-coordinate of 8.

Functions — Notation, Domain, Range & Graphs — Question 5 (Easy)



If $f(x) = 4x - 9$, what is the value of $f(5) + f(-1)$?

Desmos: Type $y = 4x - 9$ in Desmos, trace to $x = 5$ and $x = -1$ to read both y -values, then add them.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Functions — Notation, Domain, Range & Graphs — Question 6 (Medium)

If $f(x) = 3x^2 + 2$ and $f(k) = 29$ for some positive number k , what is the value of k ?

Desmos: Type $y = 3x^2 + 2$ in Desmos and trace to where $y = 29$; the positive x -value there is k .

Functions — Notation, Domain, Range & Graphs — Question 7 (Medium)

If $f(x) = x^2 - 4x$, what is $f(x + 1)$ in simplified form?

Desmos: Type $y = (x+1)^2 - 4(x+1)$ and $y = x^2 - 2x - 3$ in Desmos on the same screen — the graphs overlapping exactly confirms the simplification.

Functions — Notation, Domain, Range & Graphs — Question 8 (Medium)

What is the domain of the function $f(x) = \sqrt{x - 1} / (x - 4)$?

Desmos: Type $y = \sqrt{x-1}/(x-4)$ in Desmos; the graph starts at $x = 1$ and shows a break (undefined point) at $x = 4$.

Functions — Notation, Domain, Range & Graphs — Question 9 (Medium)

The population P , in thousands, of Patras t years after 2010 can be modeled by a linear function. The graph of P passes through the points $(0, 120)$ and $(5, 145)$, and city records also show that tourism revenue rose sharply over the same period. Based on the population model, what is $P(8)$?

Desmos: Type $y = 5x + 120$ in Desmos, then trace to $x = 8$ to read the predicted population directly.

**Functions — Notation, Domain, Range & Graphs — Question 10** (Medium)

If $f(x) = ax^2 - 5$, where a is a constant, and $f(3) = 13$, what is the value of $f(-2)$?

Desmos: Type $y = a \cdot x^2 - 5$ with a slider for a in Desmos, adjust a until the graph passes through $(3, 13)$, then trace to $x = -2$ to read $f(-2)$.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Functions — Notation, Domain, Range & Graphs — Question 11 (Hard)

A function f is defined piecewise as $f(x) = 3x + 4$ for $x \leq 1$, and $f(x) = -x^2 + 10$ for $x > 1$. What is the value of $f(1) + f(4)$?

Desmos: Type $f(x) = \{x \leq 1: 3x+4, x > 1: -x^2+10\}$ in Desmos to graph the piecewise function directly and trace to $x=1$ and $x=4$ to confirm both values.

Functions — Notation, Domain, Range & Graphs — Question 12 (Hard)



A drone's height above the ground, $h(t)$, in meters, t seconds after launch, is modeled by $h(t) = -5(t - 3)^2 + 45$. The model is only meaningful while the drone is at or above ground level. What is the range of the drone's height during its flight, from launch to landing?

Desmos: Type $y = -5(x-3)^2+45$ in Desmos, restrict the trace to $0 \leq x \leq 6$, and read the lowest and highest y -values shown on that piece of the curve.

Functions — Notation, Domain, Range & Graphs — Question 13 (Hard)

The graph of $f(x) = ax^2 + bx + 12$ has a y -intercept of 12 and passes through the points (2, 24) and (-2, 8). What is the value of $f(5)$?

Desmos: Type $y = x^2 + 4x + 12$ in Desmos, confirm it passes through (2,24) and (-2,8), then trace to $x = 5$ to read $f(5)$.

Functions — Notation, Domain, Range & Graphs — Question 14 (Hard)

If $f(x) = 1/(x - 2)$ and $g(x) = \sqrt{x + 3}$, what is the domain of $f(g(x))$?

Desmos: Type $y = 1/(\sqrt{x+3}-2)$ in Desmos and look for where the graph starts ($x = -3$) and where it breaks (a vertical asymptote near $x = 1$).

**Functions — Notation, Domain, Range & Graphs — Question 15 (Hard)**

A bakery in Ljubljana models its daily profit, $P(x)$, in euros, from selling x loaves of specialty bread as $P(x) = -2x^2 + 80x - 150$. Due to oven capacity, the bakery only ever sells between 5 and 25 loaves per day, so the model is restricted to $5 \leq x \leq 25$. What is the range of the bakery's possible daily profit?

Desmos: Type $y = -2x^2 + 80x - 150$ in Desmos, restrict the trace to $5 \leq x \leq 25$, and read the lowest and highest points shown on that piece of the curve.



15. Composite & Inverse Functions

Formula & Rules for This Skill

$(f \circ g)(x) = f(g(x))$: evaluate the inside function first. Inverse function f^{-1} undoes f : if $f(a) = b$, then $f^{-1}(b) = a$.

Fastest Way to Solve: For composite functions, always evaluate the innermost function first; for inverses, swap x and y and re-solve for y .

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Composite & Inverse Functions — Question 1 (Easy)

If $f(x) = 3x - 2$ and $g(x) = x^2 + 1$, what is the value of $f(g(2))$?

Desmos: Type $y1 = x^2 + 1$ and $y2 = 3x - 2$, then trace $y1$ at $x = 2$ to read $g(2)$, and trace $y2$ at that output value to confirm $f(g(2)) = 13$.

Composite & Inverse Functions — Question 2 (Easy)

The table below gives the values of two functions, f and g , for several inputs. x : 0, 1, 2, 3 $f(x)$: 5, 8, 11, 14 $g(x)$: 2, 0, 3, 1 What is the value of $f(g(3))$?

Desmos: A table lookup like this isn't suited to Desmos graphing — just re-scan the table row by row to double-check both lookups.

Composite & Inverse Functions — Question 3 (Easy)

The function f is defined by $f(x) = 4x + 7$. What is $f^{-1}(x)$?

Desmos: Type $y = 4x + 7$ and $y = (x-7)/4$ together — the two lines should be mirror images across $y = x$, confirming the inverse.

Composite & Inverse Functions — Question 4 (Easy)

A shipping company in Rotterdam calculates the base cost to send a package using $C(w) = 4w + 6$, where w is the package's weight in kilograms and $C(w)$ is in euros. A fuel surcharge is then added using $S(x) = x + 3$, applied to the base cost. What is $S(C(5))$, in euros?

Desmos: Type $y = 4x + 9$ (the combined function) and trace to $x = 5$ to read the output 29 directly.

Composite & Inverse Functions — Question 5 (Easy)



The table below shows values of the function f . x : 1, 2, 3, 4 $f(x)$: 6, 9, 12, 15 What is the value of $f^{-1}(12)$?

Desmos: Type $y = 3x + 3$ (the rule behind the table) and trace to where $y = 12$ to confirm $x = 3$.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Composite & Inverse Functions — Question 6 (Medium)

The inverse of a function h satisfies $h^{-1}(x) = 5x - 3$ for all real numbers x . What is the value of $h(7)$?

Desmos: Type $y = 5x - 3$ and $y = x$ together; find where the first graph crosses $y = 7$, then read its x -coordinate — that x -coordinate, 2, is $h(7)$.

Composite & Inverse Functions — Question 7 (Medium)

The functions f and g are defined by $f(x) = 2x + 7$ and $g(x) = kx - 2$, where k is a constant. If $f(g(2)) = 15$, what is the value of $g(5)$?

Desmos: Type $y = 2(k \cdot x - 2) + 7$ with a slider for k , and adjust k until the graph passes through $(2, 15)$ — that slider value is k , ready to use in $g(5)$.

Composite & Inverse Functions — Question 8 (Medium)

If $f(2x + 1) = 6x - 5$ for all real numbers x , what is the value of $f(7)$?

Desmos: Type $y = 6((x-1)/2) - 5$, which rewrites f in terms of its actual input x , then trace to $x = 7$ to read $f(7) = 13$.

Composite & Inverse Functions — Question 9 (Medium)

A mobile carrier in Lisbon launched its 'EuroTalk' plan in 2019 and calculates a customer's monthly bill using $C(m) = 0.05m + 18$, where m is the number of minutes used and $C(m)$ is in euros. Portuguese law then requires an 18% value-added tax to be applied to the bill using $T(x) = 1.18x$. If a customer uses 200 minutes in a month, what is $T(C(200))$, in euros, rounded to the nearest cent?

Desmos: Type $y = 1.18(0.05x + 18)$ and trace to $x = 200$ to read the final billed amount, 33.04.

**Composite & Inverse Functions — Question 10** (Medium)

Let $f(x) = 3x + 2$ for all real numbers x , and let g be the function defined by the table below. x : 0, 1, 2, 3 $g(x)$: 5, 8, 11, 14 What is the value of $f^{-1}(g(2))$?

Desmos: Type $y = 3x + 2$ and trace to find where the graph's y -value equals 11 — the corresponding x -value, 3, is $f^{-1}(11)$.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Composite & Inverse Functions — Question 11 (Hard)

A courier company in Zurich calculates the delivery cost for a package traveling x kilometers using $D(x) = kx + 12$, where k is a constant and $D(x)$ is in Swiss francs. For deliveries booked online, a discount function $P(y) = 0.9y$ is then applied to the delivery cost. If the online price for a 20-kilometer delivery is $P(D(20)) = 45$ francs, what is the online price, in francs, for a 50-kilometer delivery?

Desmos: Type $y = 0.9(1.9x + 12)$ and trace to $x = 50$ to confirm the online price is 96.3 francs.

Composite & Inverse Functions — Question 12 (Hard)

If $f(3t - 2) = 5t + 4$ for all real numbers t , what is the value of x for which $f(x) = 39$?

Desmos: Type $y = 5((x+2)/3) + 4$, which is f rewritten in terms of its real input x , then trace to where $y = 39$ to read $x = 19$.

Composite & Inverse Functions — Question 13 (Hard)

The function f is defined by $f(x) = (x - 3)^2 + 1$ for $x \geq 3$. This restriction makes f one-to-one, so f^{-1} exists. What is the value of $f^{-1}(17)$?

Desmos: Type $y = (x-3)^2 + 1$ restricted to $x \geq 3$ (or just graph it and only trust the right half), then trace to where $y = 17$ to confirm $x = 7$.

**Composite & Inverse Functions — Question 14** (Hard)

Functions f and g satisfy the equations $f(x) + g(x) = 4x + 9$ and $f(x) - g(x) = 2x - 1$ for all real numbers x . What is the value of $f(g(3))$?

Desmos: Type $y1 = (4x + 9 + 2x - 1)/2$ for $f(x)$ and $y2 = (4x + 9 - (2x - 1))/2$ for $g(x)$, trace $y2$ at $x = 3$ to get 8, then trace $y1$ at $x = 8$ to confirm $f(g(3)) = 28$.

Composite & Inverse Functions — Question 15 (Hard)

A function f is one-to-one, and its graph passes through the points $(2, 5)$, $(5, 11)$, $(8, 20)$, and $(11, 35)$. A city's water utility in Porto has used f since 2015 to model pipe pressure at various depths. What is the value of $f^{-1}(11) + f^{-1}(20)$?

Desmos: Plot the four points directly in Desmos, then read off the x -coordinates of the points whose y -values are 11 and 20 to confirm $f^{-1}(11) + f^{-1}(20) = 13$.



16. Function Transformations

Formula & Rules for This Skill

$f(x) + k$ shifts up k . $f(x - h)$ shifts right h . $-f(x)$ reflects over the x -axis. $a f(x)$ stretches vertically by a .

Fastest Way to Solve: Match each change in the equation to exactly one transformation, and apply horizontal shifts using the opposite sign of what appears inside the parentheses.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Function Transformations — Question 1 (Easy)

If $f(x) = x^2 - 5x + 2$ and $g(x) = f(x) + 7$, what is the value of $g(3)$?

Desmos: Type $y = x^2 - 5x + 2 + 7$ and trace to $x=3$ to read $g(3)$ directly.

Function Transformations — Question 2 (Easy)

If $f(x) = 2x + 1$ and $g(x) = f(x - 5)$, what is the value of $g(5)$?

Desmos: Type $y = 2(x-5)+1$ and trace to $x=5$ to confirm $g(5) = 1$.



Function Transformations — Question 3 (Easy)

If $f(x) = x^2 + 3x$ and $g(x) = -f(x)$, what is the value of $g(2)$?

Desmos: Type $y = -(x^2 + 3x)$ and trace to $x=2$ to check $g(2) = -10$.

Function Transformations — Question 4 (Easy)

If $f(x) = x^2 - 4x$ and $g(x) = f(-x)$, what is the value of $g(-3)$?

Desmos: Type $y = (-x)^2 - 4(-x)$ and trace to $x=-3$ to confirm $g(-3) = -3$.

Function Transformations — Question 5 (Easy)

If $f(x) = 4x - 6$ and $g(x) = (1/2)f(x)$, what is the value of $g(4)$?

Desmos: Type $y = (1/2)(4x - 6)$ and trace to $x=4$ to confirm $g(4) = 5$.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Function Transformations — Question 6 (Medium)

The graph of $g(x) = f(x) + k$ passes through the point $(3, 11)$. If $f(3) = 4$, what is the value of $g(7)$, given that $f(7) = 9$?

Desmos: There's no graph to trace here — just confirm the arithmetic by typing $11-4$ to get $k=7$, then $9+7$ to verify $g(7)=16$.

Function Transformations — Question 7 (Medium)

Let $f(x) = x^2 - 2x$ and $g(x) = f(x + 3)$. If $g(x) = 8$ and x is negative, what is the value of x ?

Desmos: Type $y = (x+3)^2 - 2(x+3)$ and $y = 8$, then find their intersection points to confirm $x = -5$ is the negative solution.

Function Transformations — Question 8 (Medium)

If $f(x) = x^2 + 3x$ and $g(x) = 3f(x - 1)$, what is the value of $g(2)$?

Desmos: Type $y = 3((x-1)^2 + 3(x-1))$ and trace to $x=2$ to confirm $g(2) = 12$.

Function Transformations — Question 9 (Medium)

A function is defined by $h(x) = x^2 - 6x + 11$. Written in the form $h(x) = (x - 3)^2 + k$, this shows h is the graph of $f(x) = x^2$ after a transformation. What is the value of $h(0) - k$?

Desmos: Type $y = x^2 - 6x + 11$ and $y = (x-3)^2 + 2$ and confirm the two graphs overlap exactly, verifying $k = 2$.

Function Transformations — Question 10 (Medium)



A weather station in Reykjavik models the daily high temperature (in degrees Celsius) t days after January 1 using the function $C(t)$. The station's director, who has worked there for 12 years, updates the model for a data correction by shifting every value up by 3 degrees to account for a recalibrated sensor, giving a new model $D(t) = C(t) + 3$. If $C(20) = -5$, what is $D(20)$?

Desmos: Type $y = -5 + 3$ to confirm $D(20) = -2$ quickly, since only the arithmetic matters here, not a graph.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Function Transformations — Question 11 (Hard)

Let $f(x) = x^2 - 4x$. Define $g(x) = a f(x) + 5$ for some constant a . If $g(1) = -4$, what is the value of $g(5)$?

Desmos: Type $y = a(x^2 - 4x) + 5$ with a slider, adjust a until the graph passes through $(1, -4)$, then trace to $x=5$ to read $g(5) = 20$.

Function Transformations — Question 12 (Hard)

A firm in Zurich models its monthly revenue (in thousands of euros) using $R(x)$, where x is the number of months since launch. Analysts define a new function $P(x) = -2R(x + 2) + 5$ to model profit after accounting for a 2-month reporting delay and a cost structure that can turn revenue into a loss. If $R(4) = 18$, what is $P(2)$?

Desmos: Type $y = -2x + 5$ and trace to $x=18$ (the value of $R(4)$) to instantly read off $P(2) = -31$.

Function Transformations — Question 13 (Hard)

The graph of $y = f(x)$ is a line with equation $f(x) = mx + 4$, for some slope m . The graph of $g(x) = f(x - 2) + 6$ is also a line. If $g(x)$ has a y -intercept of 16, what is the slope m of f ?

Desmos: Type $y = m \cdot (x-2) + 4 + 6$ with a slider for m , and adjust until the graph crosses the y -axis at 16 to confirm $m = -3$.

Function Transformations — Question 14 (Hard)

The graph of $g(x) = a(x - h)^2 + 5$ is obtained from $f(x) = x^2$ by transformations. The graph passes through $(2, 5)$ and $(5, 32)$. What is the value of $a - h$?

Desmos: Type $y = a(x-2)^2 + 5$ with a slider for a , and adjust until the graph passes through $(5, 32)$ to confirm $a = 3$.

**Function Transformations — Question 15** *(Hard)*

An architect in Ljubljana designs a decorative arch whose height profile (in meters) is modeled by $h(x) = f(bx)$, where $f(x) = 20 - x^2/5$ gives the height of a reference arch at horizontal distance x (in meters) from center, and b is a horizontal compression factor used to fit the arch into a narrower space. The compressed arch $h(x)$ reaches a height of 15 meters at a horizontal distance of $x = 2$ meters from center. What is the value of b ?

Desmos: Type $y = 20 - (2x)^2/5$ and trace to find where $y=15$ to confirm the corresponding x -value equals $b = 2.5$.



17. Polynomial Operations & Special Products

Formula & Rules for This Skill

$(a+b)^2 = a^2 + 2ab + b^2$ · $(a-b)^2 = a^2 - 2ab + b^2$ · $(a+b)(a-b) = a^2 - b^2$ · **Distribute every term when multiplying polynomials.**

Fastest Way to Solve: Recognize the special-product patterns before expanding term by term — they save an entire step of algebra.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Polynomial Operations & Special Products — Question 1 (Easy)

Multiply: $(2x + 3)(x - 5)$. Write your answer as a simplified polynomial.

Desmos: Type $y1 = (2x+3)(x-5)$ and $y2 = 2x^2-7x-15$ and confirm the two graphs sit exactly on top of each other.

Polynomial Operations & Special Products — Question 2 (Easy)

Expand: $(x - 7)^2$. Write your answer as a simplified polynomial.

Desmos: Type $y = (x-7)^2 - (x^2-14x+49)$ and confirm the graph is the horizontal line $y = 0$.

**Polynomial Operations & Special Products — Question 3** (Easy)

Multiply: $(3x - 4)(3x + 4)$. Write your answer as a simplified polynomial.

Desmos: Type $y = (3x-4)(3x+4) - (9x^2-16)$ and confirm the graph is the horizontal line $y = 0$.

Polynomial Operations & Special Products — Question 4 (Easy)

Subtract: $(4x^2 - 3x + 5) - (2x^2 + 6x - 1)$. Write your answer as a simplified polynomial.

Desmos: Type $y = (4x^2-3x+5)-(2x^2+6x-1) - (2x^2-9x+6)$ and confirm it graphs as $y = 0$.

Polynomial Operations & Special Products — Question 5 (Easy)

Multiply: $(x + 2)(x^2 - 3x + 4)$. Write your answer as a simplified polynomial.

Desmos: Type $y = (x+2)(x^2-3x+4) - (x^3-x^2-2x+8)$ and trace to confirm it is identically zero.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Polynomial Operations & Special Products — Question 6 (Medium)

If $(x + a)(x - 3) = x^2 + bx - 15$ for all values of x , what is the value of b ?

Desmos: Type $y = (x+5)(x-3)$ and trace to the y-intercept to confirm it equals -15, then confirm the graph matches $x^2+2x-15$.

Polynomial Operations & Special Products — Question 7 (Medium)

Use a special product to compute 97×103 without long multiplication. What is the product?

Desmos: Type $97*103$ directly into the calculator and confirm it returns 9991.

**Polynomial Operations & Special Products — Question 8** (Medium)

Simplify completely: $(x + 6)^2 - (x - 6)^2$. Write your answer as a simplified polynomial.

Desmos: Type $y = (x+6)^2 - (x-6)^2$ and trace any point to confirm y equals 24 times that x-value.

Polynomial Operations & Special Products — Question 9 (Medium)

A rectangular garden plot in Delft has a length of $(x + 9)$ meters and a width of $(x - 4)$ meters. The gardener also plans to spend 15 euros per meter fencing the plot's perimeter. Write a simplified polynomial expression for the area of the plot, in square meters.

Desmos: Type $y = (x+9)(x-4) - (x^2+5x-36)$ and confirm the graph is the zero line.

Polynomial Operations & Special Products — Question 10 (Medium)

For which positive value of k does $x^2 - 24x + k$ form a perfect square trinomial?

Desmos: Type $y = x^2 - 24x + 144$ and confirm the graph touches the x-axis at exactly one point ($x=12$), confirming it's a perfect square.



Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Polynomial Operations & Special Products — Question 11 (Hard)

Simplify completely: $(2x - 5)^2 - (2x - 5)(2x + 5)$. Write your answer as a simplified polynomial.

Desmos: Type $y = (2x-5)^2 - (2x-5)(2x+5) - (-20x+50)$ and confirm the graph is identically zero.

Polynomial Operations & Special Products — Question 12 (Hard)

A metal sheet used to make storage crates for a shipping company in Rotterdam measures 40 cm by 30 cm. Equal squares of side length x centimeters are cut from each of the four corners, and the remaining sides are folded up to form an open-top box. The box's volume, in cubic centimeters, equals its length times its width times its height. Find the volume of the box when $x = 5$ cm.

Desmos: Type $y = x(40-2x)(30-2x)$ and trace to $x=5$ to read the volume directly off the graph.



Polynomial Operations & Special Products — Question 13 (Hard)

If $(2x - 3)(ax + b) = 6x^2 + cx - 12$ is true for every value of x , what is the value of $a + b + c$?

Desmos: Type $y = (2x - 3)(3x + 4) - (6x^2 - x - 12)$ and confirm the graph is the zero line, verifying $a = 3$, $b = 4$, $c = -1$ all at once.

Polynomial Operations & Special Products — Question 14 (Hard)

If $(x + y)^2 = 64$ and $xy = 10$, what is the value of $(x - y)^2$?

Desmos: Create sliders for x and y satisfying $xy = 10$, then compute $(x + y)^2$ and $(x - y)^2$ as separate expressions to confirm their difference is always $4xy$.

Polynomial Operations & Special Products — Question 15 (Hard)



A tech startup in Zurich models its weekly profit, in thousands of francs, as $P(x) = (x + 8)(x - 8) - (x^2 - 100)$, where x represents the number of product units sold, in hundreds. The company's CEO also announced a one-time 12,000-franc bonus pool for employees. Simplify $P(x)$ completely, then find $P(6)$.

Desmos: Type $y=(x+8)(x-8)-(x^2-100)$ and confirm the graph is a horizontal line at $y=36$ for every x , including $x=6$.



18. Factoring Polynomials

Formula & Rules for This Skill

GCF first, always. Difference of squares: $a^2 - b^2 = (a+b)(a-b)$.

Trinomial ax^2+bx+c : find two numbers that multiply to ac and add to b .

Fastest Way to Solve: Pull out the greatest common factor before trying any other factoring method — it simplifies every step that follows.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Factoring Polynomials — Question 1 (Easy)

Factor completely: $8x^2 + 20x$

Desmos: Type $y1 = 8x^2 + 20x$ and $y2 = 4x(2x + 5)$ as two separate graphs and confirm they lie exactly on top of each other.

Factoring Polynomials — Question 2 (Easy)

Factor completely: $x^2 - 64$

Desmos: Type $y = x^2 - 64$ and confirm the graph crosses the x-axis at $x = -8$ and $x = 8$, matching the two binomial factors.

Factoring Polynomials — Question 3 (Easy)**Factor completely:** $x^2 + 9x + 20$

Desmos: Type $y = x^2 + 9x + 20$ and trace to the x-intercepts at $x = -4$ and $x = -5$ to confirm the factors $(x + 4)$ and $(x + 5)$.

Factoring Polynomials — Question 4 (Easy)**Factor completely:** $x^2 - 2x - 24$

Desmos: Type $y = x^2 - 2x - 24$ and trace to the x-intercepts at $x = 6$ and $x = -4$ to confirm the factors.

Factoring Polynomials — Question 5 (Easy)**Factor completely:** $x^2 + 14x + 49$

Desmos: Type $y = x^2 + 14x + 49$ and confirm it touches the x-axis at only one point, $x = -7$, matching the repeated root from $(x + 7)^2$.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Factoring Polynomials — Question 6 (Medium)

If $x^2 + kx - 45$ factors as $(x - 5)(x + 9)$, what is the value of k ?

Desmos: Type $y = (x - 5)(x + 9)$ and $y = x^2 + kx - 45$ as two lines and confirm the graphs coincide exactly.

Factoring Polynomials — Question 7 (Medium)

Factor completely: $3x^3 - 27x$

Desmos: Type $y = 3x^3 - 27x$ and trace the x-intercepts at $x = 0$, $x = 3$, and $x = -3$, matching the three factors $3x$, $(x - 3)$, and $(x + 3)$.

Factoring Polynomials — Question 8 (Medium)

A rectangular vineyard in Santorini has area $(6x^2 + 19x + 10)$ square meters, where x is measured in meters. The vineyard's owner also spent 450 euros last spring on fertilizer. If the length of the vineyard is $(2x + 5)$ meters, what is its width, in terms of x ?

Desmos: Type $y = (6x^2 + 19x + 10) / (2x + 5)$ and confirm the graph looks like the line $y = 3x + 2$ (with a hole at $x = -5/2$), confirming the width.

Factoring Polynomials — Question 9 (Medium)

Factor completely: $x^4 - 20x^2 + 64$

Desmos: Type $y = x^4 - 20x^2 + 64$ and trace all four x -intercepts at $x = -4, -2, 2,$ and 4 to confirm the four linear factors.

Factoring Polynomials — Question 10 (Medium)



The polynomial $x^2 + bx + 21$ factors into two binomials with integer constants. One of the binomials is $(x + 3)$. What is the value of b ?

Desmos: Type $y = (x + 3)(x + 7)$ and $y = x^2 + 10x + 21$ and confirm the graphs match to verify $b = 10$.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Factoring Polynomials — Question 11 (Hard)

For x not equal to 5, the expression $(x^2 - 3x - 10) / (x - 5)$ is equivalent to $x + k$ for a constant k . What is the value of k ?

Desmos: Type $y = (x^2 - 3x - 10) / (x - 5)$ and trace near $x = 5$ (avoiding the hole) to see the y -value approaching 7, matching $y = x + 2$ at $x = 5$.

Factoring Polynomials — Question 12 (Hard)



A rectangular plot of land near Zurich has area $(2x^2 - 2x - 40)$ square meters. The plot's width is $(x - 5)$ meters. Officials will also build a walking path around the plot with length equal to the plot's perimeter, not its area. Find an expression for the plot's length, in meters, in terms of x .

Desmos: Type $y = (2x^2 - 2x - 40) / (x - 5)$ and confirm the graph looks linear, matching $y = 2x + 8$ (with a hole at $x = 5$).

Factoring Polynomials — Question 13 (Hard)

The two solutions of $x^2 - 6x + c = 0$ are consecutive even integers. What is the value of c ?

Desmos: Type $y = x^2 - 6x + 8$ and trace the x -intercepts to confirm they land exactly at $x = 2$ and $x = 4$.

Factoring Polynomials — Question 14 (Hard)

The graph of $y = x^3 - 5x^2 - 4x + 20$ crosses the x -axis at three points. What are the three x -coordinates where the graph crosses the x -axis?

Desmos: Type $y = x^3 - 5x^2 - 4x + 20$ and trace to confirm the graph crosses the x -axis exactly at $x = -2$, $x = 2$, and $x = 5$.



Factoring Polynomials — Question 15 (Hard)

The polynomial $x^2 + bx - 36$ has a factor of $(x - 4)$, and both of its roots are integers. What is the positive difference between the two roots?

Desmos: Type $y = (x - 4)(x + 9)$ and trace the x-intercepts at $x = 4$ and $x = -9$ to read off both roots directly, then subtract to confirm the difference of 13.



19. Dividing Polynomials & Rational Expressions

Formula & Rules for This Skill

To simplify a rational expression, factor both the numerator and denominator fully, then cancel matching factors (not terms).

Fastest Way to Solve: Factor completely before canceling anything — you can only cancel whole factors, never individual terms inside a sum.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Dividing Polynomials & Rational Expressions — Question 1 (Easy)

Divide $x^2 + 9x + 20$ by $x + 4$. What is the quotient?

Desmos: Type $y1 = (x^2+9x+20)/(x+4)$ and $y2 = x+5$, then confirm the graphs overlap everywhere except at $x=-4$.

Dividing Polynomials & Rational Expressions — Question 2 (Easy)

What is the remainder when $x^3 - 2x^2 + 4x - 3$ is divided by $x - 2$?

Desmos: Type $y = x^3-2x^2+4x-3$ and trace to $x=2$ to read off the y-value directly, which is the remainder.

Dividing Polynomials & Rational Expressions — Question 3 (Easy)

Simplify $(x^2 - 16)/(x + 4)$ for $x \neq -4$.

Desmos: Type $y1=(x^2-16)/(x+4)$ and $y2=x-4$, then confirm the two graphs are identical except for a hole at $x=-4$.

Dividing Polynomials & Rational Expressions — Question 4 (Easy)

Use synthetic division to divide $2x^3 - 3x^2 - 11x + 6$ by $x - 3$. What is the quotient?

Desmos: Type $y1=(2x^3-3x^2-11x+6)/(x-3)$ and $y2=2x^2+3x-2$, and confirm the graphs match everywhere except at $x=3$.

Dividing Polynomials & Rational Expressions — Question 5 (Easy)

Simplify $((x+2)/(3x)) \times (6x^2/(x+2))$ for $x \neq 0$ and $x \neq -2$.

Desmos: Type $y1=((x+2)/(3x)) \times (6x^2/(x+2))$ and $y2=2x$ and confirm the graphs coincide except at $x=0$ and $x=-2$.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Dividing Polynomials & Rational Expressions — Question 6 (Medium)

For what value of k is $x + 3$ a factor of $x^2 + kx + 12$?

Desmos: Type $y = x^2 + 7x + 12$ (using the solved k) and confirm it crosses zero at $x = -3$.

Dividing Polynomials & Rational Expressions — Question 7 (Medium)

The polynomial $P(x) = x^3 + ax^2 - 4x + 6$ leaves a remainder of 3 when divided by $x - 1$. What is $P(2)$?

Desmos: Type $y = x^3 - 4x + 6$ (after solving for a) and trace to $x = 2$ to confirm the y -value is 6.

Dividing Polynomials & Rational Expressions — Question 8 (Medium)

Simplify $(x^2 - 1)/(x - 1) - (x + 3)$ for $x \neq 1$.

Desmos: Type $y = (x^2 - 1)/(x - 1) - (x + 3)$ and trace anywhere -- the graph should sit on the horizontal line $y = -2$.

Dividing Polynomials & Rational Expressions — Question 9 (Medium)

A packaging company in Rotterdam models the volume of a shipping crate as $V(x) = x^3 + 9x^2 + 26x + 24$ cubic meters, where x is a positive integer. The crate's height is $(x + 2)$ meters and its base width is $(x + 3)$ meters. Cardboard for the crate costs 1.50 euros per square meter. What is the length of the crate, in terms of x ?

Desmos: Type $y1 = (x^3 + 9x^2 + 26x + 24)/(x^2 + 5x + 6)$ and $y2 = x + 4$ and confirm the graphs match for $x > 0$.

Dividing Polynomials & Rational Expressions — Question 10 (Medium)



Simplify $[(x^2 - 4)/(x + 1)] \div [(x - 2)/(x^2 - 1)]$ to a single polynomial, for all excluded values of x .

Desmos: Type $y1=((x^2-4)/(x+1))/((x-2)/(x^2-1))$ and $y2=x^2+x-2$ and confirm the graphs agree except at the excluded x -values.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Dividing Polynomials & Rational Expressions — Question 11 (Hard)

The polynomial $P(x) = x^3 - 5x^2 + kx - 6$ is divisible by $(x - 2)$ with no remainder. When $P(x)$ is divided by $(x - 3)$ instead, the remainder is m . What is $k + m$?

Desmos: Type $y=x^3-5x^2+9x-6$ and trace to $x=3$ to confirm the remainder $m=3$ directly as the y -value.

Dividing Polynomials & Rational Expressions — Question 12 (Hard)



For $x \neq -3$, $(x^3 + 4x^2 - 3x - 18)/(x + 3)$ simplifies to a quadratic expression that can be factored as $(x - p)(x + q)$, where p and q are positive integers. What is $p + q$?

Desmos: Type $y = (x^3 + 4x^2 - 3x - 18)/(x + 3)$ and find its two x -intercepts -- they occur at $x = 2$ and $x = -3$, confirming the quotient's factors.

Dividing Polynomials & Rational Expressions — Question 13 (Hard)

A pharmaceutical lab in Osaka models the concentration of a drug in a patient's bloodstream, in mg/L, t hours after injection as $C(t) = 30t / (t^2 + 9)$. For what sum of all positive values of t does $C(t)$ equal 3 mg/L?

Desmos: Type $y = 30x/(x^2 + 9)$ and $y = 3$, then find both intersection points' x -coordinates and add them.

Dividing Polynomials & Rational Expressions — Question 14 (Hard)

The function $f(x) = x^3 - 2x^2 - 5x + 6$ has a zero at $x = 1$. Given that $f(x)$ can be written as $(x - 1)(x - a)(x - b)$ with $a < b$, what is $b - a$?

Desmos: Type $y = x^3 - 2x^2 - 5x + 6$ and find its three x -intercepts (-2 , 1 , and 3) directly by tracing.



Dividing Polynomials & Rational Expressions — Question 15 (Hard)

Solve for x : $\frac{2x}{x-4} = \frac{8}{x-4} + 3$. If there is a valid solution, give it; if every algebraic solution must be rejected, write 'no solution.'

Desmos: Type $y1=2x/(x-4)$ and $y2=8/(x-4)+3$ and note the graphs never actually intersect at any valid x -- they only appear to align at the hole $x=4$.



20. Solving Quadratic Equations & the Discriminant

Formula & Rules for This Skill

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. **Discriminant** $b^2 - 4ac$ tells the number of real solutions (positive: 2, zero: 1, negative: 0).

Fastest Way to Solve: Try factoring first since it's fastest; if the trinomial won't factor cleanly, move straight to the quadratic formula rather than guessing further.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Solving Quadratic Equations & the Discriminant — Question 1 (Easy)

What are the solutions to the equation $x^2 - 5x + 6 = 0$?

Desmos: Type $y = x^2 - 5x + 6$ and find where the graph crosses the x-axis; the two x-intercepts are the solutions.

Solving Quadratic Equations & the Discriminant — Question 2 (Easy)

Use the quadratic formula to solve $2x^2 - 7x + 3 = 0$.

Desmos: Type $y = 2x^2 - 7x + 3$ and check that the x-intercepts appear at $x = 1/2$ and $x = 3$.



Solving Quadratic Equations & the Discriminant — Question 3 (Easy)

Without solving, determine the number of real solutions to $x^2 + 4x + 7 = 0$ by evaluating its discriminant.

Desmos: Type $y = x^2 + 4x + 7$ and observe that the parabola never touches the x-axis, confirming no real solutions.

Solving Quadratic Equations & the Discriminant — Question 4 (Easy)

What are the solutions to $x^2 + 7x + 10 = 0$?

Desmos: Type $y = x^2 + 7x + 10$ and read the two x-intercepts directly off the graph.

Solving Quadratic Equations & the Discriminant — Question 5 (Easy)



What is the solution to $x^2 - 6x + 9 = 0$?

Desmos: Type $y = x^2 - 6x + 9$ and note the parabola only touches the x-axis at one point, $x=3$, confirming a repeated root.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Solving Quadratic Equations & the Discriminant — Question 6 (Medium)

A quadratic equation of the form $x^2 + bx + c = 0$ has solutions $x = -4$ and $x = 7$. What is the value of $b + c$?

Desmos: Type $y = (x+4)(x-7)$ and trace the intercepts at $x=-4$ and $x=7$ to confirm the equation.

Solving Quadratic Equations & the Discriminant — Question 7 (Medium)

Solve for x : $(x-2)(x+5) = 18$.

Desmos: Type $y = (x-2)(x+5) - 18$ and find where the graph crosses zero to get $x=-7$ and $x=4$.



Solving Quadratic Equations & the Discriminant — Question 8 (Medium)

For what value of k does the equation $x^2 - 8x + k = 0$ have exactly one real solution?

Desmos: Type $y = x^2 - 8x + 16$ (using $k=16$) and confirm the parabola just touches the x -axis at one point, $x=4$.

Solving Quadratic Equations & the Discriminant — Question 9 (Medium)

The equation $3x^2 - 5x + m = 0$ has $x=3$ as one of its solutions. What is the other solution?

Desmos: Type $y = 3x^2 - 5x - 12$ and trace the graph to confirm it crosses zero at $x=3$ and $x=-4/3$.

Solving Quadratic Equations & the Discriminant — Question 10 (Medium)



A ball is launched so that its height in meters after t seconds is given by $h(t) = -5t^2 + 20t$, on a day when the temperature in Vienna was 22 degrees Celsius. After how many seconds does the ball land back on the ground (other than at $t=0$)?

Desmos: Type $y = -5x^2 + 20x$ and trace to the second x-intercept to read off the landing time directly.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Solving Quadratic Equations & the Discriminant — Question 11 (Hard)

A rectangular garden in Bruges is 3 meters longer than it is wide. If the area of the garden is 108 square meters, what is the width of the garden, in meters?

Desmos: Type $y = x(x+3) - 108$ and trace to the positive x-intercept to confirm the width is 9.

Solving Quadratic Equations & the Discriminant — Question 12 (Hard)



For what positive value of k does the equation $kx^2 - 12x + 9 = 0$ have exactly one real solution, and what is that solution?

Desmos: Type $y = 4x^2 - 12x + 9$ and confirm the parabola touches the x -axis at exactly one point, $x=1.5$.

Solving Quadratic Equations & the Discriminant — Question 13 (Hard)

The equation $x^2 - 14x + m = 0$ has two real solutions that differ by 6. What is the value of m ?

Desmos: Type $y = x^2 - 14x + 40$ and confirm the two x -intercepts are $x=4$ and $x=10$, six apart.

Solving Quadratic Equations & the Discriminant — Question 14 (Hard)

Solve for x : $(x^2 - 9)/(x - 3) = 2x - 1$.

Desmos: Type $y = (x^2-9)/(x-3) - (2x-1)$ and trace to where it crosses zero, avoiding $x=3$ where the function is undefined, to confirm $x=4$.



Solving Quadratic Equations & the Discriminant — Question 15 (Hard)

For how many positive integers k less than 10 does the equation $x^2 - kx + 9 = 0$ have two distinct real solutions?

Desmos: Type $y = x^2 - 36$ and note it crosses zero at $x=6$ and $x=-6$, confirming the boundary $k=6$ is where the discriminant condition flips.



21. Quadratic Graphs, Roots & Word Problems

Formula & Rules for This Skill

Vertex form: $y = a(x-h)^2 + k$, vertex at (h,k) . **Roots are the x-intercepts**, where $y = 0$. **Axis of symmetry:** $x = -b/2a$.

Fastest Way to Solve: Identify which form the equation is already in (vertex, standard, or factored) — each form hands you a different piece of information for free.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Quadratic Graphs, Roots & Word Problems — Question 1 (Easy)

Solve for the roots of $x^2 - 2x - 15 = 0$.

Desmos: Type $y = x^2 - 2x - 15$ and look at where the graph crosses the x-axis to confirm the roots at $x = -3$ and $x = 5$.

Quadratic Graphs, Roots & Word Problems — Question 2 (Easy)

A parabola is given by $y = (x - 4)^2 - 9$. State the coordinates of its vertex and whether the vertex is a minimum or a maximum point.

Desmos: Type $y = (x - 4)^2 - 9$ and trace to the lowest point of the curve to confirm the vertex at $(4, -9)$.



Quadratic Graphs, Roots & Word Problems — Question 3 (Easy)

Find the x-intercepts of the parabola $y = x^2 - 6x + 5$.

Desmos: Type $y = x^2 - 6x + 5$ and identify where the graph crosses the x-axis to confirm intercepts at $x = 1$ and $x = 5$.

Quadratic Graphs, Roots & Word Problems — Question 4 (Easy)

Find the axis of symmetry of the parabola $y = 3x^2 - 12x + 7$.

Desmos: Type $y = 3x^2 - 12x + 7$ and trace to the vertex; the x-coordinate shown there confirms the axis of symmetry at $x = 2$.

Quadratic Graphs, Roots & Word Problems — Question 5 (Easy)



A ball is thrown upward so that its height in meters after t seconds is modeled by $h(t) = -5t^2 + 20t$. How many seconds after being thrown does the ball land back on the ground?

Desmos: Type $y = -5x^2 + 20x$ and trace to the second x-intercept to confirm the ball lands at $x = 4$.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Quadratic Graphs, Roots & Word Problems — Question 6 (Medium)

A quadratic function has roots $x = 3$ and $x = -7$, and its graph passes through the point $(0, -42)$ as its y-intercept. Write the function in standard form $y = ax^2 + bx + c$.

Desmos: Type $y = 2(x - 3)(x + 7)$ and check that the graph crosses the y-axis at -42 and the x-axis at -7 and 3.

Quadratic Graphs, Roots & Word Problems — Question 7 (Medium)

A parabola has vertex $(2, -5)$ and passes through the point $(4, 3)$. Find the value of a in its vertex form equation $y = a(x - 2)^2 - 5$.

Desmos: Type $y = 2(x - 2)^2 - 5$ and confirm the graph passes through $(4, 3)$ and has its lowest point at $(2, -5)$.



Quadratic Graphs, Roots & Word Problems — Question 8 (Medium)

Solve for all real values of x that satisfy $x^4 - 13x^2 + 36 = 0$.

Desmos: Type $y = x^4 - 13x^2 + 36$ and look for all four x -axis crossings to confirm the roots -3, -2, 2, and 3.

Quadratic Graphs, Roots & Word Problems — Question 9 (Medium)

A bakery in Rotterdam that is open 7 days a week models its weekly revenue from croissant sales, in euros, as $R(p) = -20p^2 + 160p$, where p is the price per croissant in euros. What price per croissant maximizes the bakery's weekly revenue?

Desmos: Type $y = -20x^2 + 160x$ and trace to the peak of the curve to confirm the maximum occurs at $x = 4$.

**Quadratic Graphs, Roots & Word Problems — Question 10** (Medium)

In the equation $x^2 - kx + 40 = 0$, both roots are positive integers, and one root is 3 more than the other. What is the value of k ?

Desmos: Type $y = x^2 - 13x + 40$ and confirm the graph crosses the x -axis at $x = 5$ and $x = 8$.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Quadratic Graphs, Roots & Word Problems — Question 11 (Hard)

A company in Ljubljana models its weekly profit, in thousands of euros, as $P(x) = -x^2 + 10x - 16$, where x is the number of production shifts run that week. For how many whole-number values of x is the company's weekly profit positive?

Desmos: Type $y = -x^2 + 10x - 16$ and look at which x -values between the two x -intercepts give a graph above the x -axis to confirm the interval $2 < x < 8$.

Quadratic Graphs, Roots & Word Problems — Question 12 (Hard)

A parabola has its vertex on the line $x = 5$, and one of its x -intercepts is at $(1, 0)$. What is the x -coordinate of the parabola's other x -intercept?

Desmos: Type $y = (x - 1)(x - 9)$ and confirm the graph's vertex falls exactly on the vertical line $x = 5$.

Quadratic Graphs, Roots & Word Problems — Question 13 (Hard)

The line $y = 4x + k$ is tangent to the parabola $y = x^2 + 2x + 9$, meaning the line touches the parabola at exactly one point. What is the value of k ?

Desmos: Type $y = x^2 + 2x + 9$ and $y = 4x + 8$ together and confirm the line touches the parabola at exactly one point.

Quadratic Graphs, Roots & Word Problems — Question 14 (Hard)

A drone in Valletta is launched from a rooftop so that its height above the ground, in meters, after t seconds is $h(t) = -5t^2 + 30t + 20$. At what time does the drone first reach a height of 45 meters, while it is still rising?

Desmos: Type $y = -5x^2 + 30x + 20$ and $y = 45$ together and trace to the first (leftmost) intersection point to confirm $t = 1$.



Quadratic Graphs, Roots & Word Problems — Question 15 (Hard)

A parabola crosses the x-axis at $(-2, 0)$ and $(6, 0)$, and also passes through the point $(0, -24)$. Write the parabola's equation in the form $y = ax^2 + bx + c$, then state the y-coordinate of its vertex.

Desmos: Type $y = 2(x + 2)(x - 6)$ and trace to the lowest point of the graph to confirm the vertex at $(2, -32)$.



22. Radicals & Rational Exponents

Formula & Rules for This Skill

$a^{(m/n)}$ = n-th root of (a^m) . Same base rules apply to rational exponents as to integer exponents.

Fastest Way to Solve: Rewrite radical expressions as rational exponents before combining or simplifying — it turns root operations into ordinary exponent rules.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Radicals & Rational Exponents — Question 1 (Easy)

Solve for x: $\sqrt{x + 9} = 7$.

Desmos: Type $y = \sqrt{x+9}$ and $y = 7$ as two separate graphs and find their intersection — the x-coordinate, 40, is the solution.

Radicals & Rational Exponents — Question 2 (Easy)

Simplify $(n^{(1/5)})^{15}$, writing your answer as a single power of n.

Desmos: Type $y = (x^{(1/5)})^{15}$ and $y = x^3$ for $x > 0$ and confirm the two graphs coincide.

Radicals & Rational Exponents — Question 3 (Easy)

Solve for x , where $x > 0$: $x^{3/2} = 125$.

Desmos: Type $y = x^{3/2}$ and $y = 125$ and trace to their intersection near $x = 25$ to confirm.

Radicals & Rational Exponents — Question 4 (Easy)

Rewrite the cube root of x^5 (that is, the cube root of x raised to the 5th power) as a single power of x , in the form $x^{m/n}$.

Desmos: Type $y = (x^5)^{1/3}$ and $y = x^{5/3}$ for $x > 0$ and confirm the two graphs overlap exactly.

Radicals & Rational Exponents — Question 5 (Easy)

Solve for x : $5\sqrt{x} = 30$.

Desmos: Type $y = 5\sqrt{x}$ and $y = 30$ and find where the graphs intersect — the x -coordinate, 36, is the solution.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Radicals & Rational Exponents — Question 6 (Medium)

Solve for x : $\sqrt{x+10} = x - 2$.

Desmos: Type $y = \sqrt{x+10}$ and $y = x - 2$ and look for where the two graphs actually cross — only one intersection point will appear, at $x = 6$.

Radicals & Rational Exponents — Question 7 (Medium)

Solve for x , where $x > 0$: $x^{-2/3} = 1/9$.

Desmos: Type $y = x^{-2/3}$ and $y = 1/9$ for $x > 0$ and trace to their intersection at $x = 27$.

**Radicals & Rational Exponents — Question 8** (Medium)

Simplify $\sqrt{x^3} \cdot x^{1/2}$, for $x \geq 0$, as a single power of x .

Desmos: Type $y = \sqrt{x^3} \cdot \sqrt{x}$ and $y = x^2$ for $x > 0$ and confirm the two graphs overlap.

Radicals & Rational Exponents — Question 9 (Medium)

If $a^{3/4} = 8$, what is the value of $a^{3/2}$?

Desmos: Type $y = x^{3/4}$ and $y = 8$, find their intersection to confirm $a = 16$, then type $16^{3/2}$ to confirm it evaluates to 64.

Radicals & Rational Exponents — Question 10 (Medium)

In a chemistry lab in Zurich, a technician needs to find the value of x that satisfies: the cube root of $(2x - 5)$ equals 3. (The lab's thermostat is separately set to 21 degrees Celsius for the experiment, which has no effect on this calculation.) What is x ?

Desmos: Type $y = (2x-5)^{1/3}$ and $y = 3$ and trace to their intersection at $x = 16$.



Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Radicals & Rational Exponents — Question 11 (Hard)

Solve for x : $\sqrt{x+7} = 3 + \sqrt{x-8}$.

Desmos: Type $y = \sqrt{x+7} - \sqrt{x-8}$ and $y = 3$, then trace to find where the graphs meet — the x -coordinate, 9, is the solution.

Radicals & Rational Exponents — Question 12 (Hard)

Solve for x : $x^{3/5} = -8$.

Desmos: Type $(-2)^5$ to confirm it equals -32; avoid typing $x^{3/5}$ directly for negative x , since the calculator may not evaluate a fractional power of a negative base the way the algebra does.

**Radicals & Rational Exponents — Question 13 (Hard)**

Find the sum of all real values of x that satisfy $x^{2/3} - 5x^{1/3} + 6 = 0$.

Desmos: Type $y = x^{2/3} - 5x^{1/3} + 6$ and find both x -intercepts — they should land at $x = 8$ and $x = 27$, which sum to 35.

Radicals & Rational Exponents — Question 14 (Hard)

A cube-shaped shipping container in Rotterdam has edge length e meters, where e satisfies $e^{3/2} = 512$. What is the surface area of the container, in square meters? (The surface area of a cube with edge length e is $6e^2$.)

Desmos: Type $y = x^{3/2}$ and $y = 512$, trace to confirm the intersection at $x = 64$, then type $6 \cdot 64^2$ directly to get the surface area.

Radicals & Rational Exponents — Question 15 (Hard)



For all $x > 0$, the expression $(x^{3/4} \text{ times } \sqrt{x})$ divided by $x^{1/2}$ is equivalent to x^k for some constant k . What is the value of $4k$?

Desmos: Type $y = (x^{3/4} \cdot \sqrt{x}) / \sqrt{x}$ and $y = x^{3/4}$ for $x > 0$ and confirm the graphs coincide, confirming $k = 3/4$ and thus $4k = 3$.



23. Exponential Functions — Growth & Decay

Formula & Rules for This Skill

Growth/decay: $y = a(1 \pm r)^t$, where a is the starting value and r is the rate as a decimal. Growth uses $(1+r)$; decay uses $(1-r)$.

Fastest Way to Solve: Identify a (the starting amount), r (the rate), and t (time) directly from the wording before writing the equation.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Exponential Functions — Growth & Decay — Question 1 (Easy)

A colony of bacteria in a lab dish grows according to the function $N(t) = 200(2)^t$, where t is measured in hours after the colony is first observed and $N(t)$ is the number of bacteria. How many bacteria are in the colony after 4 hours?

Desmos: Type $y = 200 \cdot 2^x$ and trace to $x = 4$ to read the y -value directly.

Exponential Functions — Growth & Decay — Question 2 (Easy)

A radioactive sample has a mass modeled by $M(t) = 80(0.5)^t$, where t is the number of years and $M(t)$ is measured in grams. What is the mass of the sample after 3 years?

Desmos: Type $y = 80 \cdot 0.5^x$ and trace to $x = 3$ to confirm the mass is 10.



Exponential Functions — Growth & Decay — Question 3 (Easy)

The value of a rare stamp collection is modeled by $V(t) = 1200(1.08)^t$, where t is the number of years since purchase and $V(t)$ is the value in dollars. By what percent does the value increase each year?

Desmos: Type $y = 1200 \cdot 1.08^x$ and compare y at $x=0$ and $x=1$ to see the value jump by exactly 8%.

Exponential Functions — Growth & Decay — Question 4 (Easy)

A city in Portugal had 12,000 registered bicycles in 2020. The number of registered bicycles has been increasing by 5% every year since then. How many bicycles were registered 2 years later?

Desmos: Type $y = 12000 \cdot 1.05^x$ and trace to $x = 2$ to read the bicycle count.

Exponential Functions — Growth & Decay — Question 5 (Easy)



A car in Munich is purchased for 24,000 euros. Its value decreases by 15% each year. What is the car's value after 2 years?

Desmos: Type $y = 24000 \cdot 0.85^x$ and trace to $x = 2$ to confirm the value is 17,340.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Exponential Functions — Growth & Decay — Question 6 (Medium)

The number of subscribers to a streaming service based in Stockholm grows according to $S(t) = a(1.12)^t$, where t is the number of months after launch. After 3 months, the service has 14,049 subscribers, rounded to the nearest whole number. What was the value of a , the number of subscribers at launch ($t = 0$), rounded to the nearest whole number?

Desmos: Type $y = x \cdot 1.404928 - 14049$ and trace to find where it crosses zero — that x -value is a .

Exponential Functions — Growth & Decay — Question 7 (Medium)



A vineyard in the Douro Valley records that its annual wine output grows exponentially. In year 1 the output was 800 cases, and in year 3 the output was 1,152 cases. Assuming the growth continues at the same constant rate, what will the output be in year 5?

Desmos: Type $y = 800 \cdot 1.2^x$ and trace to $x = 5$ to read the year-5 output directly.

Exponential Functions — Growth & Decay — Question 8 (Medium)

The amount of a certain medication remaining in a patient's bloodstream is reduced to $\frac{3}{4}$ of its previous amount every 6 hours. If a patient is given a 320 mg dose, how many milligrams remain in the bloodstream after 18 hours?

Desmos: Type $y = 320 \cdot \left(\frac{3}{4}\right)^{(x/6)}$ and trace to $x = 18$ to read the remaining amount directly in original hours.

Exponential Functions — Growth & Decay — Question 9 (Medium)

A tech startup in Tallinn, founded by three university friends, had 50 users when it launched its app. The number of users has been tripling every month since launch. The startup's office has 12 desks. How many users does the app have after 4 months?

Desmos: Type $y = 50 \cdot 3^x$ and trace to $x = 4$ to confirm 4,050 users.

**Exponential Functions — Growth & Decay — Question 10** (Medium)

A forest research team finds that the number of a certain beetle species in a reserve is modeled by an exponential function $B(t) = 600(b)^t$, where t is measured in years. After 2 years, the population has grown to 864. Find the value of b , then use it to predict the population after 4 years.

Desmos: Type $y = 600 \cdot 1.2^x$ and trace to $x = 4$ to confirm the population is about 1,244.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Exponential Functions — Growth & Decay — Question 11 (Hard)

A cup of coffee is initially at 90 degrees Celsius in a room that stays at a constant 20 degrees Celsius. Every 10 minutes, the difference between the coffee's temperature and the room temperature is cut in half. What is the coffee's temperature after 30 minutes?

Desmos: Type $y = 20 + 70 \cdot 0.5^{(x/10)}$ and trace to $x = 30$ to confirm the temperature is 28.75.



Exponential Functions — Growth & Decay — Question 12 (Hard)

An investment account earns an annual interest rate of 12%, compounded quarterly (4 times per year). If the account starts with 5,000 euros and no deposits or withdrawals are made, what is the balance after 2 years, rounded to the nearest euro?

Desmos: Type $y = 5000 * 1.03^x$ and trace to $x = 8$ (8 quarters) to read the balance after 2 years.

Exponential Functions — Growth & Decay — Question 13 (Hard)

Colony A starts with 100 bacteria and doubles every 3 hours. Colony B starts with 400 bacteria and doubles every 6 hours. Both colonies are only counted at hours that are multiples of 6. At the first such hour when Colony A's count exceeds Colony B's count, what is that hour?

Desmos: Type $y = 100 * 2^{(x/3)}$ and $y = 400 * 2^{(x/6)}$ as two graphs, then trace to the multiple of 6 where the first graph rises above the second.



Exponential Functions — Growth & Decay — Question 14 (Hard)

The concentration of a chemical tracer in a water sample decreases to 40% of its previous concentration every 2 hours. If the initial concentration is 625 parts per million (ppm), after how many hours will the concentration first drop to 16 ppm?

Desmos: Type $y = 625 \cdot 0.4^{(x/2)}$ and trace to where $y = 16$ to read the number of hours directly.

Exponential Functions — Growth & Decay — Question 15 (Hard)

A population model can be written as either $P(t) = 250(4)^t$ or $P(t) = 250(2)^{(kt)}$ for all values of t , where k is a constant. What is the value of k ?

Desmos: Type $y = 4^x$ and $y = 2^{(2x)}$ on the same graph and confirm they overlap perfectly, confirming $k = 2$.



24. Radical & Rational Equations

Formula & Rules for This Skill

To solve a radical equation, isolate the radical, then raise both sides to the matching power. Always check solutions in the original equation for extraneous roots.

Fastest Way to Solve: After solving, plug every solution back into the original equation — radical and rational equations often introduce extraneous solutions that must be rejected.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Radical & Rational Equations — Question 1 (Easy)

Solve for x: $\sqrt{3x - 5} = 7$

Desmos: Type $y = \sqrt{3x - 5}$ and $y = 7$ as two separate lines and find their intersection point; the x-coordinate is the answer.

Radical & Rational Equations — Question 2 (Easy)

Solve for x: $\frac{6}{x+1} = 3$

Desmos: Type $y = \frac{6}{x+1}$ and $y = 3$, then trace to their intersection to confirm $x = 1$.



Radical & Rational Equations — Question 3 (Easy)

Solve for x: $\sqrt{2x + 7} - 3 = 2$

Desmos: Type $y = \sqrt{2x+7} - 3$ and $y = 2$, then find where the two graphs cross.

Radical & Rational Equations — Question 4 (Easy)

Solve for x: $(x - 1)/(x + 5) = 1/3$

Desmos: Type $y = (x-1)/(x+5)$ and $y = 1/3$, then trace to the intersection point to confirm $x = 4$.

Radical & Rational Equations — Question 5 (Easy)

Solve for x: $\sqrt{x + 16} = 5$

Desmos: Type $y = \sqrt{x+16}$ and $y = 5$, then trace to find where they intersect.



Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Radical & Rational Equations — Question 6 (Medium)

The square root of $(x + 9)$ is 3 less than x . What is the value of x ?

Desmos: Type $y = \sqrt{x+9}$ and $y = x - 3$, then look at where the graphs actually intersect — only one crossing point is real.

Radical & Rational Equations — Question 7 (Medium)

A cyclist named Elena rides the first 240 km leg of a race at a steady speed of x km/h. For the second, identical 240 km leg, she increases her speed by 4 km/h and finishes that leg 2 hours faster than the first. Elena started the race at 7:15 in the morning. What was her original speed x , in km/h?

Desmos: Type $y = 240/x - 240/(x+4)$ and $y = 2$, then trace to find the positive x -value where they intersect.

**Radical & Rational Equations — Question 8** (Medium)

In the equation $(x + c)/(x - 4) = 3$, $x = 10$ is a solution. What is the value of x that satisfies $(x + c)/(x - 1) = 3$, using this same value of c ?

Desmos: Type $y = (x+8)/(x-1)$ and $y = 3$, then trace to confirm the intersection lands at $x = 5.5$.

Radical & Rational Equations — Question 9 (Medium)

Solve for x : $(2x - 1)^{1/2} + 3 = 8$

Desmos: Type $y = (2x-1)^{0.5} + 3$ and $y = 8$, then trace to find their intersection.

Radical & Rational Equations — Question 10 (Medium)



Solve for x: $\sqrt{x+5} = (x+2)/2$

Desmos: Type $y = \sqrt{x+5}$ and $y = (x+2)/2$ as two separate graphs; only their real intersection point, not the extra algebraic root, is the valid answer.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Radical & Rational Equations — Question 11 (Hard)

Solve for x: $4/\sqrt{x} = \sqrt{x} - 3$

Desmos: Type $y = 4/\sqrt{x}$ and $y = \sqrt{x} - 3$, then trace to the single point, for x greater than 0, where the two graphs actually cross.

Radical & Rational Equations — Question 12 (Hard)

In Porto, pump A alone would fill a tank in x hours. Pump B, a newer model with a blue casing, alone would fill the same tank in $(x - 3)$ hours. Working together, the two pumps fill the tank in 2 hours. What is x , in hours?

Desmos: Type $y = 1/x + 1/(x-3)$ and $y = 0.5$, then trace to the intersection where x is greater than 3, so both individual times stay positive.

Radical & Rational Equations — Question 13 (Hard)

For what value of x does $(x^2 - 9)/(x - 3) = 6$?

Desmos: Type $y = (x^2 - 9)/(x - 3)$ and $y = 6$; the graphs never actually meet at $x = 3$ because of the hole in the first graph there, confirming no solution.

Radical & Rational Equations — Question 14 (Hard)

A geologist in Reykjavik uses the formula $t = 1.5 + \sqrt{h/5}$ to estimate the total time t , in seconds, for a rock to hit the ground after being dropped from height h meters, where the 1.5 seconds accounts for equipment lag. The measuring equipment costs 220 euros to calibrate before each use. If a measured total time is 6.5 seconds, what is h , in meters?

Desmos: Type $y = 1.5 + \sqrt{x/5}$ and $y = 6.5$, then trace to find the x -value (height) where they intersect.



Radical & Rational Equations — Question 15 *(Hard)*

Solve for x: $\frac{x}{x-2} - \frac{1}{x+2} = \frac{8}{x^2-4}$

Desmos: Type $y = \frac{x}{x-2} - \frac{1}{x+2}$ and $y = \frac{8}{x^2-4}$, then trace to the intersection at $x = -3$; the graphs don't actually meet at $x = 2$, since both have a hole or asymptote there.

PART 4 — GEOMETRY AND TRIGONOMETRY**25. Angles, Parallel Lines & Triangle Fundamentals****Formula & Rules for This Skill**

Triangle angles sum to 180 degrees. Parallel lines cut by a transversal: alternate interior angles are equal, co-interior angles sum to 180. Exterior angle = sum of the two remote interior angles.

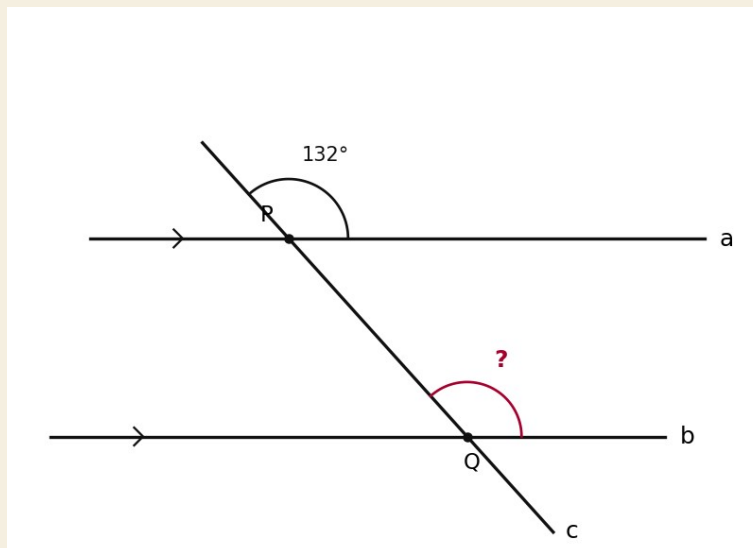
Fastest Way to Solve: Mark every angle you can find from parallel lines and straight lines first — most triangle problems solve themselves once enough angles are filled in.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

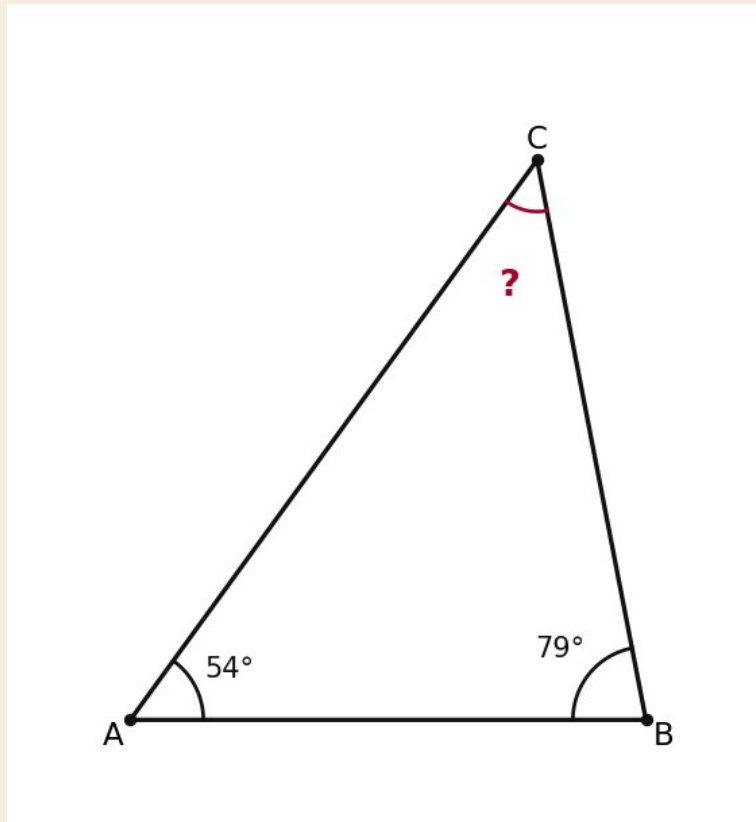
Angles, Parallel Lines & Triangle Fundamentals — Question 1 (Easy)

Line a and line b are parallel, and line c is a transversal crossing both of them: line c crosses line a at point P and crosses line b at point Q. At point P, the angle formed on the upper-right side of the intersection (above line a and to the right of line c) measures 132° . What is the measure of the angle at point Q that sits in the same upper-right position (above line b and to the right of line c) — the angle corresponding to it?



Angles, Parallel Lines & Triangle Fundamentals — Question 2 (Easy)

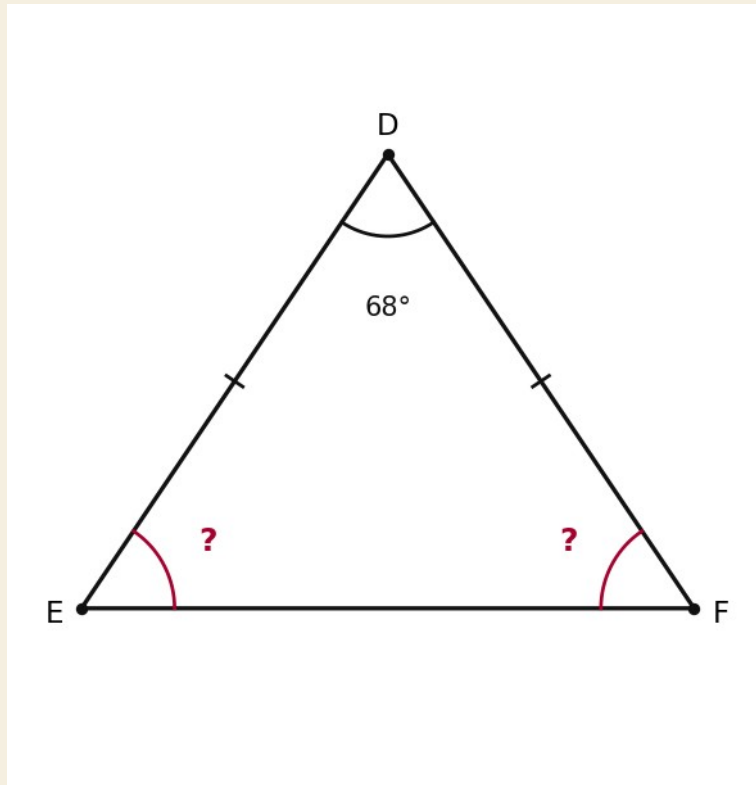
In triangle ABC, angle A measures 54° and angle B measures 79° . What is the measure of angle C, in degrees?



Desmos: Type $180-54-79$ directly into Desmos as a calculation to get angle C instantly.

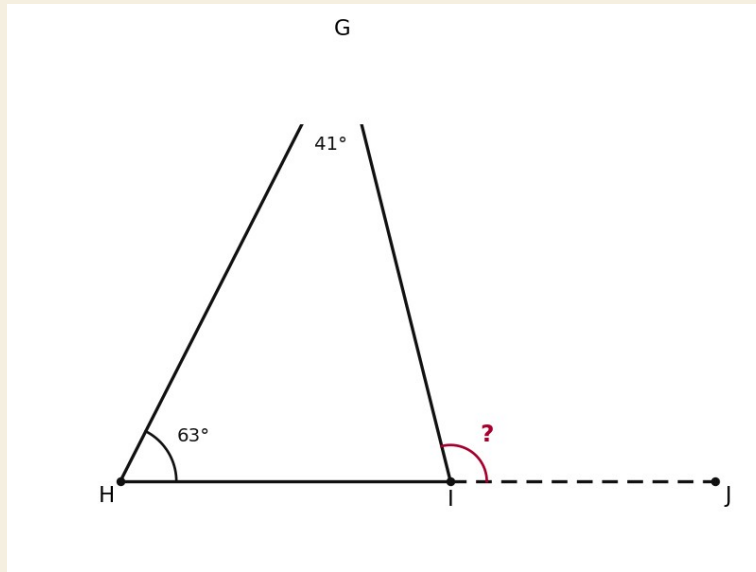
Angles, Parallel Lines & Triangle Fundamentals — Question 3 (Easy)

Triangle DEF is isosceles with $DE = DF$, and the vertex angle at D measures 68° . What is the measure of each base angle, angle E and angle F?



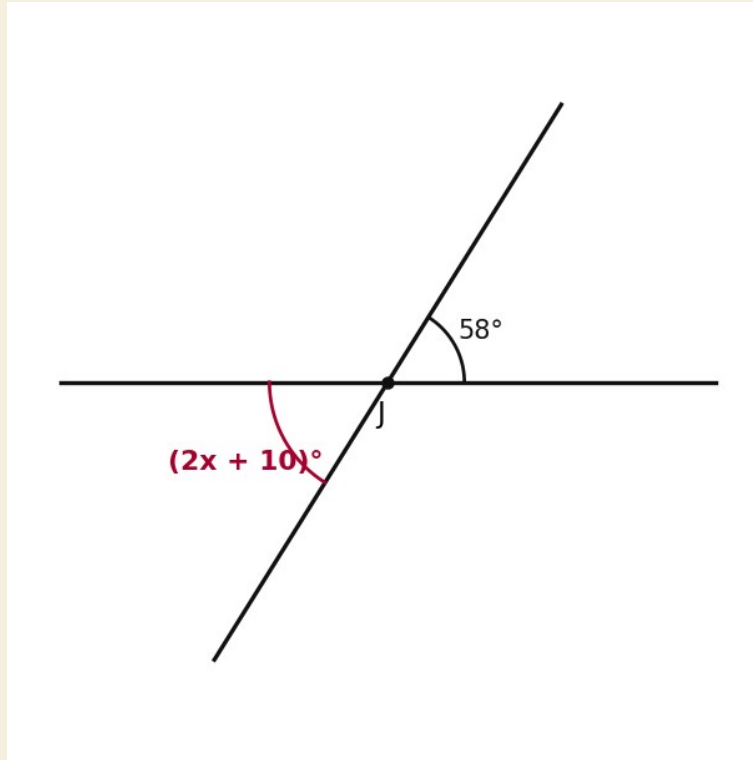
Desmos: Type $(180-68)/2$ into Desmos to get each base angle directly.

In triangle GHI, the two angles not adjacent to vertex I — angle G and angle H — measure 41° and 63° . Side HI is extended beyond I to a point J. What is the measure of the exterior angle GIJ (between side IG and the extension IJ)?



Desmos: Type $41+63$ into Desmos to confirm the exterior angle in one step.

Two lines intersect at point J, forming four angles. One of the angles measures $(2x + 10)^\circ$, and the angle vertical to it measures 58° . What is the value of x ?



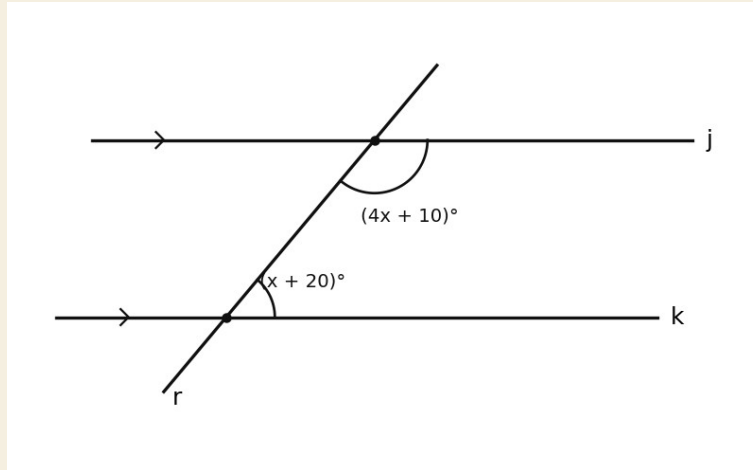
Desmos: Type $2x+10=58$ into Desmos — it will solve for x immediately.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Angles, Parallel Lines & Triangle Fundamentals — Question 6 (Medium)

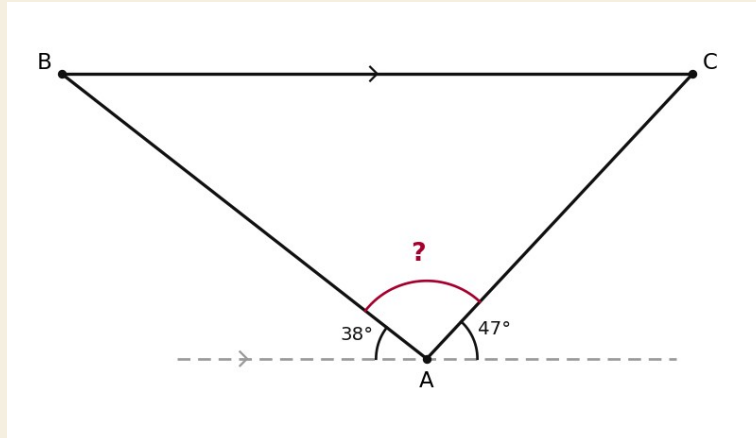
Line j and line k are parallel and are cut by a transversal, r . At the point where r crosses j , one of the same-side interior angles (also called a co-interior angle) — the one on the interior side of j , on the same side of r as line k — measures $(4x + 10)^\circ$. At the point where r crosses k , the same-side interior angle that pairs with it measures $(x + 20)^\circ$. What is the measure of the larger of these two angles?



Desmos: Type $4x+10+x+20=180$ into Desmos to solve for x directly, then substitute back into each expression.

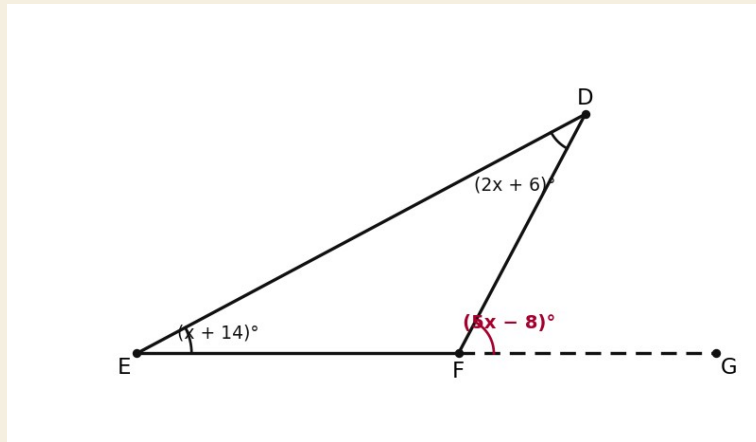
Angles, Parallel Lines & Triangle Fundamentals — Question 7 (Medium)

In triangle ABC, a line is drawn through vertex A parallel to side BC. This parallel line splits the straight angle at A into three adjacent angles: the angle between the parallel line and side AB (on the B side) measures 38° , the angle between the parallel line and side AC (on the C side) measures 47° , and the angle between them is angle BAC itself. What is the measure of angle BAC?



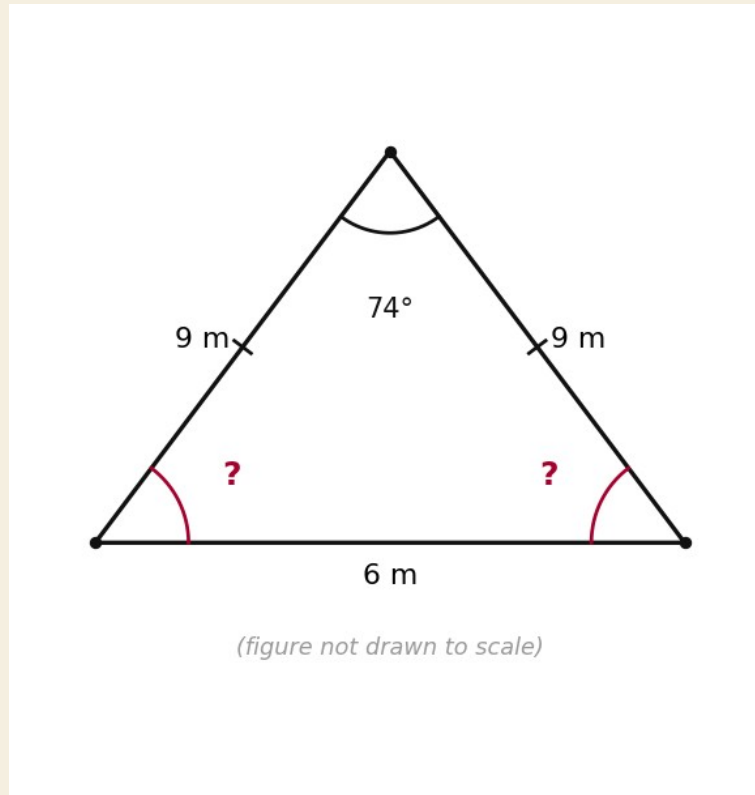
Desmos: Type $180-38-47$ into Desmos as a direct calculation to get angle BAC.

In triangle DEF, side EF is extended beyond F to a point G. Angle DFG (between side FD and the extension FG) measures $(5x - 8)^\circ$. The two interior angles at D and E measure $(2x + 6)^\circ$ and $(x + 14)^\circ$ respectively. What is the value of x ?



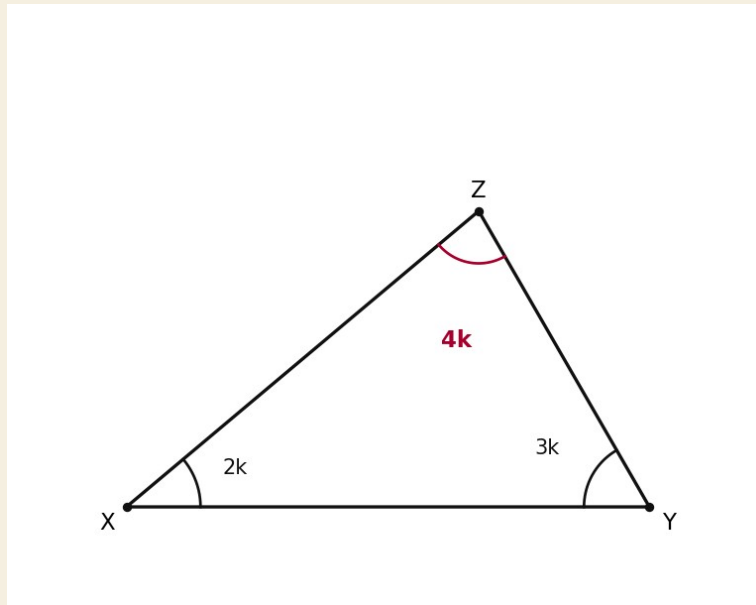
Desmos: Type $5x - 8 = 2x + 6 + x + 14$ into Desmos to solve for x directly.

A triangular flowerbed in a park in Rotterdam is isosceles, with two equal sides of length 9 meters each and a base of 6 meters, giving it a perimeter of 24 meters. The angle at the apex (between the two equal sides) measures 74° . What is the measure of each base angle?



Desmos: Type $(180-74)/2$ into Desmos to get each base angle directly.

The three interior angles of triangle XYZ are in the ratio 2:3:4. What is the measure of the largest angle?



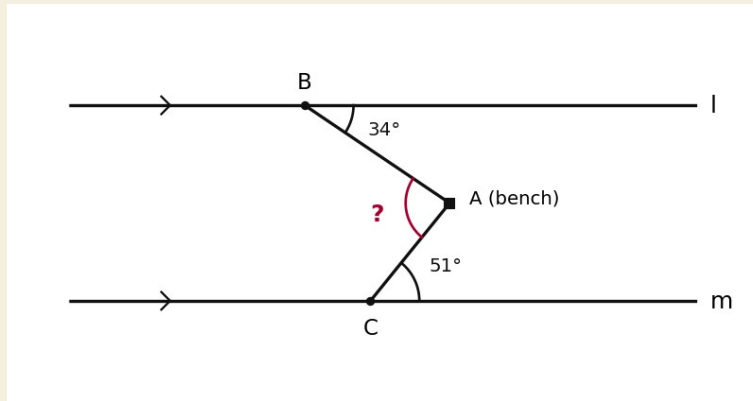
Desmos: Type $180/9 \cdot 4$ into Desmos to get the largest angle directly without solving for k as a separate step.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

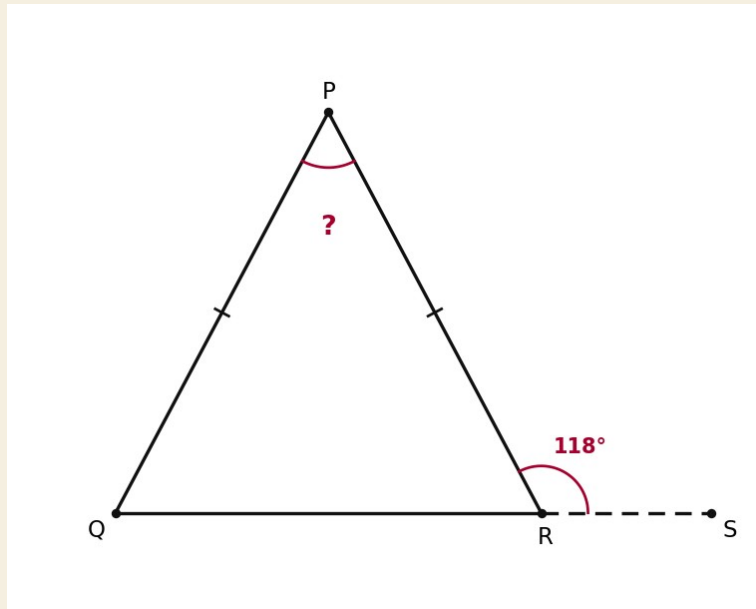
Angles, Parallel Lines & Triangle Fundamentals — Question 11 (Hard)

In a park in Vienna, two straight garden paths, path l and path m , run parallel to each other, with path l to the north of path m . A bench sits at point A , in the grassy area between the two paths (not on either path). A straight stone walkway runs from a point B on path l down to the bench at A , meeting path l at a 34° angle, measured on the side of the walkway that faces into the grassy area between the paths. A second straight stone walkway runs from the bench at A down to a point C on path m , meeting path m at a 51° angle, measured on the same side (facing into the grassy area between the paths). What is the measure of angle BAC , the angle between the two walkways at the bench?



Desmos: Type $34+51$ into Desmos to confirm the combined bend angle once you've identified both alternate interior parts.

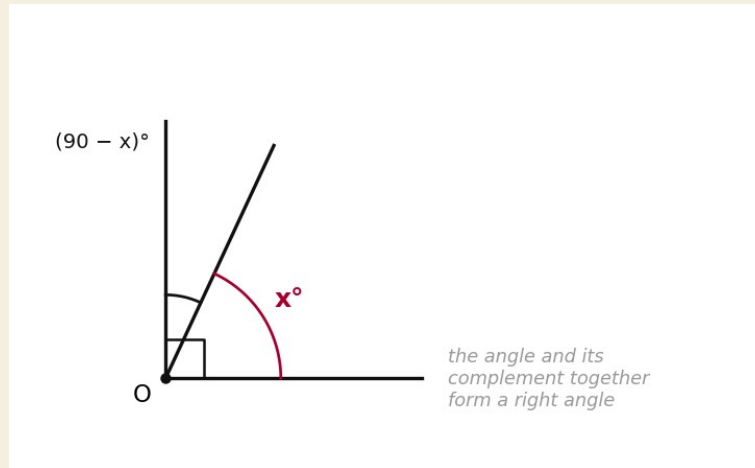
Triangle PQR is isosceles with $PQ = PR$, so angle $Q =$ angle R . Side QR is extended beyond R to a point S. The exterior angle PRS measures 118° . What is the measure of angle P, the vertex angle of the triangle?



Desmos: Type $180 - 2 * (180 - 118)$ into Desmos to get the vertex angle in one calculation.

Angles, Parallel Lines & Triangle Fundamentals — Question 13 (Hard)

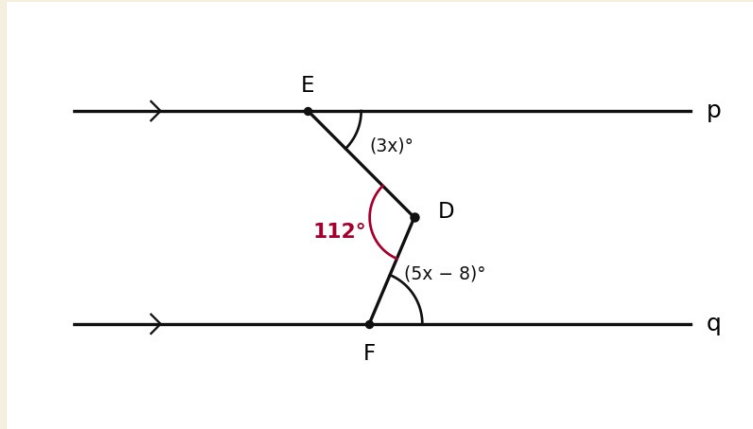
An angle's measure is 15° more than twice the measure of its complement. What is the measure of the angle?



Desmos: Type $x = 2 * (90 - x) + 15$ into Desmos and let it solve for x directly.

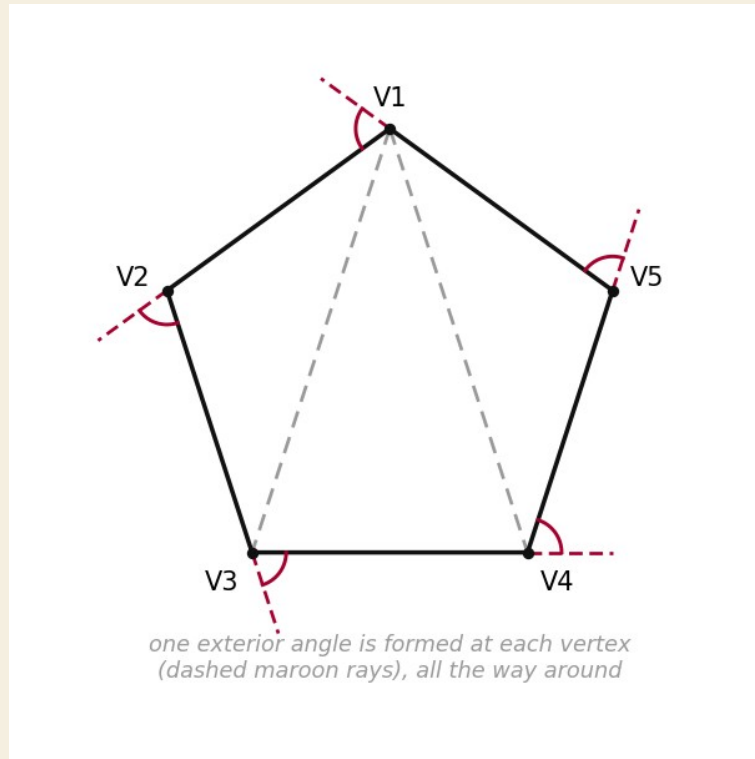
Angles, Parallel Lines & Triangle Fundamentals — Question 14 (Hard)

Line p and line q are parallel. Point D lies in the region between them (not on either line). Point E lies on line p , and segment ED is drawn from E to D , meeting line p at an angle of $(3x)^\circ$, measured on the side of the segment that faces into the region between p and q . Point F lies on line q , and segment DF is drawn from D to F , meeting line q at an angle of $(5x - 8)^\circ$, measured on the same side (facing into the region between p and q). If angle EDF — the angle at D between segments DE and DF — measures 112° , what is the value of x ?



Desmos: Type $3x+5x-8=112$ into Desmos to solve for x directly.

A pentagon (five-sided polygon) is divided into three triangles by drawing two diagonals from a single vertex. Using this triangulation, and the fact that one exterior angle is formed at each vertex of the pentagon by extending each side in turn, what is the sum of the five exterior angles of the pentagon (one at each vertex), taken in order all the way around?



26. Right Triangles & Trigonometry

Formula & Rules for This Skill

SOH-CAH-TOA: $\sin = \text{opposite/hypotenuse}$, $\cos = \text{adjacent/hypotenuse}$, $\tan = \text{opposite/adjacent}$. Pythagorean theorem: $a^2 + b^2 = c^2$.

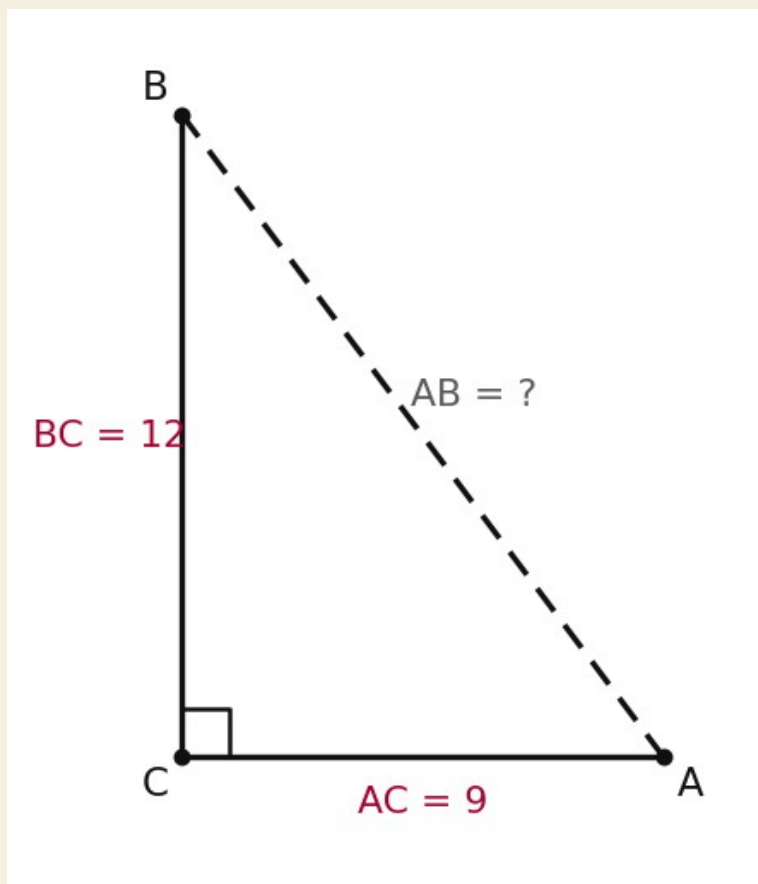
Fastest Way to Solve: Label which side is opposite, adjacent, and the hypotenuse relative to the angle you're using before choosing sin, cos, or tan.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Right Triangles & Trigonometry — Question 1 (Easy)

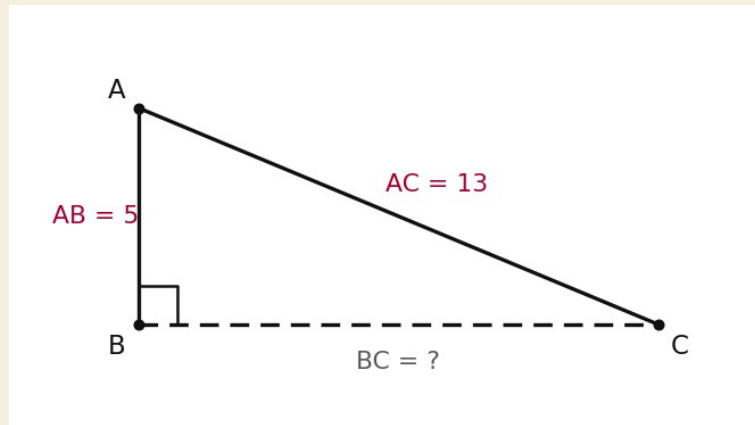
In right triangle ABC, the right angle is at C. Leg AC = 9 and leg BC = 12. Find the length of hypotenuse AB.



Desmos: Type $\text{sqrt}(9^2+12^2)$ directly into Desmos and read the evaluated result.

Right Triangles & Trigonometry — Question 2 (Easy)

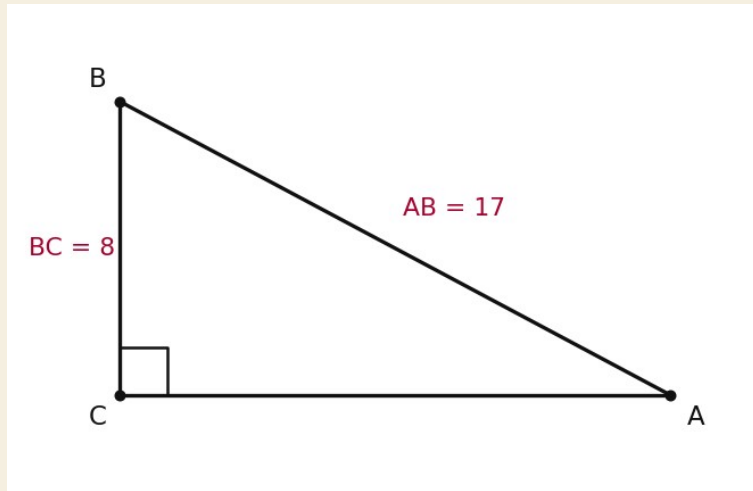
In right triangle ABC, the right angle is at B. Hypotenuse AC = 13 and leg AB = 5. Find the length of leg BC.



Desmos: Type $\sqrt{13^2 - 5^2}$ into Desmos and read the evaluated result.

Right Triangles & Trigonometry — Question 3 (Easy)

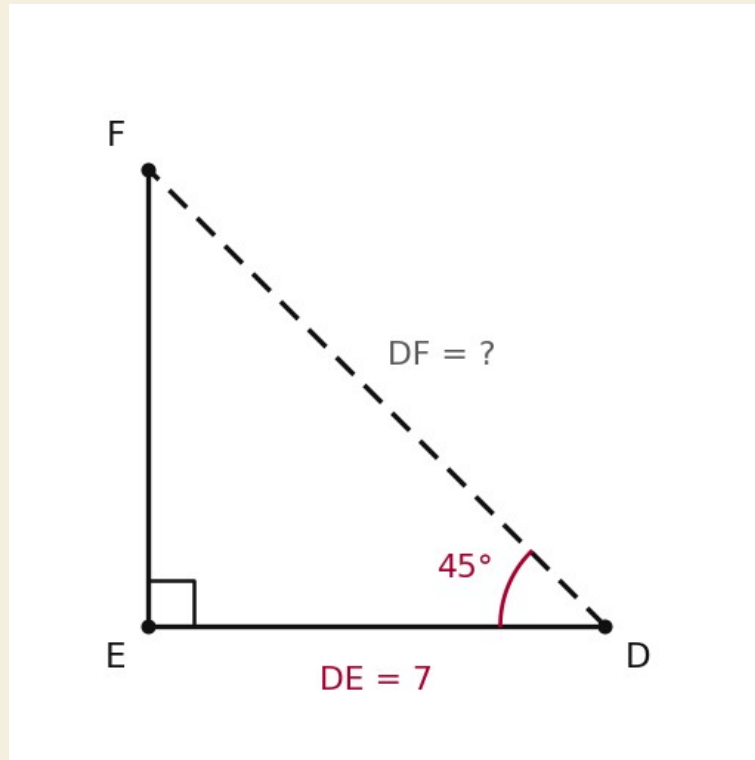
In right triangle ABC, the right angle is at C. Side BC (opposite angle A) = 8, and hypotenuse AB = 17. Find $\sin(A)$.



Desmos: Type $\text{asin}(8/17)$ into Desmos to see the angle A in degrees, confirming the ratio is a sensible acute angle.

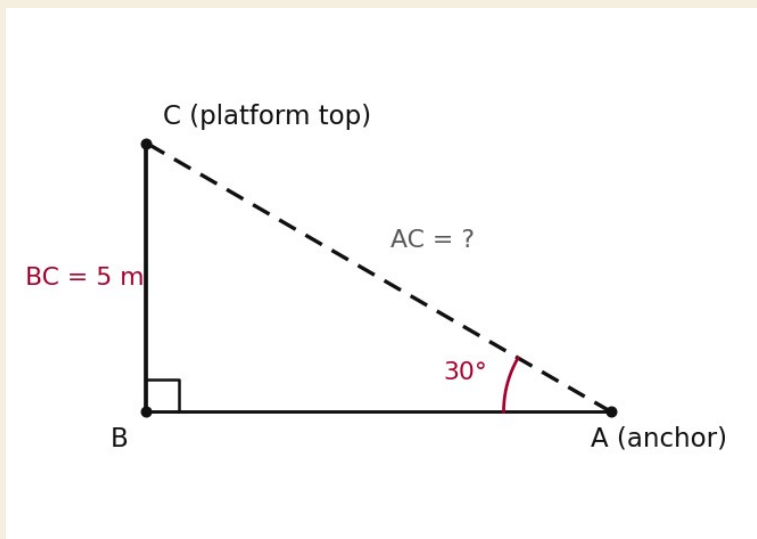
Right Triangles & Trigonometry — Question 4 (Easy)

In right triangle DEF, the right angle is at E. Angle D = 45 degrees, and leg DE = 7. Find the length of hypotenuse DF.



Desmos: Type $\sqrt{7^2+7^2}$ into Desmos and confirm it matches $7\sqrt{2}$.

A zipline support tower near Corfu forms a right triangle with the ground. The right angle is at point B, the base of the tower, where the vertical support meets the ground. Angle A, at the anchor point on the ground where the support cable meets the ground, measures 30 degrees. The vertical height of the tower, BC, is 5 meters. Find the length of the support cable, AC (the hypotenuse), in meters.



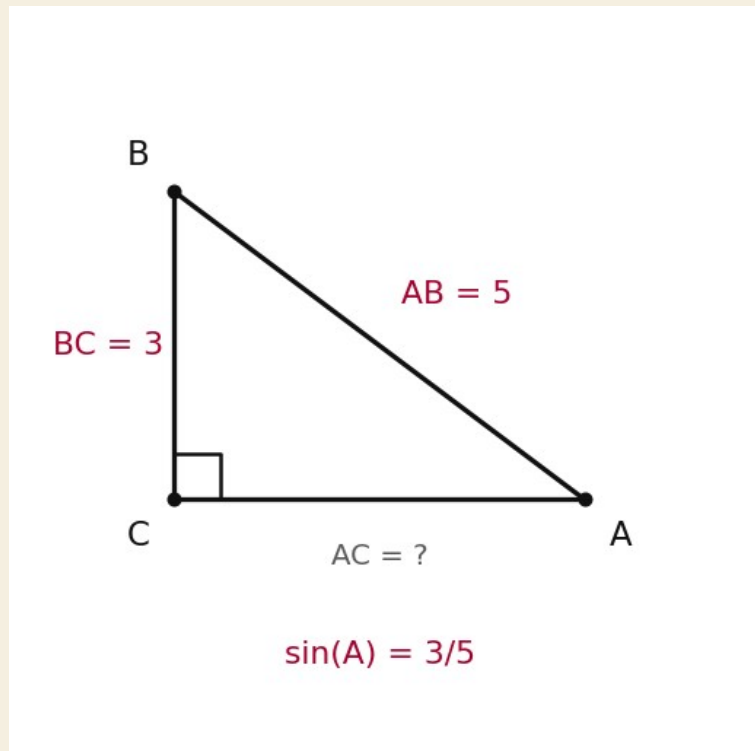
Desmos: Type $5/\sin(30)$ into Desmos (degree mode) to directly compute the hypotenuse using the sine ratio as a check.

Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

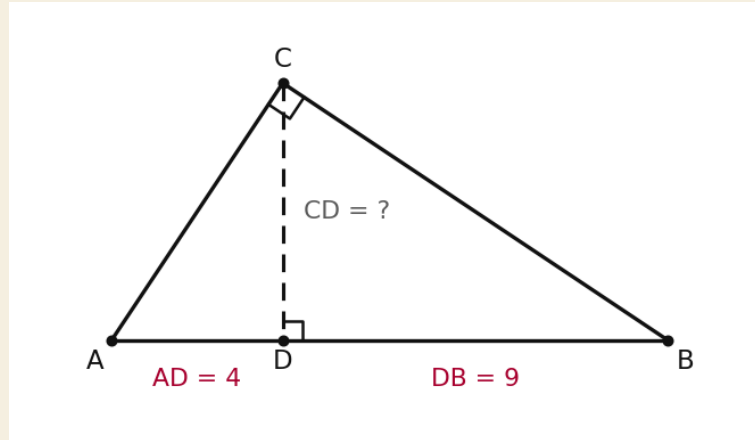
Right Triangles & Trigonometry — Question 6 (Medium)

In right triangle ABC, the right angle is at C. Angle A is such that $\sin(A) = 3/5$. Find $\cos(A)$.



Desmos: Type $\text{asin}(3/5)$ into Desmos to find angle A in degrees, then type $\cos$ of that angle value to confirm $\cos(A) = 0.8$.

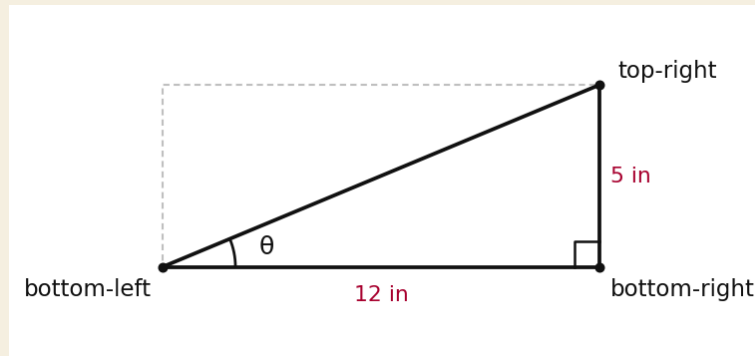
In right triangle ABC, the right angle is at C. An altitude is drawn from C, meeting hypotenuse AB at point D, so that CD is perpendicular to AB. This altitude splits AB into two segments: $AD = 4$ and $DB = 9$. Find the length of the altitude CD.



Desmos: Type $\sqrt{4 \cdot 9}$ into Desmos and read the evaluated result.

Right Triangles & Trigonometry — Question 8 (Medium)

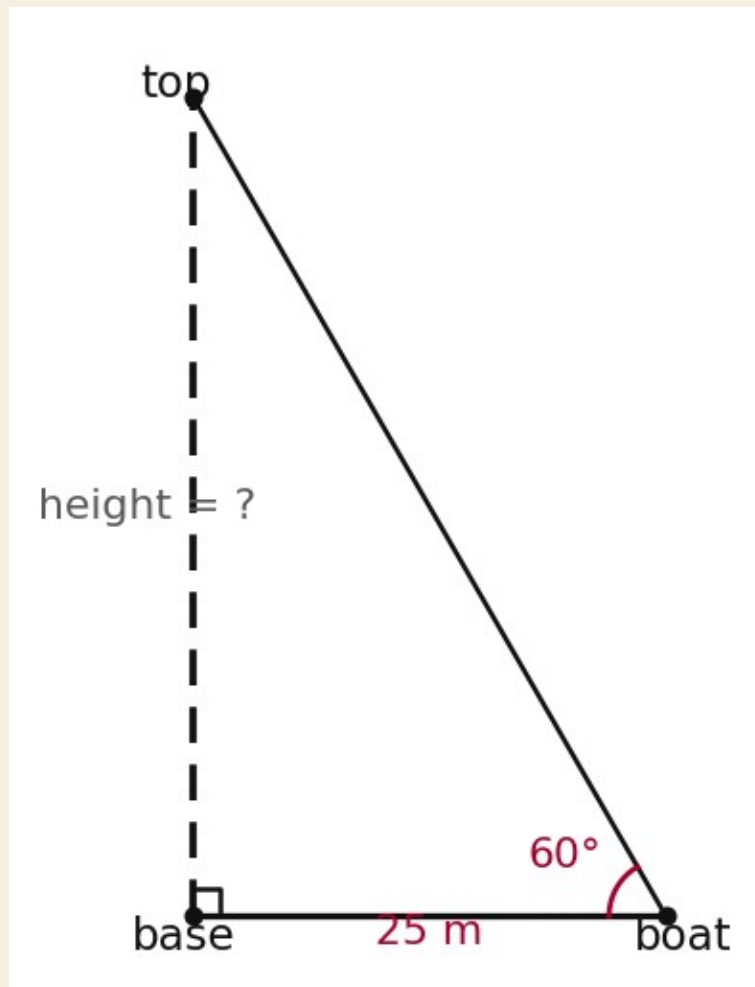
A rectangular garden plaque measures 12 inches wide and 5 inches tall. A diagonal is drawn from the bottom-left corner to the top-right corner, splitting the rectangle into two right triangles. Find the tangent of the angle the diagonal makes with the bottom (12-inch) side of the plaque.



Desmos: Type $\tan(5/12)$ into Desmos to see the actual angle in degrees as a sanity check that it's a reasonable acute angle.

Right Triangles & Trigonometry — Question 9 (Medium)

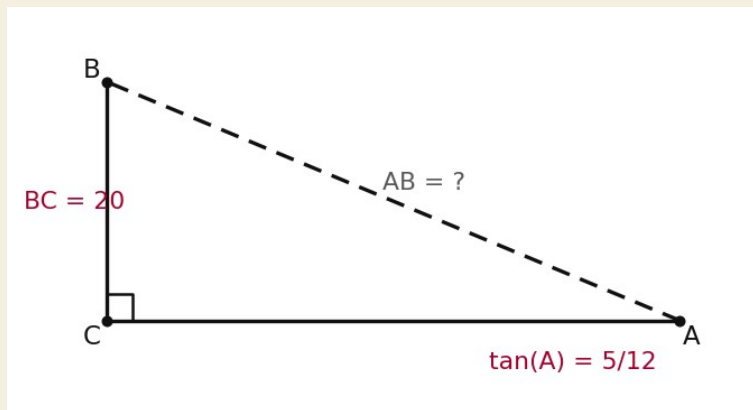
A lighthouse in Malta stands with its base at sea level. From a boat, the angle of elevation to the top of the lighthouse is 60 degrees, and the boat is 25 meters from the base of the lighthouse along the water. The lighthouse's beam rotates once every 15 seconds. Find the height of the lighthouse, to the nearest meter.



Desmos: Type $25 \cdot \tan(60)$ into Desmos (with the calculator in degree mode) and read the evaluated height directly.

Right Triangles & Trigonometry — Question 10 (Medium)

In right triangle ABC, the right angle is at C. $\tan(A) = 5/12$, and leg BC (opposite angle A) = 20. Find the length of hypotenuse AB.



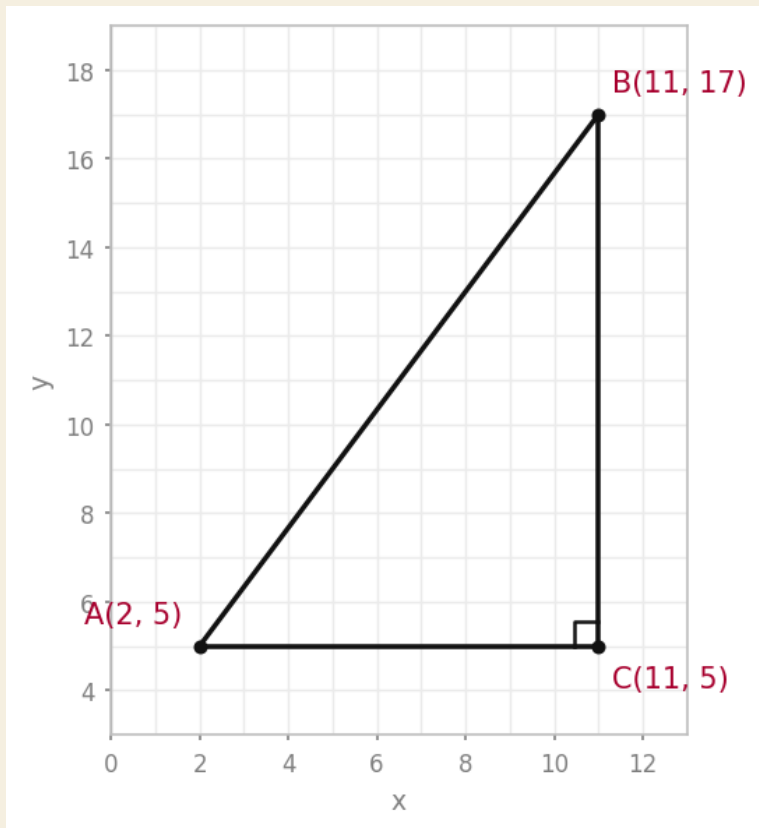
Desmos: Type $20/5 * 13$ into Desmos to compute the scaled hypotenuse in one step.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

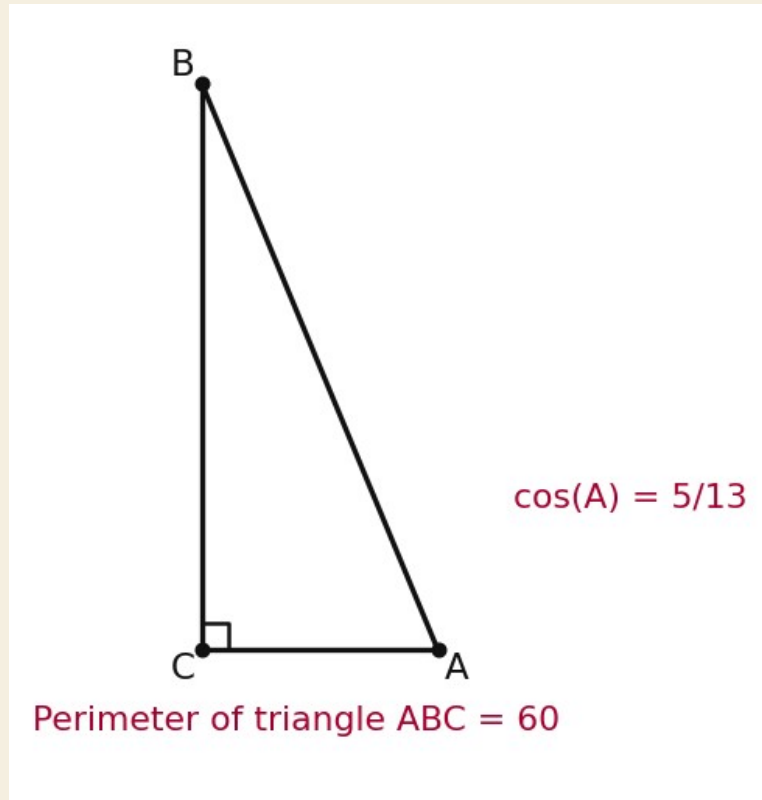
Right Triangles & Trigonometry — Question 11 (Hard)

Points A(2, 5) and B(11, 17) are graphed on a coordinate plane. A third point, C(11, 5), is also graphed, and segments AC, CB, and AB are drawn, forming right triangle ABC with the right angle at C (since AC is horizontal and CB is vertical). Find $\tan(\text{angle A})$, the tangent of the angle at vertex A.



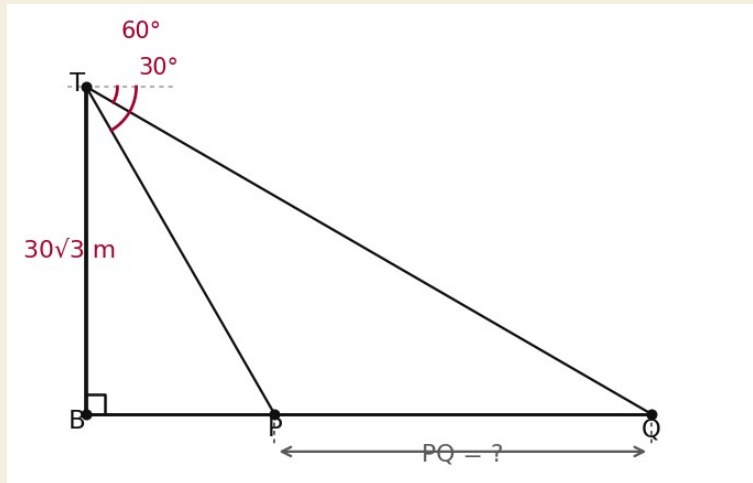
Desmos: Type $(17-5)/(11-2)$ into Desmos and simplify the result to confirm it matches $4/3$.

In right triangle ABC, the right angle is at C. $\cos(A) = 5/13$, and the perimeter of the triangle is 60. Find the area of the triangle.



Desmos: Type $(1/2) * (5 * 60/30) * (12 * 60/30)$ into Desmos to compute the area in one line as a check.

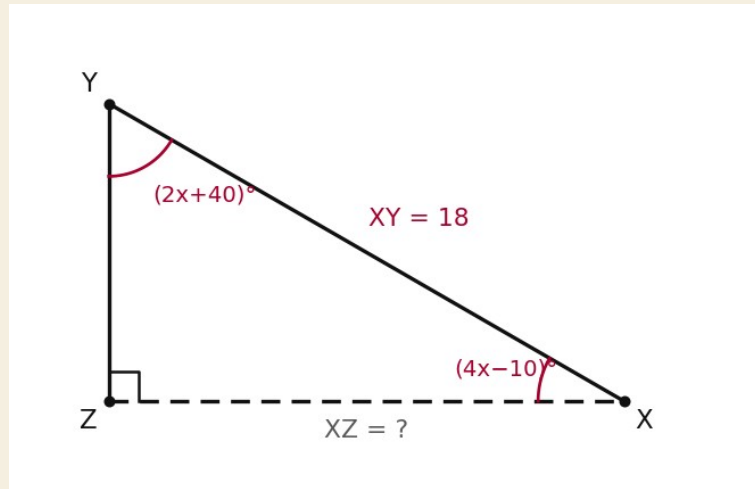
An observer stands atop a cliff near Reykjavik at point T, directly above point B at the base of the cliff, so that TB is vertical and $TB = 30\sqrt{3}$ meters. Two boats, P and Q, are on the water in a straight line with B, on the same side of the cliff, with P closer to the cliff than Q. The angle of depression from the observer T down to boat P is 60 degrees, and the angle of depression from T down to boat Q is 30 degrees. Find the distance PQ between the two boats, in meters.



Desmos: Type $30\sqrt{3}/\tan(30) - 30\sqrt{3}/\tan(60)$ into Desmos (in degree mode) and read the result directly.

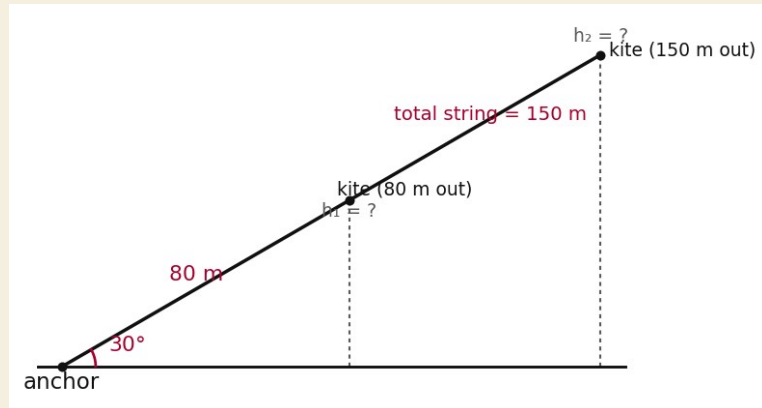
Right Triangles & Trigonometry — Question 14 (Hard)

In right triangle XYZ, the right angle is at Z. Angle X measures $(4x - 10)$ degrees, and angle Y measures $(2x + 40)$ degrees. The hypotenuse $XY = 18$. Find the length of side XZ, which is opposite angle Y.



Desmos: Type $18 \cdot \sin(60)$ into Desmos (degree mode) and confirm the decimal result matches $9 \cdot \sqrt{3}$, approximately 15.59.

A child is flying a kite near Zurich. The kite string is anchored at ground level, and the string is taut and straight, making a constant angle of 30 degrees with the ground throughout the problem. First, with 80 meters of string let out, the kite reaches a certain height above the ground. Then more string is let out, for a total of 150 meters, and the kite rises to a new height (still at the same 30-degree angle). How much higher is the kite after the additional string is let out, in meters?



Desmos: Type $\sin(30) \cdot (150 - 80)$ into Desmos (degree mode) and read the result directly.

27. Radians & the Unit Circle

Formula & Rules for This Skill

Radians = degrees $\times$ ($\pi/180$). On the unit circle, a point at angle θ is $(\cos \theta, \sin \theta)$, radius 1.

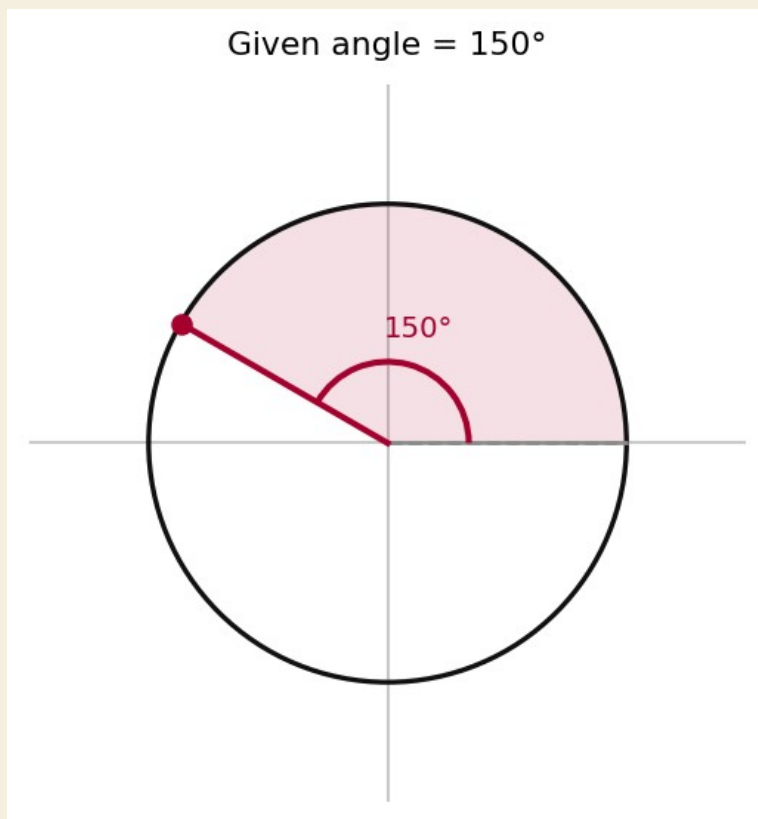
Fastest Way to Solve: Convert between degrees and radians immediately if the problem mixes both, before doing any further trig.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Radians & the Unit Circle — Question 1 (Easy)

Convert an angle of 150 degrees to radians.

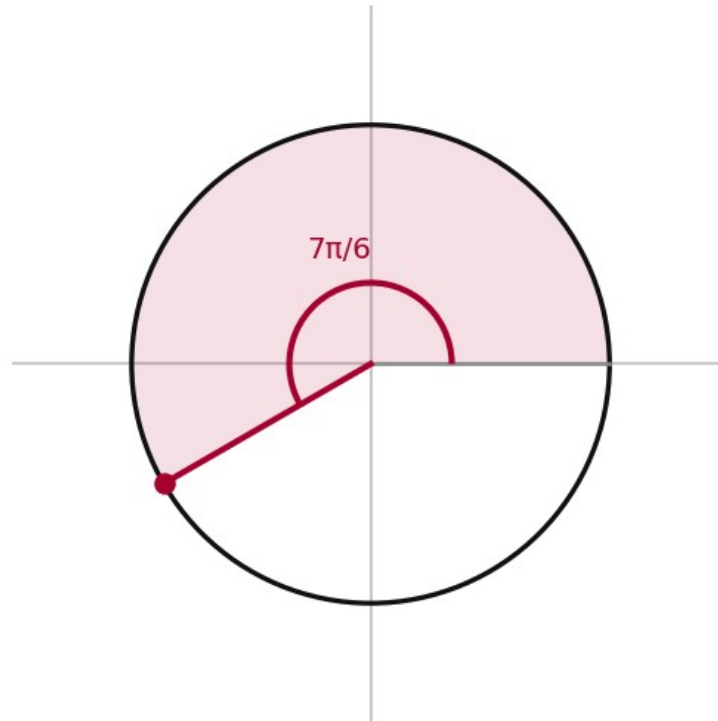


Desmos: Type $y = x(\pi/180)$, then trace to $x = 150$ to read the radian value $y = 5\pi/6$ directly.

Radians & the Unit Circle — Question 2 (Easy)

Convert an angle of $7\pi/6$ radians to degrees.

Given angle = $7\pi/6$ radians

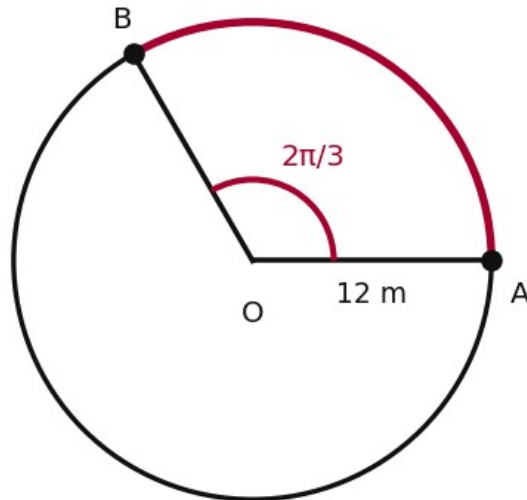


Desmos: Type $y = x(180/\pi)$, then trace to $x = 7\pi/6$ (entered as $7 * \pi/6$) to read $y = 210$.

Radians & the Unit Circle — Question 3 (Easy)

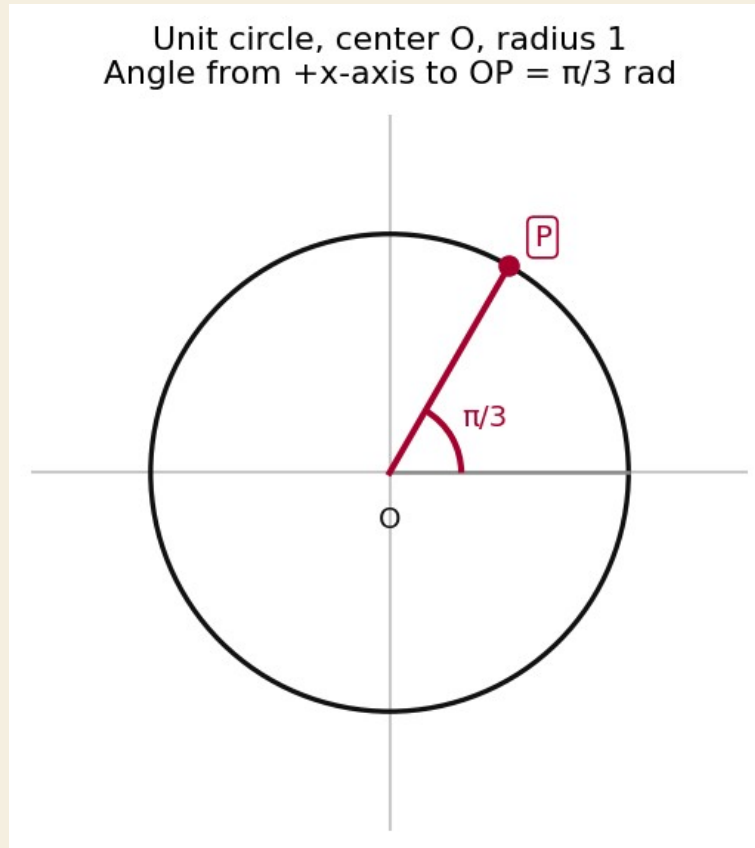
A circular fountain in Nafplio, Greece has center O and radius 12 meters. Points A and B lie on the edge of the fountain so that central angle AOB measures $2\pi/3$ radians. What is the length of arc AB , in meters, in terms of π ?

Fountain: center O , radius 12 m
Central angle $AOB = 2\pi/3$ rad



Desmos: Type $s = 12 * (2\pi/3)$ directly into the expression list to confirm $s = 8\pi$ (approximately 25.13).

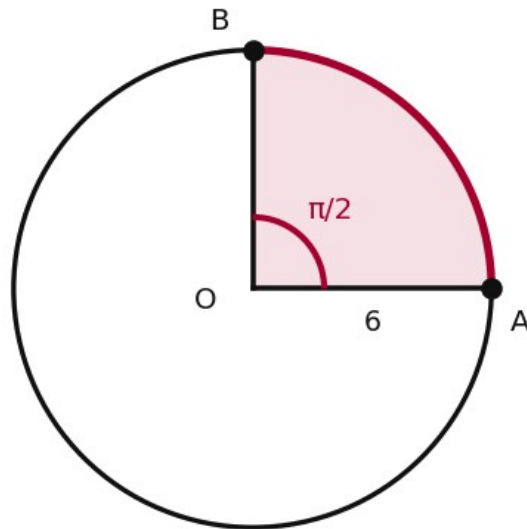
A unit circle is centered at the origin O with radius 1. Point P lies on the circle so that the angle measured counterclockwise from the positive x -axis to segment OP is $\pi/3$ radians. What are the coordinates of point P ?



Desmos: Type $(\cos(\pi/3), \sin(\pi/3))$ to plot P directly and read its coordinates from the graph.

Circle O has radius 6 and a sector formed by central angle AOB measuring $\pi/2$ radians. What is the area of sector AOB, in terms of π ?

Circle O, radius 6
Sector AOB, central angle = $\pi/2$ rad



Desmos: Type $A = 0.5 \cdot 6^2 \cdot \theta$ with $\theta = \pi/2$ directly in the expression list to confirm $A = 9\pi$ (approximately 28.27).

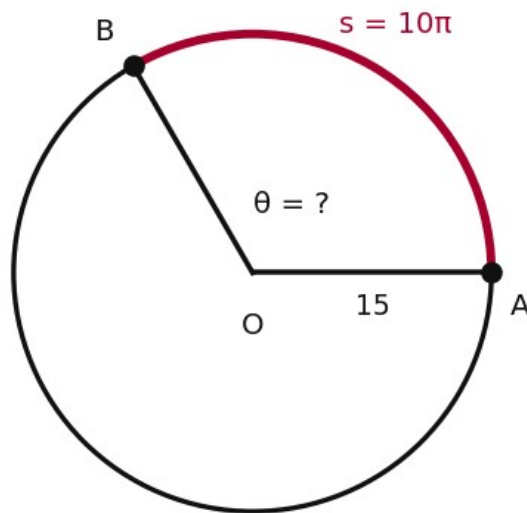
Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Radians & the Unit Circle — Question 6 (Medium)

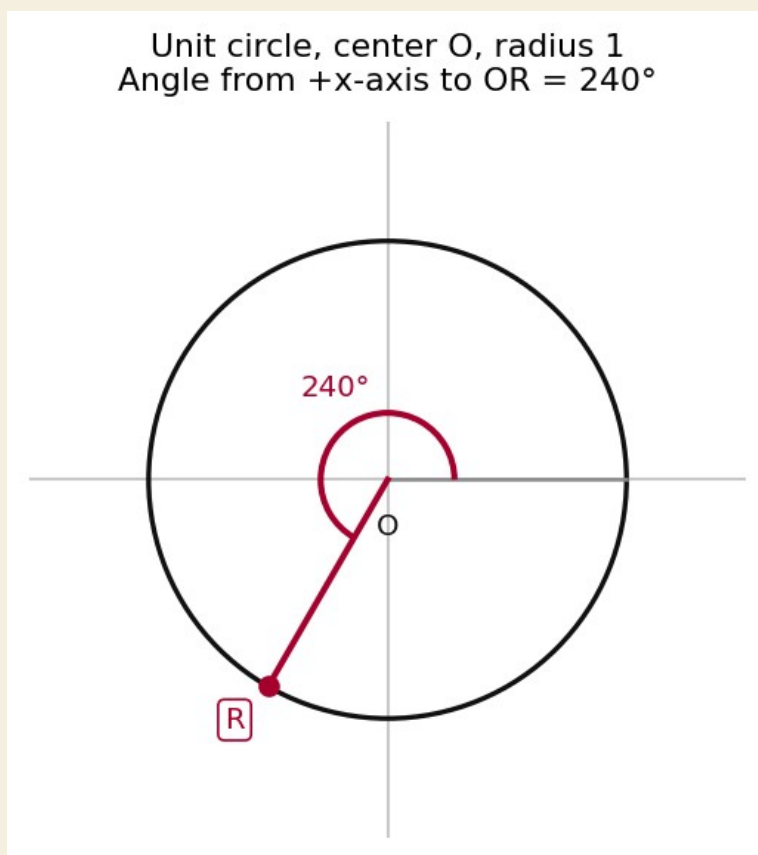
Circle O has radius 15. Points A and B lie on the circle so that arc AB has length 10π . What is the measure of central angle AOB, in radians?

Circle O, radius 15
Arc AB has length 10π



Desmos: Type $\theta = (10\pi)/15$ directly in the expression list to confirm $\theta = 2\pi/3$.

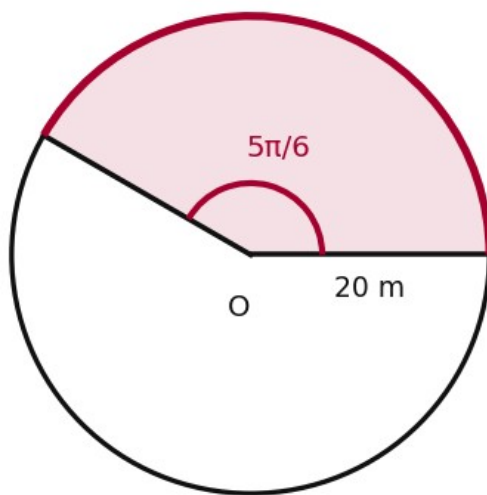
A unit circle is centered at the origin O with radius 1. Point R lies on the circle so that the angle measured counterclockwise from the positive x -axis to segment OR is 240° . What are the coordinates of point R ?



Desmos: Type $(\cos(240\pi/180), \sin(240\pi/180))$ to plot R directly and confirm both coordinates are negative.

A circular garden plaza in Ljubljana has center O and radius 20 meters. The plaza is paved with light gray stone and costs 40 euros per square meter to maintain each year. A decorative sector of the plaza is marked off by a central angle of $5\pi/6$ radians. What is the area of the decorative sector, in square meters and in terms of π ?

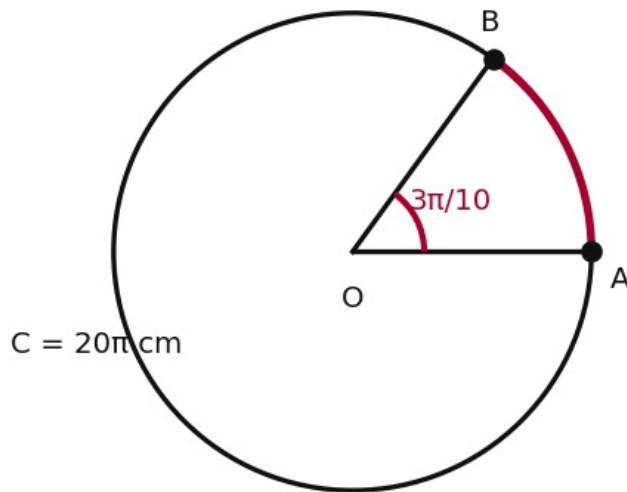
Garden plaza, center O , radius 20 m
Decorative sector, central angle = $5\pi/6$ rad



Desmos: Type $A = 0.5 * 20^2 * \theta$ with $\theta = 5\pi/6$ directly in the expression list to get $A = 500\pi/3$ (approximately 523.6), ignoring the cost entirely.

A circle has circumference 20π centimeters. Points A and B lie on the circle so that central angle AOB measures $3\pi/10$ radians. What is the length of arc AB, in centimeters and in terms of π ?

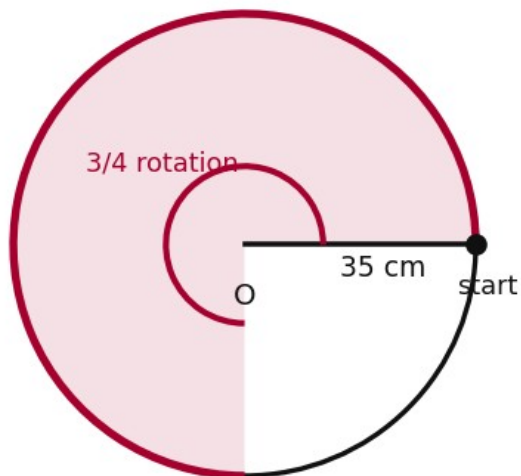
Circle O, circumference = 20π cm
Central angle AOB = $3\pi/10$ rad



Desmos: Type $r = 20\pi/(2\pi)$ to confirm $r = 10$, then type $s = r \cdot (3\pi/10)$ to confirm $s = 3\pi$.

A bicycle wheel in Rotterdam has radius 35 centimeters. As the bike moves forward, the wheel completes exactly $\frac{3}{4}$ of one full rotation. What distance, in centimeters and in terms of π , does a point on the rim of the wheel travel during this $\frac{3}{4}$ rotation?

Bicycle wheel, radius 35 cm
Wheel sweeps $\frac{3}{4}$ of one full rotation



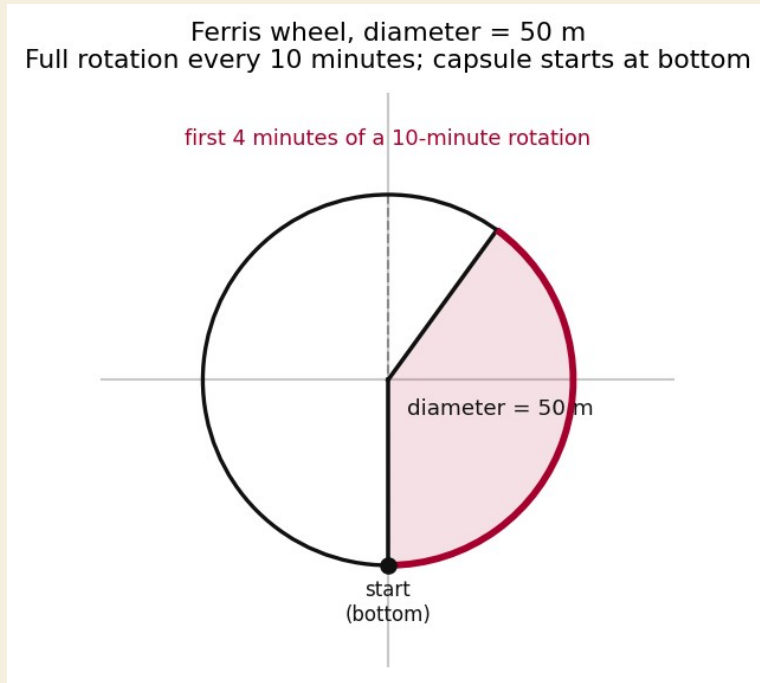
Desmos: Type $s = 35 \cdot (0.75 \cdot 2\pi)$ directly in the expression list to confirm $s = 105\pi/2$ (approximately 164.9 cm).

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Radians & the Unit Circle — Question 11 (Hard)

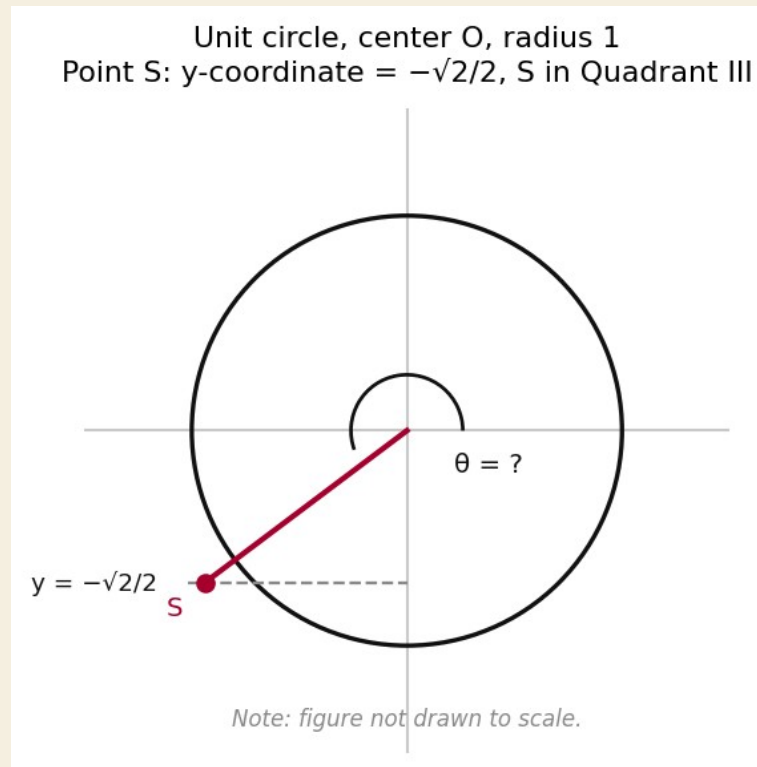
A Ferris wheel at a fair in Vienna has a diameter of 50 meters and rotates at a constant rate, completing one full rotation every 10 minutes. A capsule starts at the bottom of the wheel at the moment it begins rotating. What arc length, in meters and in terms of π , does the capsule travel along the rim of the wheel during the first 4 minutes of the ride?



Desmos: Type $s = 25 * ((2\pi/10) * 4)$ in the expression list to confirm $s = 20\pi$ (approximately 62.8 meters).

Radians & the Unit Circle — Question 12 (Hard)

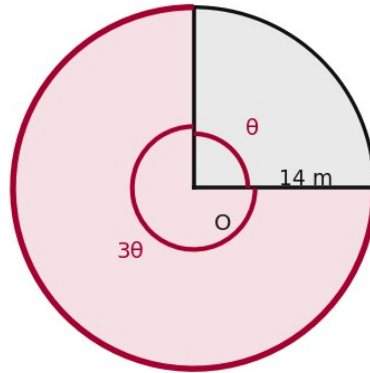
Point S lies on a unit circle centered at the origin O, with radius 1. The y-coordinate of S is $-\sqrt{2}/2$, and S is located in Quadrant III (both coordinates negative). What is the angle theta, with $0 \leq \theta < 2\pi$, measured counterclockwise from the positive x-axis to segment OS, in radians?



Desmos: Type $(\cos(5\pi/4), \sin(5\pi/4))$ to plot the point and confirm both coordinates are negative with y approximately -0.707.

A circular garden bed in Valletta has center O and radius 14 meters. Two straight paths from the center divide the garden into exactly two sectors: one with central angle θ and the other with central angle $3 \times \theta$, together making up the entire circle. What is the area, in square meters and in terms of π , of the larger of the two sectors?

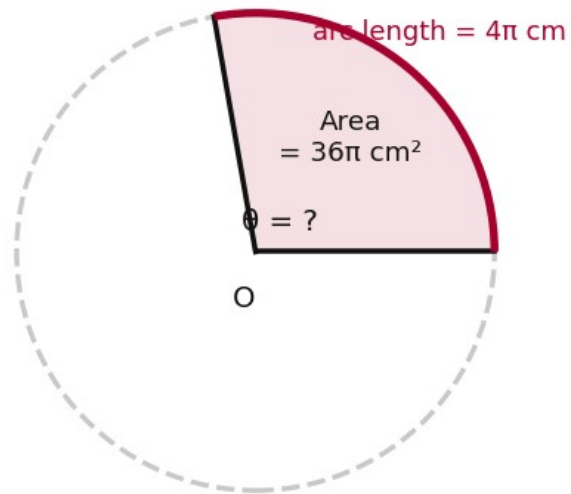
Garden bed, center O, radius 14 m
Two sectors: central angles θ and 3θ (together = full circle)



Desmos: Type $\theta = 2\pi/4$ to confirm $\theta = \pi/2$, then type $A = 0.5 \cdot 14^2 \cdot (3 \cdot \theta)$ to confirm $A = 147\pi$ (approximately 461.8).

A circular sector has area 36π square centimeters and arc length 4π centimeters. What is the measure of its central angle, in radians?

Circular sector
Area = $36\pi \text{ cm}^2$, arc length = $4\pi \text{ cm}$

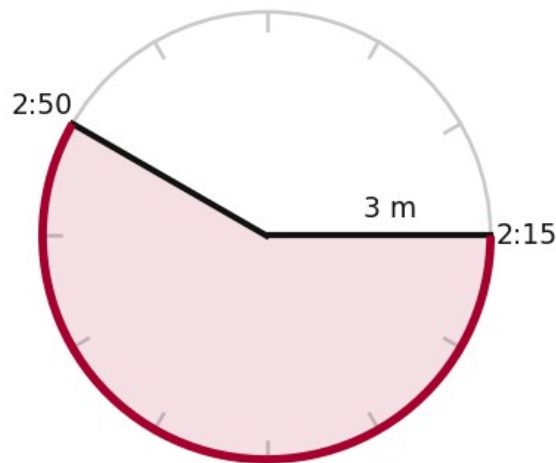


Note: figure not drawn to scale.

Desmos: Type $r = 2 \cdot 36\pi / (4\pi)$ to confirm $r = 18$, then type $\theta = 4\pi / r$ to confirm $\theta = 2\pi / 9$.

The clock tower in Bern has a minute hand that is 3 meters long. Between 2:15 and 2:50 on the clock face, what is the exact distance, in meters and in terms of pi, traveled by the tip of the minute hand?

Clock tower minute hand, length 3 m
Sweeps from 2:15 to 2:50



Desmos: Type $s = 3 * ((35/60) * 2\pi)$ directly in the expression list to confirm $s = 7\pi/2$ (approximately 10.996 meters).

28. Circles — Circumference, Arcs, Sectors, Equations & Theorems

Formula & Rules for This Skill

Circumference = $2 \pi r$. **Area** = πr^2 . **Standard equation:** $(x-h)^2 + (y-k)^2 = r^2$, center (h,k) , radius r . **Arc length** = $(\text{angle}/360) \times 2 \pi r$.

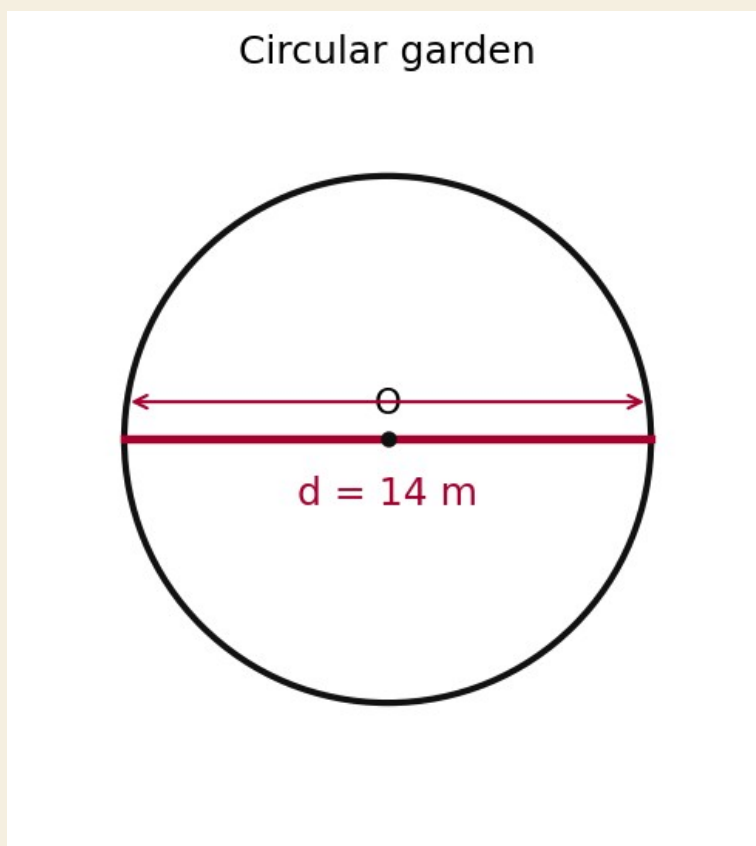
Fastest Way to Solve: Identify the center and radius first — from either the graph or by completing the square on the equation — before answering anything else about the circle.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Question 1 (Easy)

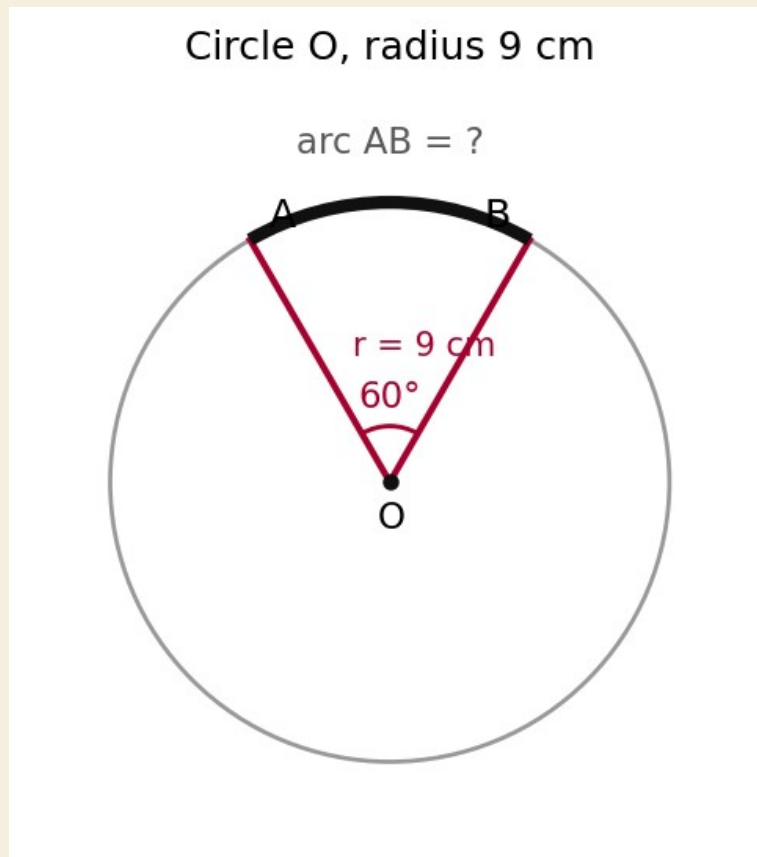
A circular garden in Vienna has a diameter of 14 meters. What is the circumference of the garden, in meters, in terms of π ?



Desmos: Type $14 \cdot \pi$ into the calculator to get the decimal value (about 43.98) and confirm it matches 14π .

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Question 2 (Easy)

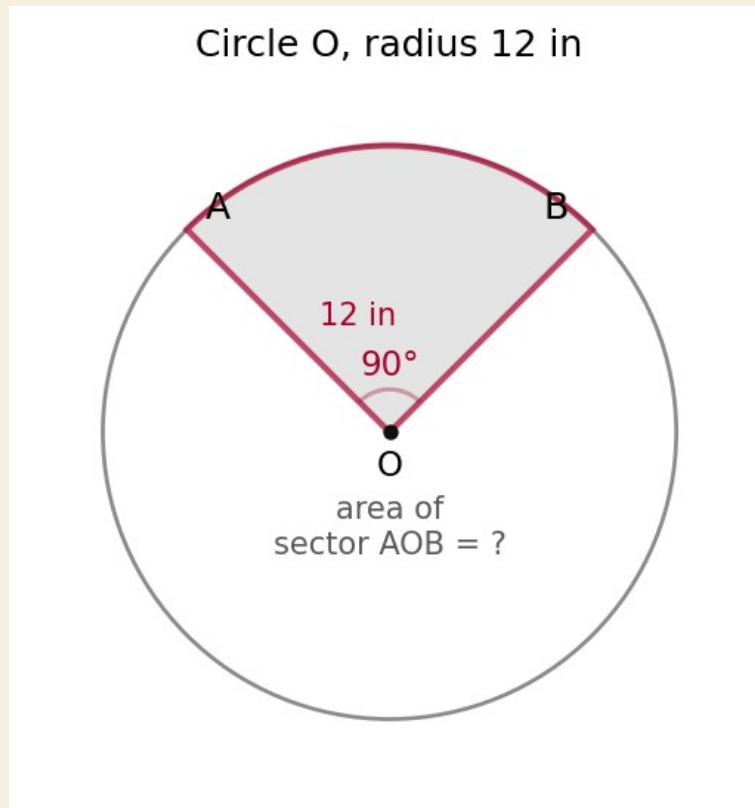
A circle has center O and radius 9 cm. Points A and B lie on the circle so that central angle AOB measures 60° . What is the length of arc AB , in centimeters, in terms of π ?



Desmos: Type $(60/360) \cdot 2 \cdot \pi \cdot 9$ directly into the calculator to get the decimal value and check it equals $3\pi \approx 9.42$.

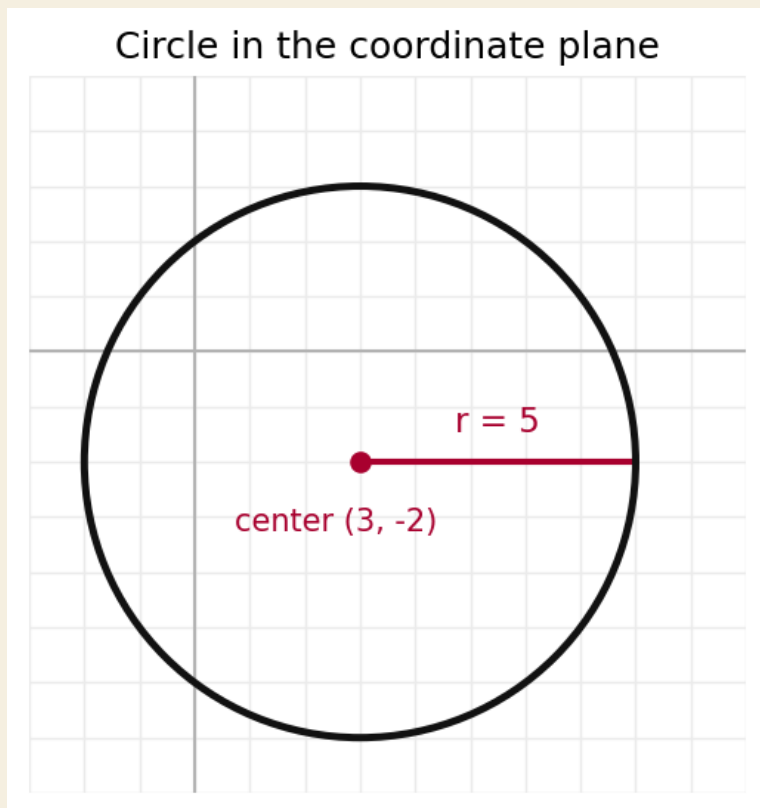
Circles — Circumference, Arcs, Sectors, Equations & Theorems — Question 3 (Easy)

A circle has center O and radius 12 inches. Central angle AOB measures 90° . What is the area of sector AOB, in square inches, in terms of π ?



Desmos: Type $(90/360) \cdot \pi \cdot 12^2$ into the calculator to confirm the decimal matches $36\pi \approx 113.1$.

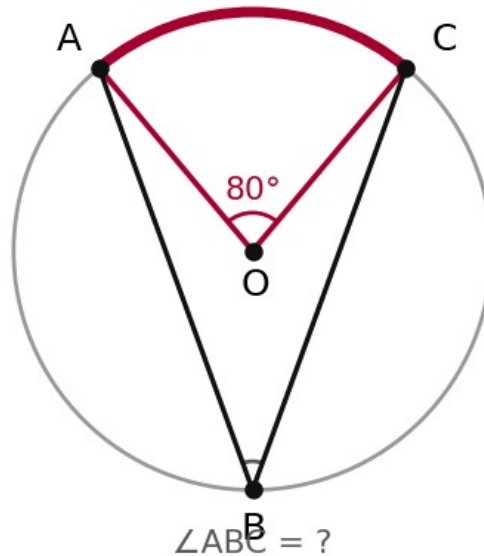
A circle has center $(3, -2)$ and radius 5. Which equation represents this circle in standard (center-radius) form?



Desmos: Type $(x-3)^2+(y+2)^2=25$ into the calculator and confirm the graph is a circle centered at $(3, -2)$ with radius 5.

Points A, B, and C lie on a circle with center O. Central angle AOC (the angle at the center between radii OA and OC) measures 80° and intercepts the minor arc AC. Point B lies on the circle on the major arc AC, so that inscribed angle ABC also intercepts that same minor arc AC. What is the measure of angle ABC?

Circle O with inscribed angle ABC

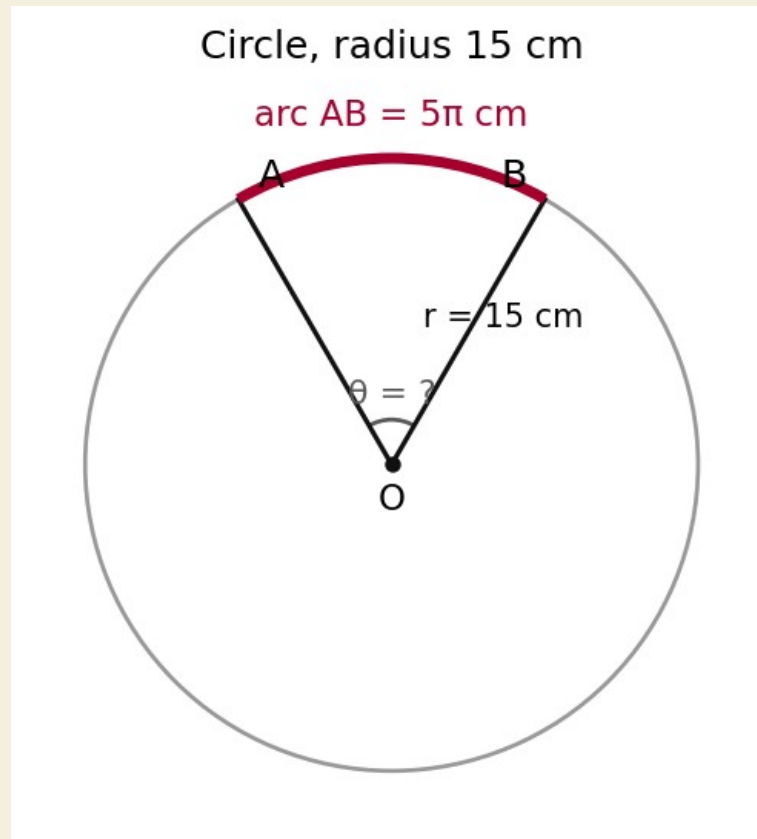


Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

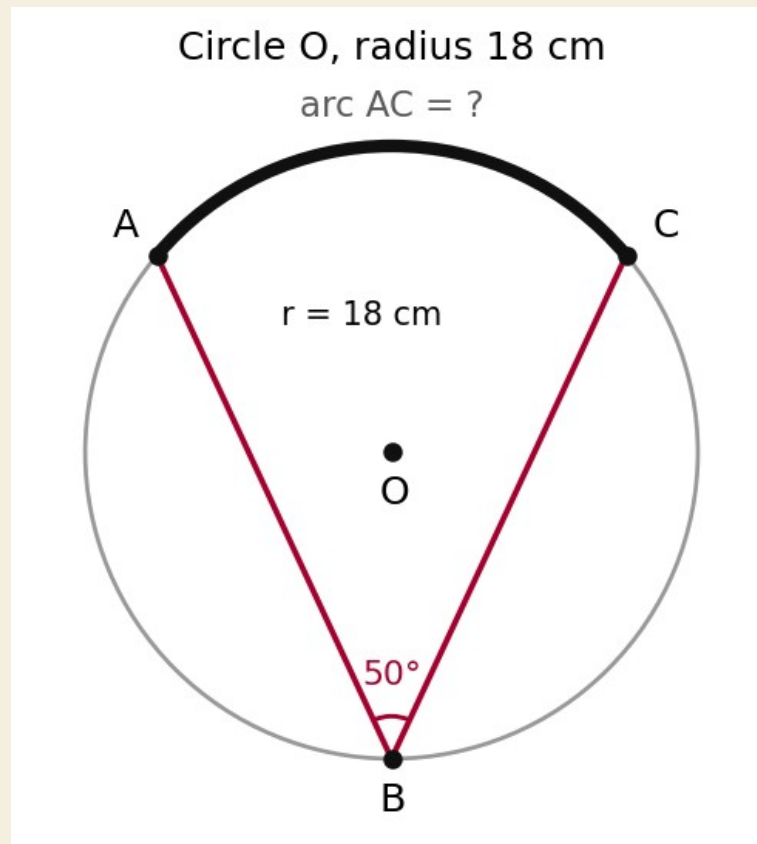
Circles — Circumference, Arcs, Sectors, Equations & Theorems — Question 6 (Medium)

A circle has radius 15 cm. An arc on this circle has length 5π cm. What is the measure of the central angle that intercepts this arc, in degrees?



Desmos: Type $360 \cdot (5 \cdot \pi) / (2 \cdot \pi \cdot 15)$ into the calculator to get 60 directly, confirming the central angle in degrees.

In a circle with center O and radius 18 cm, points A , B , and C lie on the circle. Inscribed angle ABC (with vertex B) measures 50° and intercepts arc AC that does not contain B . What is the length of arc AC , in centimeters, in terms of π ?

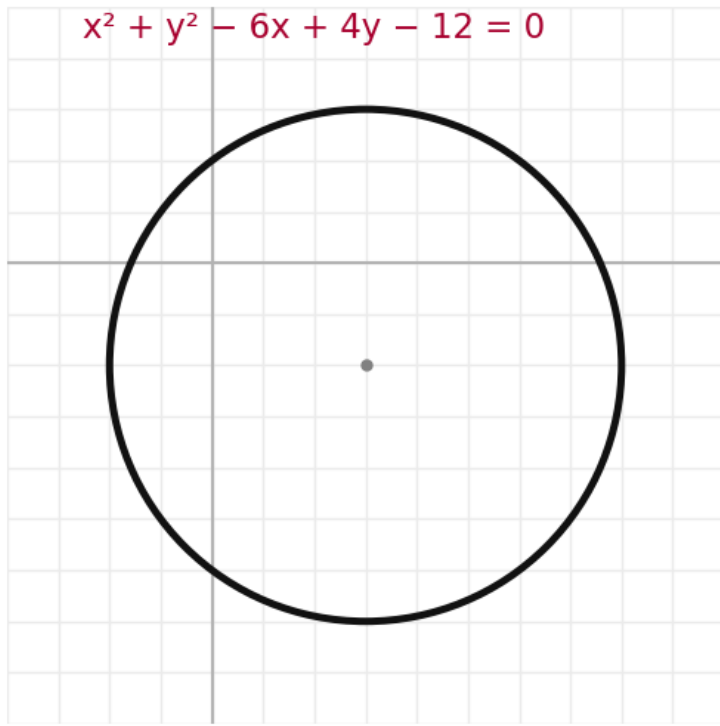


Desmos: Type $(2 \cdot 50/360) \cdot 2 \cdot \pi \cdot 18$ into the calculator to confirm the result equals 10π .

A circle is given by the equation $x^2 + y^2 - 6x + 4y - 12 = 0$. What is the radius of this circle?

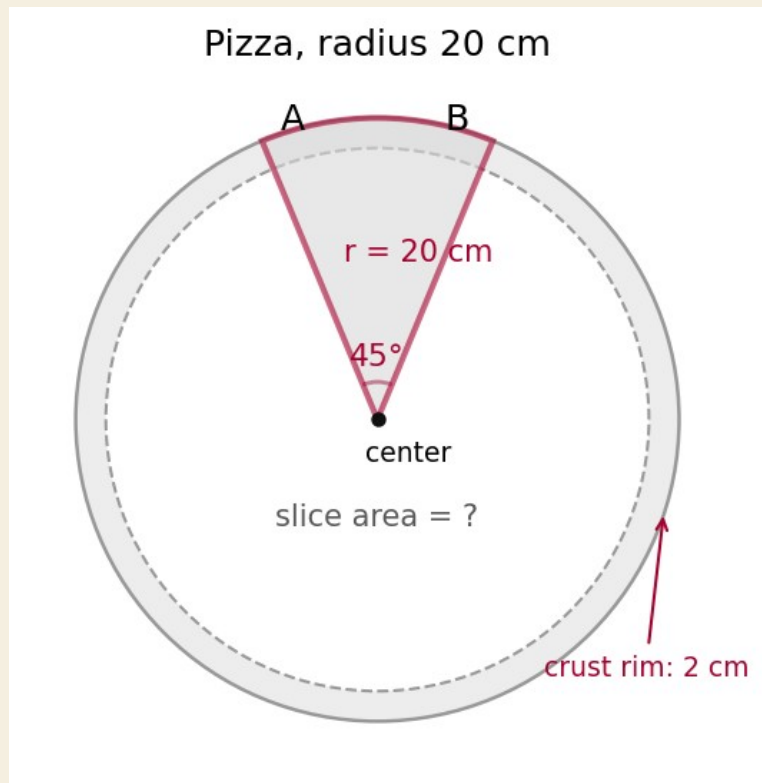
A circle in the coordinate plane

$$x^2 + y^2 - 6x + 4y - 12 = 0$$



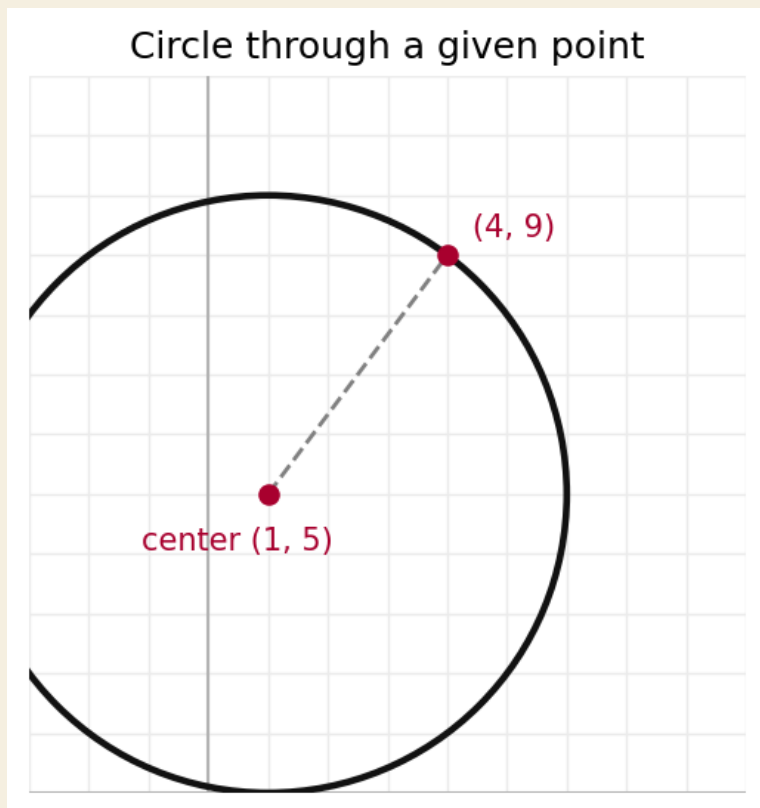
Desmos: Type $x^2 + y^2 - 6x + 4y - 12 = 0$ into the calculator, let it graph the circle, and use the graph's visible radius to confirm $r = 5$.

A circular pizza in Naples has a radius of 20 cm and a crust rim that is 2 cm thick around its outer edge. The pizza is cut into slices from the center, and one slice has a central angle of 45° . What is the area of that slice, in square centimeters, in terms of π ?



Desmos: Type $(45/360) \cdot \pi \cdot 20^2$ into the calculator to confirm the decimal matches $50\pi \approx 157.1$.

A circle is centered at $(1, 5)$ and passes through the point $(4, 9)$. What is the circumference of the circle, in terms of π ?



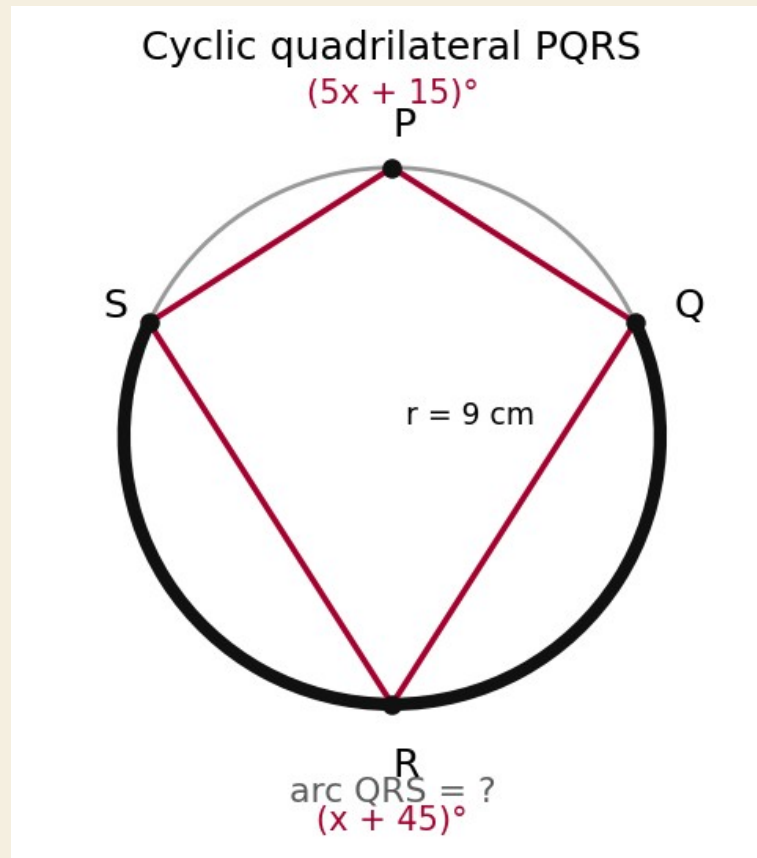
Desmos: Type $\text{sqrt}((4-1)^2+(9-5)^2)$ into the calculator to instantly get the radius, 5, then multiply by 2π for the circumference.

Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Circles — Circumference, Arcs, Sectors, Equations & Theorems — Question 11 (Hard)

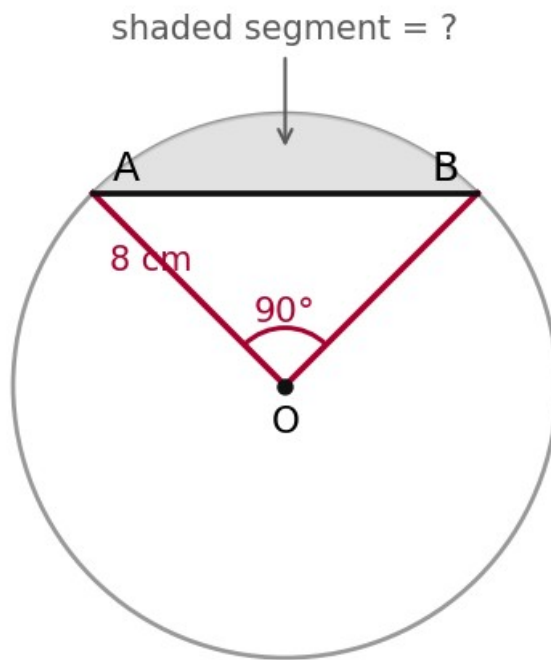
Quadrilateral PQRS is inscribed in a circle, with the vertices appearing in that order around the circle, and the circle has radius 9 cm. Angle P and angle R are opposite angles of the quadrilateral. Angle P measures $(5x + 15)^\circ$ and angle R measures $(x + 45)^\circ$. First find x , then use angle P as an inscribed angle to find the length of arc QRS — the arc from Q to S that passes through R, which is the arc intercepted by angle P — in terms of π .



Desmos: Type $(230/360) \cdot 2 \cdot \pi \cdot 9$ into the calculator to confirm the decimal value matches $11.5\pi \approx 36.13$.

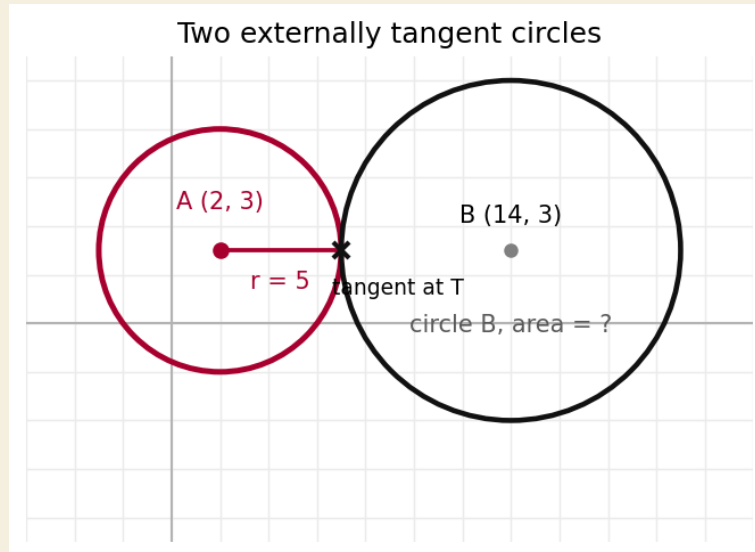
A circle has center O and radius 8 cm. Points A and B lie on the circle so that central angle AOB measures 90° . Chord AB is drawn, splitting the disk into the minor segment (the region between chord AB and the minor arc AB) and the rest of the circle. What is the area of the minor segment, in square centimeters, in terms of π ?

Circle O , radius 8 cm, chord AB



Desmos: Type $(90/360) \cdot \pi \cdot 8^2 - 0.5 \cdot 8 \cdot 8$ into the calculator to get the segment's area directly as a decimal, about 18.27.

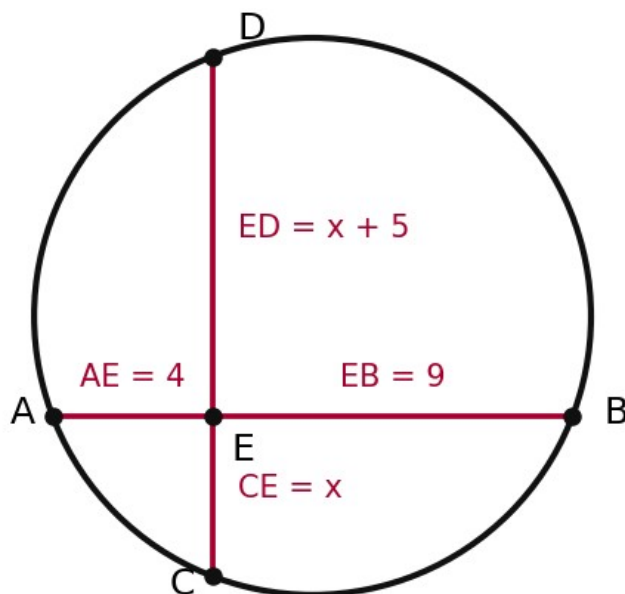
Circle A has center $(2, 3)$ and radius 5. Circle B has center $(14, 3)$ and is externally tangent to Circle A, meaning the two circles touch at exactly one point and neither circle's interior overlaps the other's. Circle B's equation is $(x - 14)^2 + (y - 3)^2 = r^2$ for some positive radius r . What is the area of Circle B, in terms of π ?



Desmos: Type $(x-2)^2+(y-3)^2=25$ and $(x-14)^2+(y-3)^2=49$ together into the calculator and confirm the two circles touch at exactly one point.

Inside a circle, two chords AB and CD intersect at point E. $AE = 4$, $EB = 9$, $CE = x$, and $ED = x + 5$, where all lengths are in centimeters. What is the length of chord CD?

Intersecting chords AB and CD



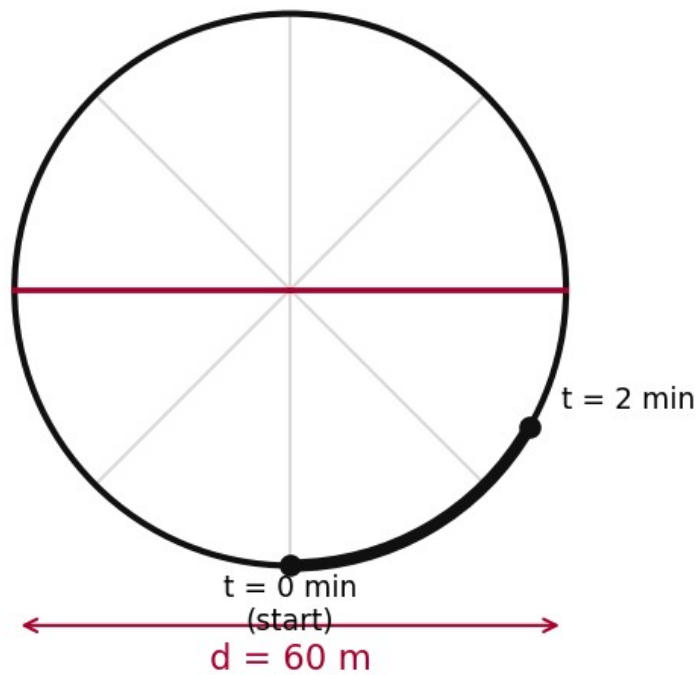
Desmos: Type $y = x^2 + 5x - 36$ into the calculator and trace to find its positive root at $x = 4$, confirming the algebra.

A Ferris wheel in Vienna has a diameter of 60 meters and completes one full rotation every 12 minutes at a constant speed. A capsule starts at the bottom of the wheel. What distance does the capsule travel along the circular path in the first 2 minutes, in meters, in terms of π ?

Ferris wheel, diameter 60 m

one full rotation = 12 min

distance traveled = ?



Desmos: Type $(2/12) * 2 * \pi * 30$ into the calculator to confirm the decimal value matches $10\pi \approx 31.42$.

29. Volume, Surface Area & Similarity of Solids

Formula & Rules for This Skill

Prism/cylinder volume = base area $\times$ height. Sphere volume = $(4/3) \pi r^3$. Cone volume = $(1/3) \pi r^2 h$. Similar solids: volume ratio = (linear scale factor)³.

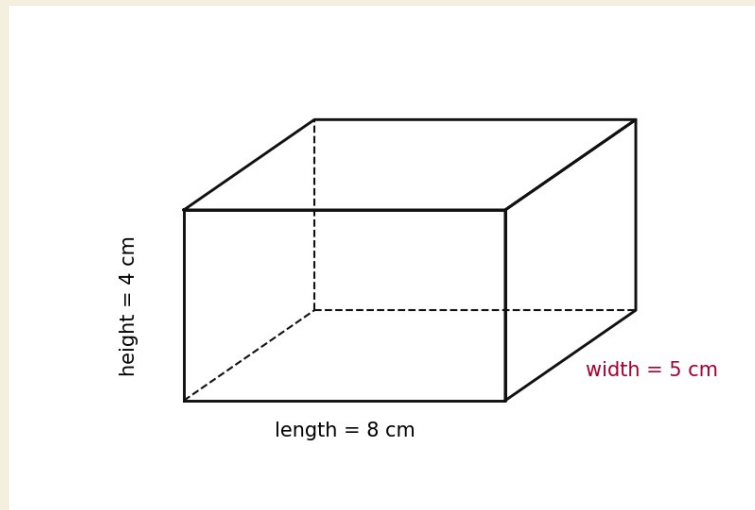
Fastest Way to Solve: Identify which solid you have and pull its exact formula before substituting — do not blend formulas from different solids.

Easy — Questions 1-5

Start here — one clean application of the rule per question, to build confidence.

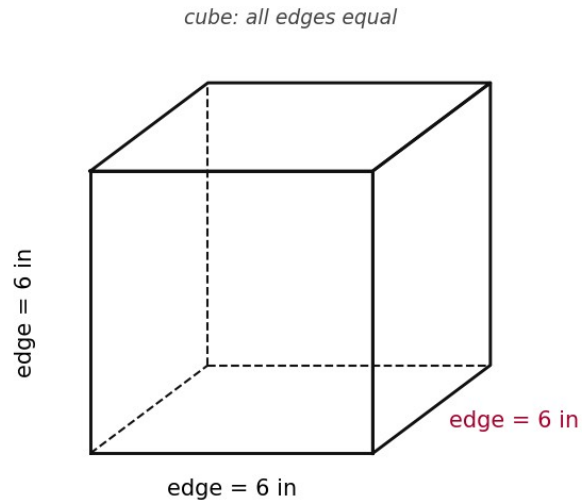
Volume, Surface Area & Similarity of Solids — Question 1 (Easy)

A rectangular shipping box has a length of 8 centimeters, a width of 5 centimeters, and a height of 4 centimeters. What is the volume of the box, in cubic centimeters?

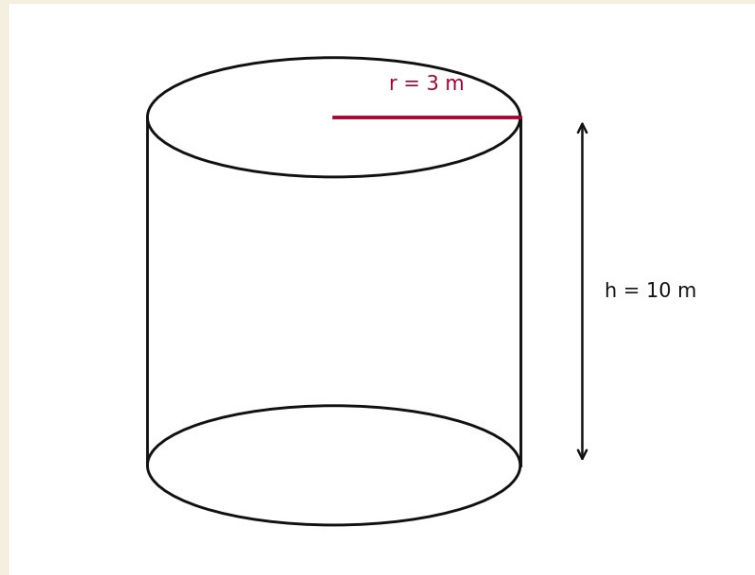


Volume, Surface Area & Similarity of Solids — Question 2 (Easy)

A cube-shaped gift box has an edge length of 6 inches. What is the total surface area of the box, in square inches?



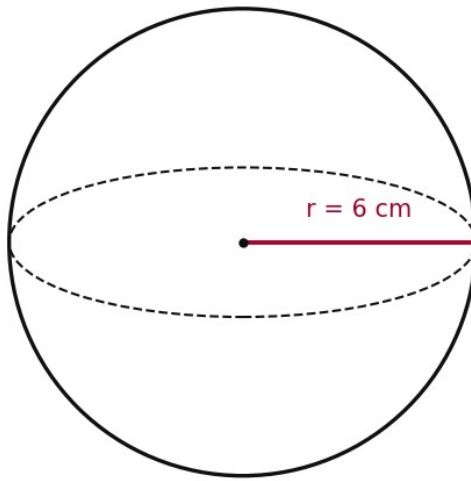
A cylindrical water tank has a radius of 3 meters and a height of 10 meters. What is the volume of the tank, in cubic meters, in terms of π ?



Desmos: Type $y = \pi \cdot 3^2 \cdot 10$ in the calculator to instantly see the numeric value of 90π .

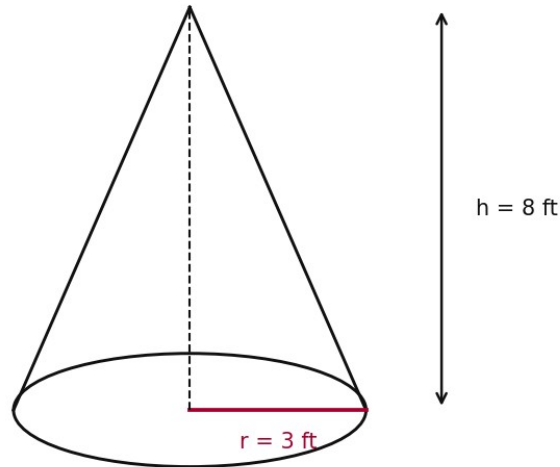
Volume, Surface Area & Similarity of Solids — Question 4 (Easy)

A solid sphere has a radius of 6 centimeters. What is the volume of the sphere, in cubic centimeters, in terms of π ?



Volume, Surface Area & Similarity of Solids — Question 5 (Easy)

A right circular cone has a base radius of 3 feet and a height of 8 feet. What is the volume of the cone, in cubic feet, in terms of π ?

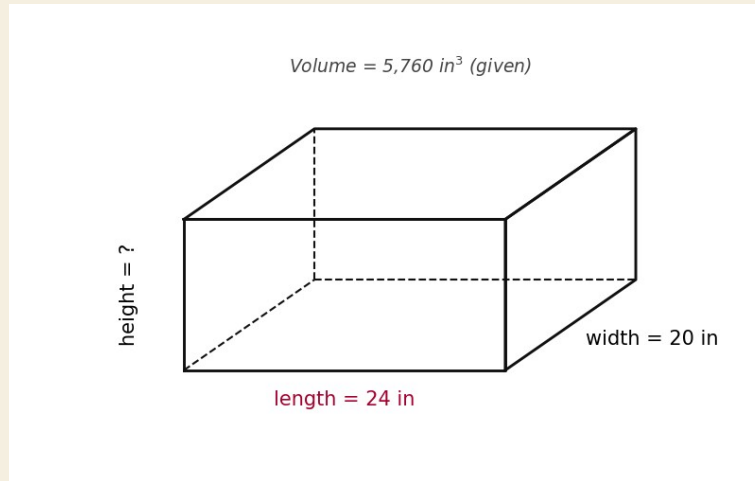


Medium — Questions 6-10

Each question adds one genuine twist. Slow down and re-read before you commit to a method.

Volume, Surface Area & Similarity of Solids — Question 6 (Medium)

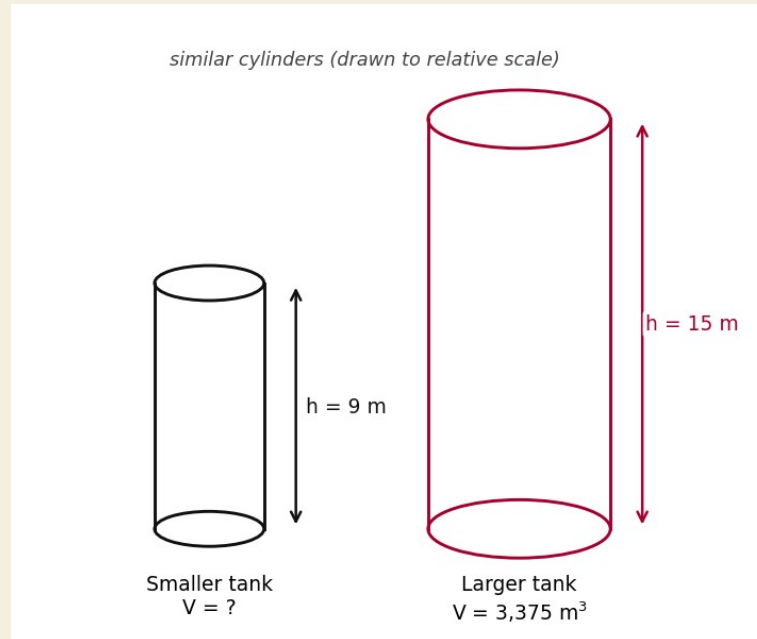
A rectangular fish tank has a base measuring 24 inches by 20 inches and a volume of 5,760 cubic inches. What is the height of the tank, in inches?



Desmos: Type $y = 480x - 5760$ and trace to where it crosses zero — that x-value is the height.

Volume, Surface Area & Similarity of Solids — Question 7 (Medium)

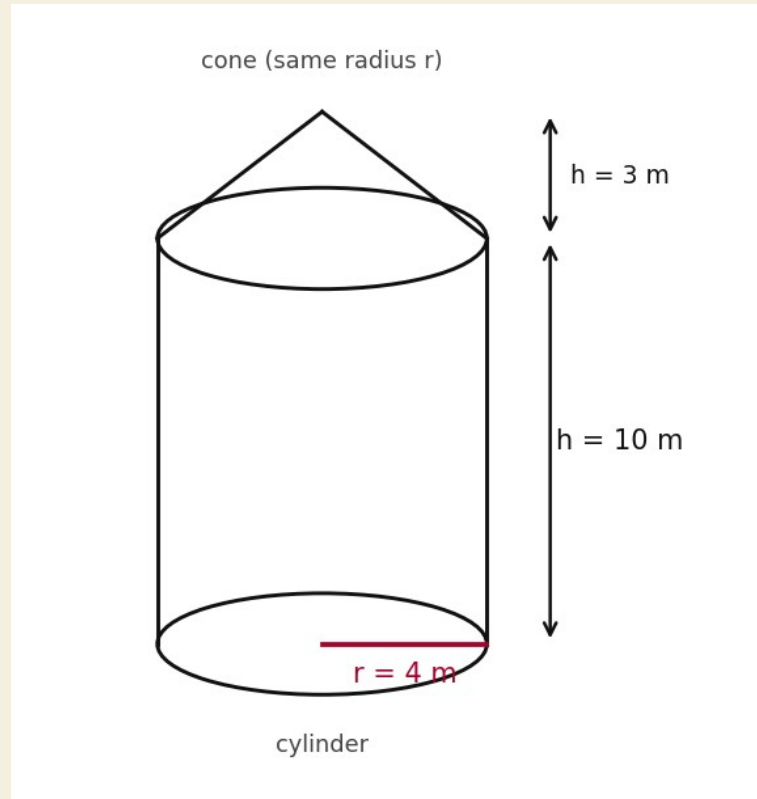
Two similar cylindrical fuel tanks are used at a refinery in Rotterdam. The larger tank has a height of 15 meters and the smaller tank has a height of 9 meters. If the volume of the larger tank is 3,375 cubic meters, what is the volume of the smaller tank, in cubic meters?



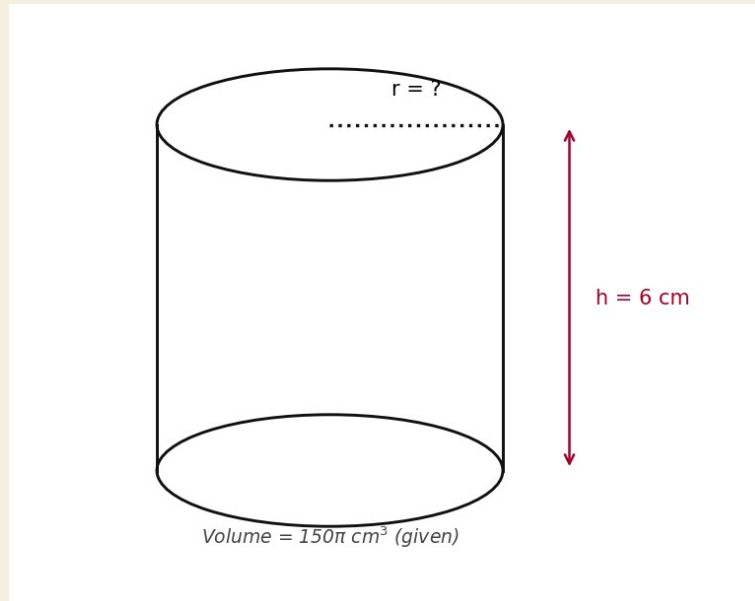
Desmos: Type $y = (9/15)^3 * 3375$ to directly evaluate the smaller tank's volume.

Volume, Surface Area & Similarity of Solids — Question 8 (Medium)

A grain silo near Krakow is shaped like a right circular cylinder topped with a right circular cone that shares the same base radius. The cylindrical part has a radius of 4 meters and a height of 10 meters, and the conical top has a height of 3 meters. What is the total volume of the silo, in cubic meters, in terms of π ?



A cylindrical candle sold in a shop in Vienna has a volume of 150π cubic centimeters and a height of 6 centimeters. The candle is lavender-scented and, once lit, burns for about 40 hours. What is the radius of the candle, in centimeters?

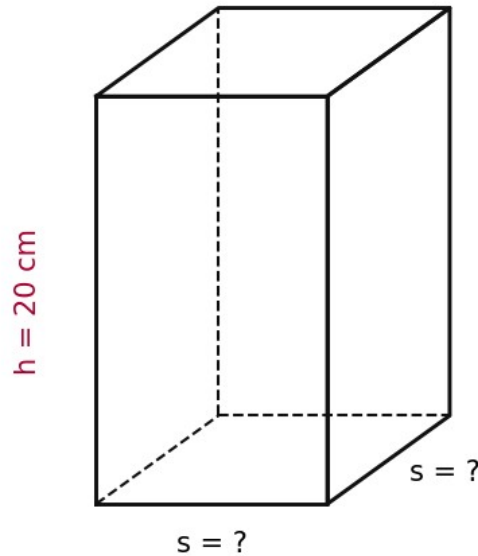


Desmos: Type $y = 6\pi x^2 - 150\pi$ and trace to the positive x-intercept — that's the radius.

Volume, Surface Area & Similarity of Solids — Question 10 (Medium)

A rectangular storage box has a square base and a volume of 720 cubic centimeters. The height of the box is 20 centimeters. What is the total surface area of the box, in square centimeters?

square base, Volume = 720 cm^3 (given)

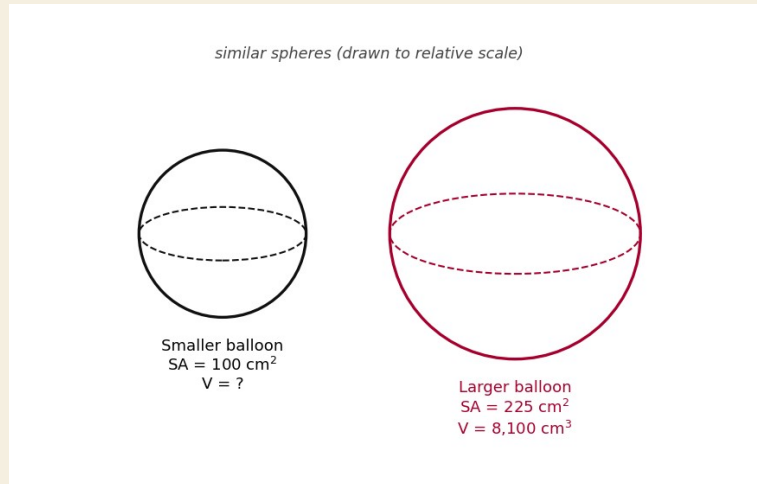


Hard — Questions 11-15

The hardest tier a well-prepared Foundation student should expect before the PSAT — usually two ideas combined in one question.

Volume, Surface Area & Similarity of Solids — Question 11 (Hard)

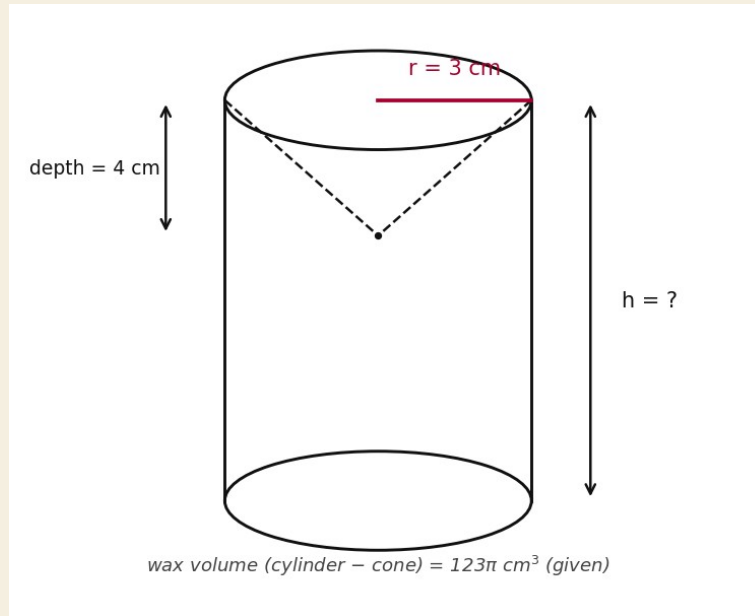
Two similar spherical balloons are being filled with helium for a festival in Valencia. The larger balloon has a surface area of 225 square centimeters, and the smaller balloon has a surface area of 100 square centimeters. If the larger balloon holds 8,100 cubic centimeters of helium, how much helium, in cubic centimeters, does the smaller balloon hold?



Desmos: Type $y = \sqrt{100/225}$ to confirm the linear ratio $2/3$, then cube that result and multiply by 8100 to get 2400.

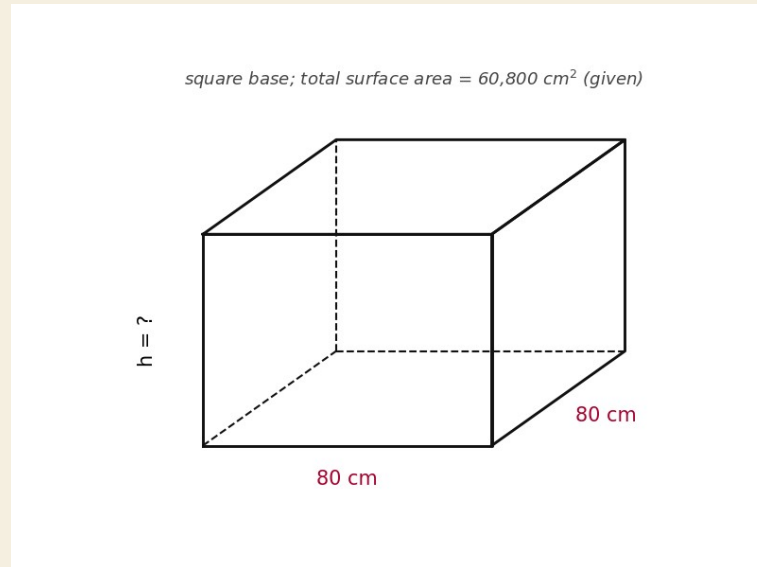
Volume, Surface Area & Similarity of Solids — Question 12 (Hard)

A candle mold used in a workshop in Lyon produces a solid candle shaped like a cylinder with a cone-shaped cavity drilled out of the top. Both the cylinder and the cone-shaped cavity share the same radius of 3 centimeters. The cylinder has an unknown height h centimeters, and the cone-shaped cavity has a depth of 4 centimeters. The volume of wax needed to fill the mold (the cylinder's volume minus the cone cavity's volume) is 123π cubic centimeters. What is the height h of the cylinder, in centimeters?



Desmos: Type $y = 9x - 12 - 123$ and trace to where it crosses zero — that x -value is h .

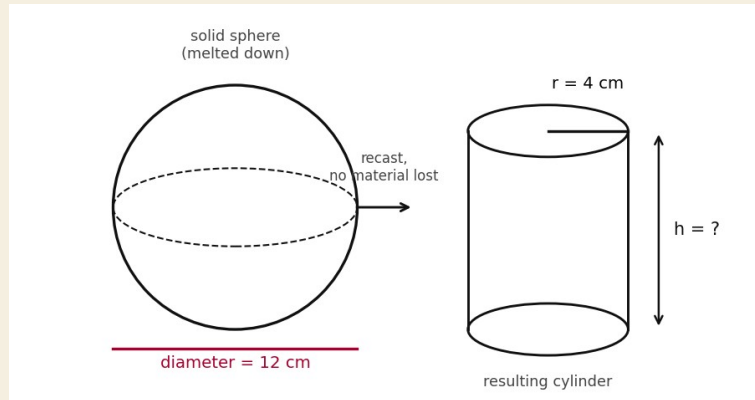
A rectangular water tank installed in Zurich has a square base with side length 80 centimeters. The tank's total surface area, counting all six faces, is 60,800 square centimeters. What is the volume of the tank, in cubic meters?



Desmos: Type $y = 320x - 48000$ and trace to where it crosses zero to get h in centimeters, then finish the unit conversion by hand.

Volume, Surface Area & Similarity of Solids — Question 14 (Hard)

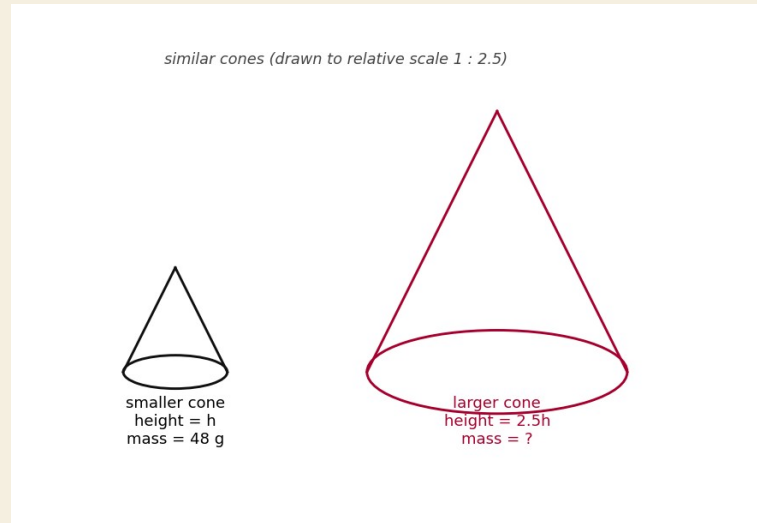
A solid metal sphere with a diameter of 12 centimeters is melted down and recast, with no loss of material, into a right circular cylinder with a base radius of 4 centimeters. What is the height of the resulting cylinder, in centimeters?



Desmos: Type $y = 16\pi x - 288\pi$ and trace to the zero to confirm $h = 18$.

Volume, Surface Area & Similarity of Solids — Question 15 (Hard)

A metalworking shop in Espoo produces two similar solid cone-shaped counterweights from the same uniform metal. The height of the larger counterweight is 2.5 times the height of the smaller counterweight. If the smaller counterweight has a mass of 48 grams, what is the mass of the larger counterweight, in grams?



Desmos: Type $y = 2.5^3 * 48$ to directly evaluate the larger counterweight's mass.



You're Ready

Before Test Day

- Check your work against the full-solution Workbook Practice or with your tutor — never guess whether an answer was right.
- Redo any Hard question you got wrong — the combined twist it tested will very likely reappear in some form.
- Time yourself once you're comfortable, to build speed as well as accuracy.
- This Workbook pairs with the Foundation Course Companion (theory) and Presentation (slides) — use all three together.